

UNIVERSITY OF COPENHAGEN

Master's Thesis

Applying Cointegration Analysis to Model Brain Connections in iEEG Data: Possibilities and Limitations

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Abstract

Understanding brain connectivity through EEG often relies on the assumption of data stationarity, yet real neural data is sometimes non-stationary. In this thesis, I use the Cointegration Analysis, a method which does not require stationarity, to model functional brain networks using intracranial EEG(iEEG) data. The data come from a working memory task where participants either recalled or manipulated a musical melody in their minds. The main goal is to observe how brain region interactions change during the mental manipulation condition compared to simple recall.

First, after viewing EEG data from both signal-processing and time-series perspectives, I chose to work with raw voltages instead of extracted phase. I modeled these voltages as a continuous Ornstein-Uhlenbeck process and then discretized this continuous process into a time series framework to derive the Vector Error Correction Model(VECM).

A key parameter in this model is the cointegration rank. Because traditional rank estimation methods fail for high-dimensional iEEG data, I applied a scree plot, cross-validated matrix angle and mean squared error methods, and a rank selection criterion to estimate this parameter, followed by a sensitivity analysis to ensure its robustness.

After applying the VECM model to the iEEG data, I derived the electrode-level interaction matrices for the recall and manipulation conditions and converted them into region-of-interest (ROI) interactions. Following a cross-subject comparison and correction for multiple testing, the results showed that, after excluding baseline effects, the mental manipulation condition did not significantly change the brain interaction network compared to the recall condition.

Finally, I discuss the possibilities and limitations of interpreting the model matrices as biological interactions between brain regions. I explain why we cannot interpret cointegration vectors in neuroscience as economists do, examine the conflict between cointegration and 0.1Hz high-pass filtering in cognitive neuroscience, and suggest what types of neural data may be best suited for cointegration analysis.

Keywords: Intracranial EEG , Cointegration Analysis, Vector Error Correction Model , Working Memory, Functional Brain Networks, Rank Estimation

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1 Introduction

Working Memory is defined as a mental function that allows people to keep and process information without direct external stimulation[7]. The information in the brain can be either extracted accurately (**recall**) or it can be changed into other forms (**manipulation**). Such mental manipulation of information is central to human higher cognition, but we still know little about how the brain actually performs this transformation process.

Quiroga-Martinez et al. recently [16] studied auditory working memory to investigate the neural dynamics of mental manipulation. In the experiment, participants either recalled a short melody or mentally manipulated it, and their brain activity was recorded using intracranial EEG (iEEG). This iEEG dataset makes it possible to study how interactions between brain regions may differ when information is simply maintained or actively changed in the mind.

Traditional EEG data analysis usually assumes that brain signals are stationary. In practice, the natural physical limits of EEG recordings make long-lasting drifts unlikely, but temporary deviations can still appear in real signals. To relax this stationarity assumption, this thesis introduces a statistical method called cointegration analysis, which can also be applied to non-stationary EEG data.

2 Experimental Cognitive Task and Data Description

2.1 Task Description

According to [16], The subjects are 30 patients with epilepsy who were waiting for surgery. In each trial, they followed a similar procedure, as illustrated in Table 1.

2.2 Data Description

The 30 patients have some electrodes implanted inside their brains to help diagnose epilepsy(see figure 18). Similar to scalp EEG, intracranial EEG(iEEG) records the electrical activity of cortical neuronal populations[21]. The location of their electrodes depends on the specific lesions in each patient, and the number of electrodes also varies a lot.

Our raw iEEG data has three dimensions. The first is the channels, which represent the electrodes inside the patient's brain. The second is the time period, which corresponds to the different task periods described in the above table. The third dimension is the trials. In each trial, subjects may have answered correctly, incorrectly, or given a null

Table 1: Experiment timeline and task descriptions

Period	Condition	Task Description
Baseline Period (-2s to 0s)	Both Conditions	subjects see the trial number and task condition (recall or manipulation) in the center of the screen.
Listening Period (0s to 2s)	Both Conditions	subjects listen to a 3-note melody (1: A-C#-E or 2: E-C#-A)*.
Delay Period (2s to 4s)	Recall	Imagine the previous melody vividly in their brains.
	Manipulation	Imagine the previous melody backwards vividly in their brains.
Test Period (4s+)	Both Conditions	subjects will hear another melody, which may be the same as the previous one from the listening period (50%), its backward version (25%), or a distractor melody (25%). They need to determine whether it is the same as the previous one (recall block) or its backward version (manipulation block).

*Note: Two distractor melodies (A-E-C# and E-A-C#) have the same first note as melody 1 or 2, so that subjects cannot make a choice based only on the first note.

response. The latter two mean that they were either distracted or failed to complete the required working memory task. Therefore, in order to make sure the brain signal data truly reflect the working memory task, we only kept the trials in which subjects answered correctly.

2.3 Preliminary Preprocessing

First, specific filters were applied to decrease the influence of some electrical power noise and the data were downsampled to 400Hz.

Second, by inspecting all the channels carefully, the channels located in the white matter were excluded. In central nervous system, grey matter consists of the cell body and the synapses of neuron, which will generate the signal of interest, while white matter mostly consist of nerve fibers and will not generate such strong signal.

Third, I decide to use bipolar referencing for the EEG data, which is obtained by calculating the difference between neighboring channels. For example, if there are four electrodes inside the participant’s amygdala (AMY), say (A’1, A’2, A’3, and A’4), A’1–A’2, A’2–A’3, and A’3–A’4 will be calculated to generate 3 new bipolar channels. The number of channels has decreased by one because the last electrode has no neighboring channel to subtract. Other brain regions will follow a similar method to subtract and discard channels.

Bipolar referencing has some disadvantages. For example, if the activity of electrode A’2 increases, the voltage of the bipolar channel A’1–A’2 will decrease, while the voltage of A’2–A’3 will increase. These opposite signs can make the signal harder to interpret. Also, if A’1 and A’2 both show similar strong activity, the bipolar signal A’1–A’2 may become small because the two signals cancel each other out [21].

Despite these disadvantages, I still use bipolar referencing in this project. The main reason is that our goal is to model region-to-region interactions, so the signal should reflect local brain activity as much as possible. If we cannot tell where a signal mainly comes from, then a detected interaction between two signals may not really mean an interaction between two specific brain regions. This is especially important in this dataset, because the subjects were patients with epilepsy, and some channels may contain noisy or abnormal waves. With common average reference, abnormal activity from one channel can enter the average reference and then influence many other channels. In contrast, bipolar referencing compares one electrode with its immediate surroundings, so it can reduce common noise from distant brain areas and provide signals with higher spatial resolution.

3 What to Model

Instead of using the time-frequency analysis to infer network changes during the manipulation period, as in the original study [16], I apply cointegration analysis to infer the brain’s functional network from the data. The advantage of this method is that we do not need to assume the EEG signals are stationary, and we can get the interaction network for all channels at once.

When modeling EEG signals, there are generally two different approaches. We can use voltage-based modeling, as shown in [9], or phase-based modeling, as discussed in [13].

3.1 Phase Extraction by Hilbert Transform?

In [13], they first consider the signal $x(t) = A(t) \cos(\phi(t))$, where $\phi(t)$ is the phase angle process (or called the "carrier"), $A(t)$ is the amplitude process (or the envelope), $x(t) \in \mathbb{R}$, and $t \in [0, \infty)$. If $\phi(t)$ increases monotonically, $x(t)$ can be called an "oscillator".

Consider a simple case: $x(t) = A(t) \cos(\omega_0 t)$, where $\omega_0 > 0$. If $x_H(t) = \mathcal{H}\{x(t)\}$ is the Hilbert transform of the original signal $x(t)$, then we can construct its analytical signal: $z(t) = x(t) + i x_H(t)$. According to Bedrosian's theorem [12], if the spectra (i.e., the frequency ranges they cover) of $A(t)$ and $\cos(\omega_0 t)$ do not overlap, we have $x_H(t) = \mathcal{H}\{A(t) \cos(\omega_0 t)\} = A(t) \mathcal{H}\{\cos(\omega_0 t)\} \stackrel{*}{=} A(t) \sin(\omega_0 t)$.

Then, $z(t) = x(t) + i x_H(t) = A(t) (\cos(\omega_0 t) + i \sin(\omega_0 t)) = A(t) e^{i\omega_0 t}$. And the extracted phase is $\theta(t) = \arg z(t) = \arctan\left(\frac{x_H(t)}{x(t)}\right)$.

Based on the simple case discussed above, if we apply the Hilbert transform to a narrowband signal, we can interpret the extracted phase similarly as the instantaneous phase of the central frequency (or carrier frequency). However, for a wideband signal, its Hilbert transform does not have a clear meaning. See the detailed discussion in A.2. From [16] we know the iEEG signal of the cognitive task in this thesis has a frequency range from 1Hz to 160Hz, so it is not a good idea to extract the phase angle directly.

3.2 Filtering?

The following background knowledge is mainly from [11] [10]

When we say "filtering", we usually refer to it as the filtering of a discrete signal. Assume the discretized version of $x(t)$ is $x[n]$ (where n is the n -th sampling point), and its discrete-time Fourier transform (DTFT) is defined as $X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-i\omega n}$, $\omega \in [-\pi, \pi)$. Assume the filtered signal is $y[n]$ and its spectrum is $Y(\omega)$. The filtering in the frequency domain can be represented as $Y(\omega) = H(\omega)X(\omega)$, where $H(\omega)$ is the frequency response.

According to the inverse DTFT of $Y(\omega)$, the filtering in the time domain can be derived as:

$$\begin{aligned} y[n] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} Y(\omega) e^{i\omega n} d\omega \\ &= \sum_{k=-\infty}^{\infty} x[k] \underbrace{\left[\frac{1}{2\pi} \int_{-\pi}^{\pi} H(\omega) e^{i\omega(n-k)} d\omega \right]}_{\triangleq h[n-k]} \\ &= \sum_{k=-\infty}^{\infty} x[k] h[n-k] \triangleq (h * x)[n], \end{aligned} \tag{1}$$

** See detailed derivations in A.1.

If we consider the simplest unit impulse sequence $\delta[n]$, then $y[n] = (h * \delta)[n] = \sum_{m=-\infty}^{\infty} h[m]\delta[n-m] = h[n]$, so $h[n]$ is also called the impulse response. From the definition of $h[n]$, $h[n]$ and $H(\omega)$ form a Fourier transform pair. Although the theoretical formula in (1) uses an infinite sum, we only have finite data in real world, thus the length of $h[n]$ is finite.

Based on the uncertainty principle of the Fourier transform [26], the more concentrated a function is in the frequency domain (e.g., $\sin(2t)$), the wider its range is in the time domain (i.e., $\sin(2t)$ spans from $-\infty$ to $+\infty$). By contrast, if it is a delta function $\delta(t)$ in the time domain, its Fourier transform $X(f) = 1$, which spans all frequency ranges. A similar principle also imposes some restrictions on how we design filters. If we want to distinguish between 2Hz and 2.1Hz, the filter must observe the signal over a longer time window compared to distinguishing between 2Hz and 10Hz. This is exactly what is said in the [28]. " The product of the width of the frequency function and the width of the impulse response cannot be smaller than a specific constant. "

So, if we only select a very narrow frequency range to make the extracted phase angle have a clear physical meaning, the required length of the impulse response will be very large, and the time resolution of the original signal will become lower. Consider an extreme case: if we choose $H(\omega) = 2\pi\delta(\omega)$ (only allowing 0 Hz to pass), the ideal $h[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H(\omega)e^{i\omega n}d\omega = 1$, and $y[n] = \sum_{k=-\infty}^{\infty} x[k]$. For finite data, this corresponds to replacing every sampling point with the sum of the entire sequence. The structure of the noise in the original time series is seriously destroyed because of such smoothing, which will violate some assumptions of some time series statistical modeling.

After these discussions, I decide to model with the original voltage as in [9]

3.3 Other Supplementary Preprocessing Step

0.1Hz high pass filtering

From Figure 1, we can see a continuous slow drift across different trials. Of course, trials do not need to look exactly the same, because brain activity naturally changes from trial to trial. But if this were a cognitive signal related to the working memory task, it should still show some similar patterns in trials with the same condition and similar behavioral performance, such as accuracy and vividness. However, in this plot, the signal mainly changes with the trial number. It slowly moves from one trial to the next, instead of showing a clear event-related pattern, such as a similar increase after the listening period starts, or a delay-period change that appears in the manipulation task but not in the recall task. So the components below 0.1Hz (very slow change) in the iEEG signal are most likely not cognitive event-related, but rather:

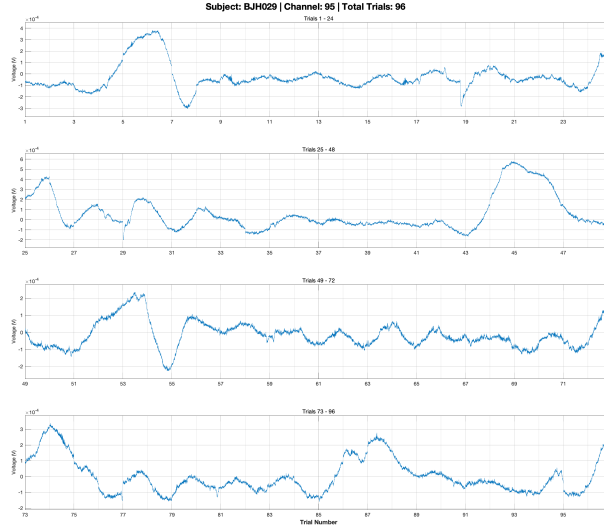


Figure 1: Original iEEG voltage curve before 0.1Hz high-pass filtering. The x-axis shows consecutive trial numbers, and the y-axis shows the recorded voltage in volts. The four panels show the same channel split into groups of 24 trials. The blue curve represents the original voltage signal, where slow continuous changes can be observed across neighboring trials, indicating very low-frequency drift in the recording.

1. Brain background signals, which may influence the performance of the cognitive task, but are not of our interest. We only have a 2-second time window in each delay period; to observe 0.1 Hz, we need at least 10s.
2. Some other slow physiological changes (such as vascular changes).
3. Slow changes of electrodes.

In order for our model to truly capture the cognitive signal, rather than irrelevant noise, I decide to apply 0.1Hz **high**-pass filtering to the signal, as commonly recommended in cognitive neuroscience studies, which will not affect most components of the original signal. Still, it will influence the cointegration analysis, which will be discussed in section 9.2

Manually Inspect and Discard Some Bad Channels and Trials

Even after applying 0.1Hz filtering to all subjects and all channels, there are still some very weird channels, as shown in Figure 2. Then, I check the voltage waveform of each channel and each subject and manually discard some weird channels. I am not sure if this is due to electrical problems or epileptic seizures in certain regions.

Because in our study, the subjects have a relatively small number of trials (each condition has around 30-50), I will discard a trial only if many electrodes behave abnormally at one time point, such as during a massive epileptic seizure.

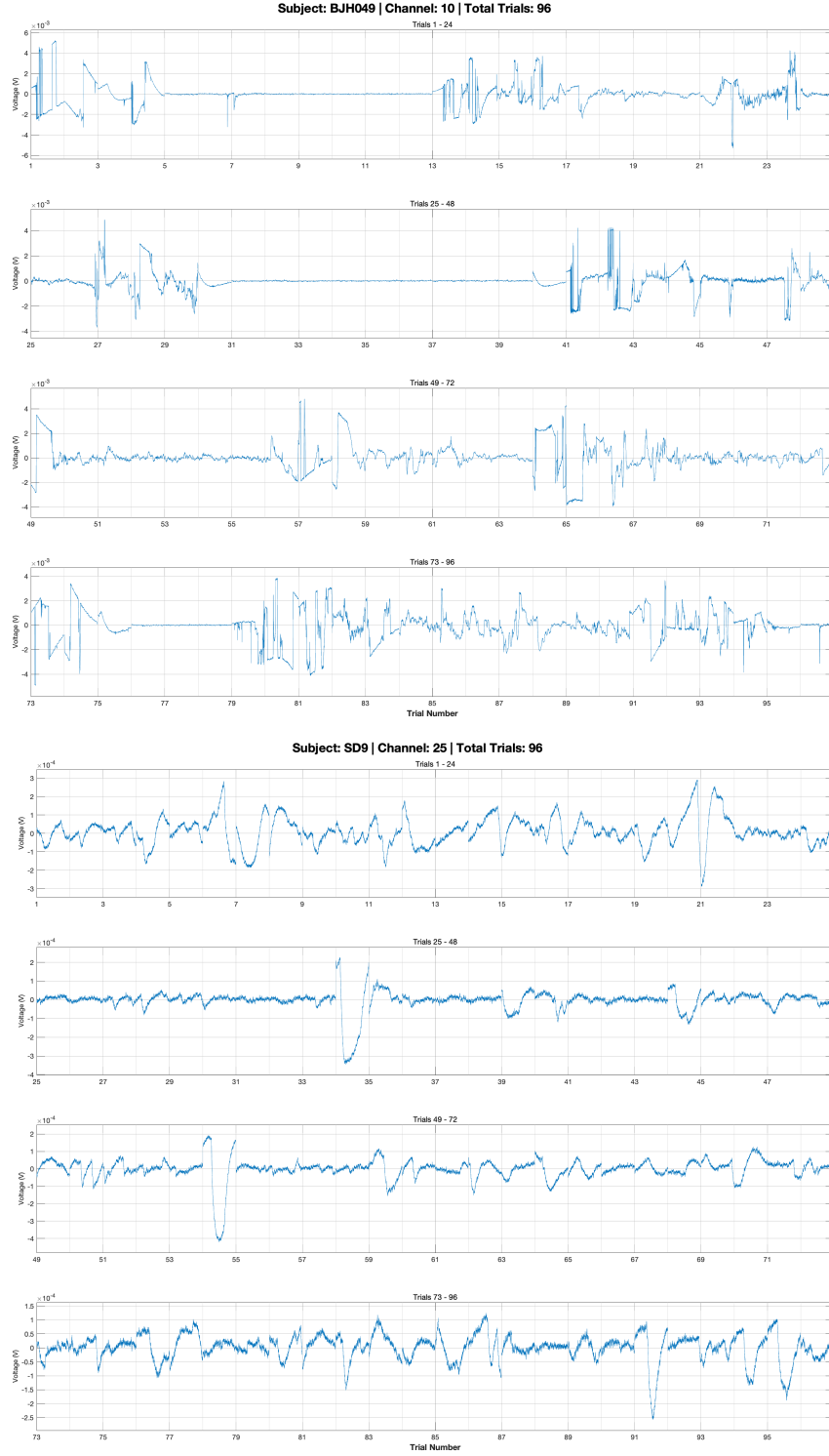


Figure 2: Examples of abnormal channels after applying the 0.1 Hz high-pass filter. The axis definitions and trial grouping are the same as in Figure 1. Even after filtering, these channels still show abrupt jumps, flat segments, and large abnormal fluctuations across trials, suggesting that high-pass filtering alone cannot remove all noisy channels.

Two Special subjects

As shown in Figure 3, two subjects, OS68 and OS70, have very abnormal high voltages. Some channels may reach 0.02 V, while other channels and all other subjects have a range of $10^{-5} \sim 10^{-4}$ V. Even after I apply 0.1Hz filtering to these two subjects, they still show very weird patterns on most channels. So I guess there may be some serious electrical issues in these two subjects, and I decided not to include their data in my model.

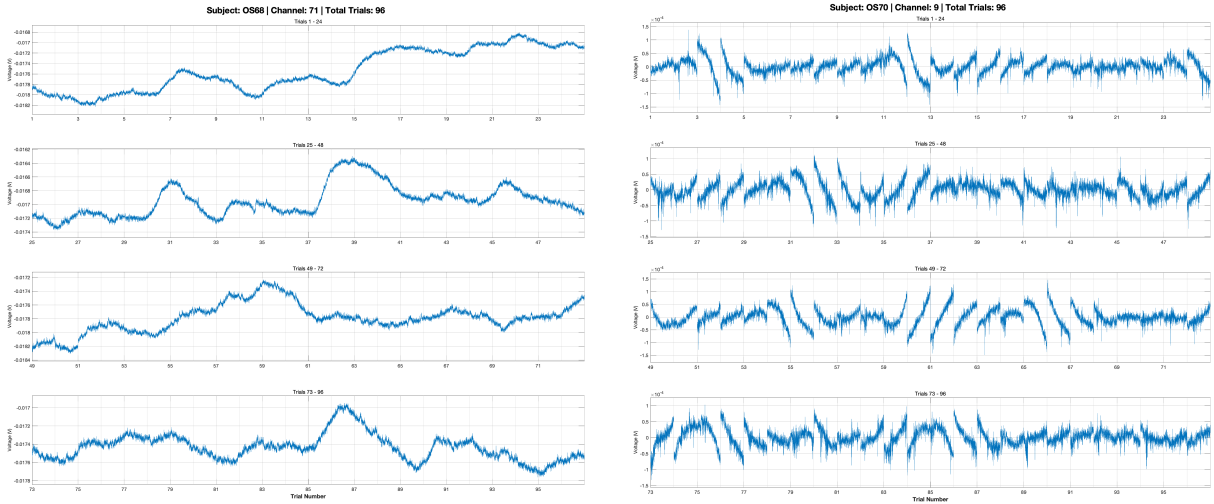


Figure 3: Examples of abnormal signals from subjects OS68 and OS70. The axis definitions and trial grouping are the same as in Figure 1. The left panel shows OS68 Channel 71 before 0.1Hz filtering, where the voltage stays around -0.018 V and is much larger than the normal range of most other channels. The right panel shows OS70 Channel 9 after 0.1Hz filtering, where large sharp changes and repeated abnormal patterns can still be observed. This suggests that the abnormal signals in these two subjects are not only caused by very slow drift, but may come from other problems.

Normalization

For each subject, different channels may have different voltage ranges. Different subjects also have very different voltage distributions. In order to better compare the results between subjects, I decide to normalize the original signal for each channel across all trials to make it have a mean value of 0 and a standard deviation of 1.

4 From Continuous Stochastic Process to VAR and VECM

4.1 Model Definition and Restrictions

As discussed in [9], we can model EEG voltage as the stochastic process. At time t , assume the vector of EEG voltage follow a distribution of multidimensional Ornstein-Uhlenbeck (OU) process.

$$dx_t = P(x_t - m)dt + DdW_t \quad (2)$$

Here is the physical meaning of each symbol:

- $x_t \in \mathbb{R}^p$: EEG voltage vector at time t , p is the number of the electrodes);
- $m \in \mathbb{R}^p$: p dimensional constant mean vector.
- $P \in \mathbb{R}^{p \times p}$: $p \times p$ matrix, represent the velocity of returning to the mean.
- $D \in \mathbb{R}^{p \times d}$: $p \times d$ dimensional matrix, related to the covariance matrix
- $W_t \in \mathbb{R}^d$ is d dimensional Brownian motion,

Assume $0 \leq \text{rank}(P) = r \leq p$. It can be proven that the matrix P can be decomposed into the product of two matrices of rank r ; the proof is shown in B.1:

$$P = \alpha_c \beta_c^\top \quad (3)$$

where $\alpha_c \in \mathbb{R}^{p \times r}$ and $\beta_c \in \mathbb{R}^{p \times r}$ are both column full-rank matrices.

Theorem 1 (The Analytical Solution of O-U process). *Given the initial value x_0 , at $t \geq 0$, the equation (2) has the solution:*

$$x_t = m + e^{Pt}(x_0 - m) + \int_0^t e^{P(t-s)} D dW(s) \quad (4)$$

Proof. The detailed derivation can be seen in B.3 and the discussion of its stability can be seen in B.4 □

Before using this model for EEG data, I first checked what the simulated paths look like under different choices of P . At the very least, the simulated signal should look like real EEG in its overall appearance: on the time scale of the task, the simulated voltage should not explode, and it should not drift away too quickly. Figure 4 shows four examples. Case 3 is the most obvious failure case, where P has one positive eigenvalue and one component grows very fast. Case 2 is also problematic. Although it does not have

a positive nonzero eigenvalue, $\det(\beta_c^\top \alpha_c) = 0$, and the simulated paths still drift away too quickly. In contrast, Case 1 behaves much more reasonably: $\det(\beta_c^\top \alpha_c) \neq 0$, and the two nonzero eigenvalues of P are both negative. Case 4 gives a full-rank comparison, where all eigenvalues are negative and the process stays stable.

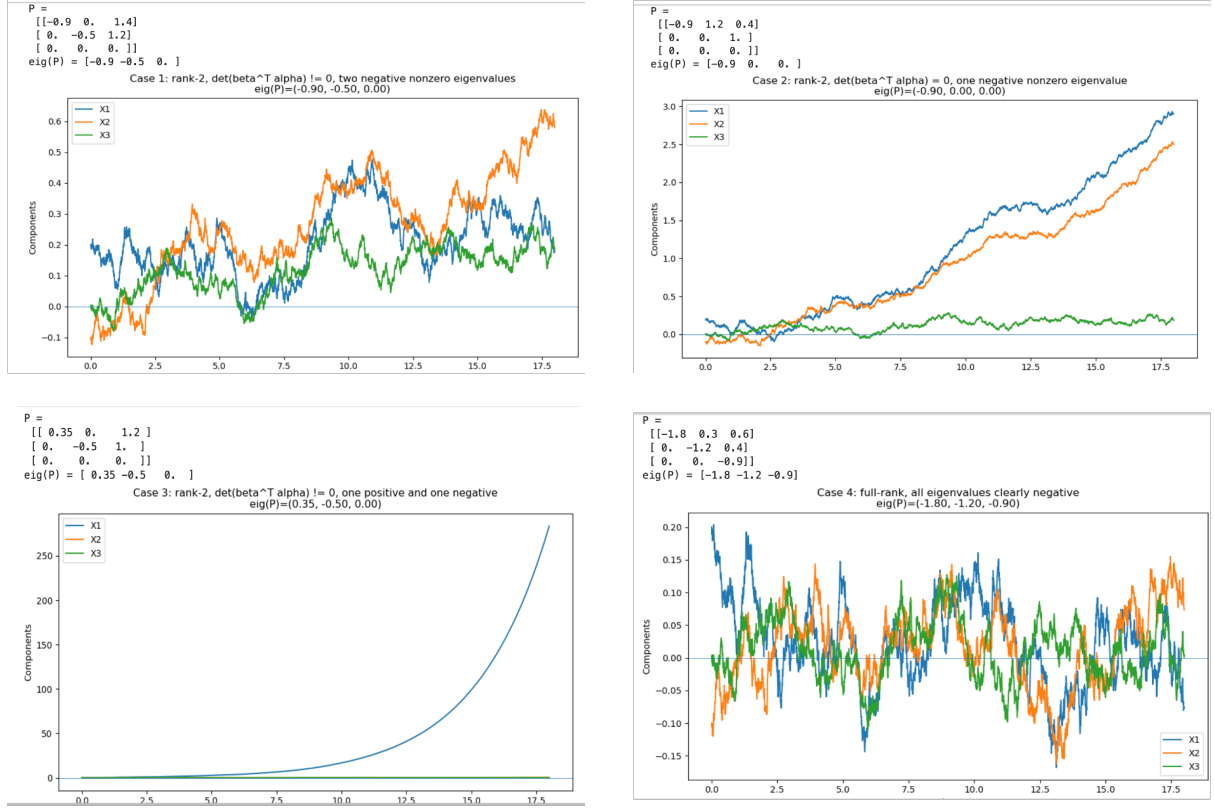


Figure 4: Simulated trajectories under four different choices of P . **Upper left: Case 1**, a reduced-rank example with $\det(\beta_c^\top \alpha_c) \neq 0$ and two negative nonzero eigenvalues, which gives a reasonable-looking path. **Upper right: Case 2**, where $\det(\beta_c^\top \alpha_c) = 0$; the paths can still drift away too quickly even without a positive nonzero eigenvalue. **Lower left: Case 3**, where P has a positive eigenvalue and the process becomes explosive. **Lower right: Case 4**, a full-rank comparison where all eigenvalues are clearly negative and the simulated process remains stable.

Therefore, in the remaining part of the thesis, some restrictions will be imposed on P .

Assumption 1 (The condition of stability). *In order for the voltage to return to the mean value, we assume the real parts of all **nonzero** eigenvalues of P to be negative.*

Assumption 2 (Additional Condition). *We further assume that $\det(\beta_c^\top \alpha_c) \neq 0$;*

Theorem 2 (The equivalent conditions for Assumption 2). *Assume $\alpha_{c\perp}, \beta_{c\perp}$ are orthogonal-complement basis matrices of α_c, β_c respectively, or in other words, let $\alpha_{c\perp} \in \mathbb{R}^{p \times (p-r)}$ and $\beta_{c\perp} \in \mathbb{R}^{p \times (p-r)}$ be basis matrices of the orthogonal complements of $\mathcal{R}(\alpha_c)$ and $\mathcal{R}(\beta_c)$,*

where $\mathcal{R}(\beta_c)$ denotes the subspace spanned by β_c .i.e.,

$$\mathcal{R}(\alpha_{c\perp}) = \mathcal{R}(\alpha_c)^\perp, \quad \mathcal{R}(\beta_{c\perp}) = \mathcal{R}(\beta_c)^\perp.$$

Then the following three conditions (which will be used and explained in later sections) are equivalent:

1. $\det(\beta_c^\top \alpha_c) \neq 0$;
2. $\det(\alpha_{c\perp}^\top \beta_{c\perp}) \neq 0$.
3. $\mathcal{R}(\alpha_c) \cap \mathcal{R}(\beta_{c\perp}) = \{0\}$

The detailed proof of the equivalence of the above conditions is shown in B.2.

4.2 From Continuous O-U Process to Vector Autoregression (VAR)

From the solution 1 of the continuous O-U process, assume we sample the stochastic process at time points $t_0 < t_1 < \dots < t_N$, where the time points have equal intervals $\Delta = t_n - t_{n-1}$. After this, we can derive the following VAR(1):

$$x_n = \mu + Ax_{n-1} + \varepsilon_n \tag{5}$$

Here is the physical meaning of each symbol:

- $x_n \in \mathbb{R}^p$: the observed voltage vector at time t_n (i.e., x_{t_n});
- $\mu \in \mathbb{R}^p$: intercept term, a constant vector;
- $A \in \mathbb{R}^{p \times p}$: the coefficient matrix of VAR;
- $\varepsilon_n \in \mathbb{R}^p$: Gaussian noise vector, and $\varepsilon_n \sim N(\mathbf{0}_p, \Sigma)$.

The relationship between the parameters of the discrete model (μ, A, Σ) and the continuous model (m, P, D) is:

$$\begin{aligned} A &= e^{P\Delta} \in \mathbb{R}^{p \times p} \\ \mu &= (I_p - A)m \in \mathbb{R}^p \\ \Sigma &= \int_0^\Delta e^{Pu} D D^\top e^{P^\top u} du \in \mathbb{R}^{p \times p} \end{aligned} \tag{6}$$

The terms ε_n are independent and identically distributed, and are independent of past (x_{n-1}) . And the Σ is at least semi-positive definite matrix.

(The detailed derivation can be seen in B.5.)

4.3 From O-U,VAR to VECM

From 5 , we have

$$\Delta x_n := x_n - x_{n-1} = \mu + (A - I_p)x_{n-1} + \varepsilon_n. \quad (7)$$

Define:

$$\Pi := A - I_p = e^{P\Delta} - I_p, \quad (8)$$

then we can get VECM models

$$\Delta x_n = \mu + \Pi x_{n-1} + \varepsilon_n. \quad (9)$$

As discussed in [13], regarding the relationship between the continuous process P and the discrete matrix Π from 8, if we want to estimate $\hat{P}$ from $\hat{\Pi}$:

$$\hat{P} = \frac{1}{\Delta} \log(I_p + \hat{\Pi}) \quad (10)$$

where $\log(\cdot)$ denotes the matrix logarithm. If $p = 1$ and $\hat{\Pi}$ is a real-valued number, this logarithm can give us a unique value.

However, for $p > 1$, even if $\hat{\Pi}$ is still a real-valued matrix, the logarithm function $\log(\cdot)$ is a multi-valued function (see Theorem 1.27 in [6]), so there may be an infinite number of $\hat{P}_k$ such that $e^{\hat{P}_k \Delta} = I_p + \hat{\Pi}$. But because the Δ in EEG signals is usually very small, in such a case, we can still get a roughly unique estimate for $\hat{P}$.

But when $\Delta \rightarrow 0$, from 6 with $\Psi_{cont} := -Pm$ and $\Omega_{cont} := DD^\top$:

$$\begin{aligned} P &\approx \frac{1}{\Delta}(A - I_p) \\ \Psi_{cont} &\approx \frac{1}{\Delta}\mu \\ \Omega_{cont} &\approx \frac{1}{\Delta}\Sigma \end{aligned} \quad (11)$$

The detailed derivations and discussions can be seen in Appendix B.6.

4.4 Derived Conditions for VAR and VECM

If $\text{rank}(\Pi) = r$, as proved in B.1 we can have two matrices of rank r $\alpha \in \mathbb{R}^{p \times r}$ and $\beta \in \mathbb{R}^{p \times r}$,

$$\Pi = \alpha\beta^T. \quad (12)$$

and thus VECM model

$$\Delta x_n = \mu + \alpha\beta^T x_{n-1} + \varepsilon_n. \quad (13)$$

As discussed in [13], combining with the fact that continuous matrix $P = \alpha_c \beta_c^\top$, under some weak conditions on Δ , we also have

$$\begin{aligned}
\mathcal{R}(\alpha) &= \mathcal{R}(\alpha_c) \\
\mathcal{R}(\beta) &= \mathcal{R}(\beta_c) \\
\text{rank}(\Pi) &= \text{rank}(P)
\end{aligned} \tag{14}$$

Based on our Assumption 2 on continuous matrix P and Theorem 2, we can get immediately

$$\begin{aligned}
\det(\beta^\top \alpha) &\neq 0 \\
\det(\alpha_\perp^\top \beta_\perp) &\neq 0 \\
\mathcal{R}(\alpha) \cap \mathcal{R}(\beta_\perp) &= \{0\}
\end{aligned} \tag{15}$$

Assume λ_P is an eigenvalue of the continuous matrix P , and λ_Π is an eigenvalue of the discrete matrix Π and λ_A is an eigenvalue of the discrete matrix A . According to [6] Theorem. 1.13, they satisfy:

$$\begin{aligned}
\lambda_\Pi &= e^{\lambda_P \Delta} - 1 \\
\lambda_A &= e^{\lambda_P \Delta}
\end{aligned}$$

Based on Assumption 1 on P , we can derive the following, for any non-zero eigenvalue of the three matrices:

$$\text{Re}(\lambda_P) < 0 \implies |e^{\lambda_P \Delta}| < 1 \implies |\lambda_\Pi + 1| < 1 \implies \text{Re}(\lambda_\Pi) < 0. \tag{16}$$

5 Cointegration Analysis of VECM

5.1 The Stability of VAR

For the VAR(1) model (5), if we omit the deterministic intercept and denote L as the lag operator, where $Lx_n = x_{n-1}$, we can rewrite it as:

$$(I_p - AL)x_n = \varepsilon_n. \tag{17}$$

Theorem 3. Denote the characteristic polynomial of (17) as $\det(I_p - A\zeta)$, where $\zeta \in \mathbb{C}$. This VAR(1) is stable if all eigenvalues of matrix A satisfy $|\lambda_i(A)| < 1$ (i.e., the spectral radius $\rho(A) < 1$). The condition is equivalent to all the roots of $\det(I_p - A\zeta) = 0$ satisfying $|\zeta_i| > 1$ for $i = 1, \dots, p$.

Proof. See the discussion before Equation (2.1.7) in Lütkepohl [22]. $\square$

If there exist roots $\zeta = 1$ such that $\det(I_p - A\zeta) = 0$, then the process does not satisfy the condition in the above theorem (see for example Equation (6.1.2) in Lütkepohl [22]) and we have the following :

Theorem 4. *If $\det(I_p - A\zeta)$ has d unit roots $\zeta = 1$ and all other roots satisfy $|\zeta_i| > 1$, then we can decompose the operator $\det(I_p - AL) = \alpha(L)(1 - L)^d = \alpha(L)\Delta^d$, where $\alpha(L)$ is an invertible operator. $\Delta^d x_n$ is a stable process. Hence, each component becomes stable upon differencing d times.*

Proof. See the discussion in Lütkepohl [22], Section 6.2. $\square$

In the remaining part of this thesis, we only focus on the case where there is a single unit root $\zeta = 1$, and we call such a process $I(1)$ (where $\Delta x_n = x_n - x_{n-1}$ is stable).

Also, as discussed in [13], the order of integration is a stochastic property. Therefore, including deterministic terms in a model does not change the order of integration and the above theorems can also be applied to our model (5).

5.2 Three cases of Cointegration Matrix

From VECM (9), we have:

$$x_n = (I_p + \Pi)x_{n-1} + \mu + \varepsilon_n = Ax_{n-1} + \mu + \varepsilon_n. \quad (18)$$

If we iteratively expand (18), we have:

$$\begin{aligned} x_n &= A^n x_0 + \sum_{i=1}^n A^{n-i} \mu + \sum_{i=1}^n A^{n-i} \varepsilon_i \\ &= A^n x_0 + \left(\sum_{j=0}^{n-1} A^j \right) \mu + \sum_{i=1}^n A^{n-i} \varepsilon_i. \end{aligned} \quad (19)$$

So the long-term behavior of x_n is determined by the spectrum of the eigenvalues of A .

Set $\text{rank}(\Pi) = r$.

(i) $r = 0$ then $\Pi = \mathbf{0}$ and $A = I_p$. Substituting this into (19):

$$x_n = x_0 + n\mu + \sum_{i=1}^n \varepsilon_i, \quad (20)$$

and $\Delta x_n = \mu + \varepsilon_n$. That is, x_n is a p -dimensional random walk with a drift term $n\mu$.

(ii) $r = p$ Π has no zero eigenvalues. From equation (16), $\rho(A) < 1$, so $A^n \rightarrow \mathbf{0}$ and:

$$\sum_{j=0}^{\infty} A^j = (I_p - A)^{-1}.$$

Let $n \rightarrow \infty$. According to [22] (Appendix A, Section A.9.1), the last term in (19) converges. So we obtain:

$$x_n = (I_p - A)^{-1} \mu + \sum_{j=0}^{\infty} A^j \varepsilon_{n-j}, \quad (21)$$

and

$$\mathbb{E}[x_\infty] = (I_p - A)^{-1}\mu$$

(iii) $0 < r < p$ Recall the decomposition: $\Pi = \alpha\beta^\top$. According to condition (15), Π has exactly $p - r$ zero eigenvalues, so $A = I_p + \Pi$ has exactly $p - r$ unit eigenvalues of 1.(see detailed derivations in B.7) . Also, based on (16), all other eigenvalues of A are less than 1.

On the one hand,

Let $\alpha_\perp \in \mathbb{R}^{p \times (p-r)}$ be the orthogonal complement basis of α (with full column rank), such that:

$$\alpha_\perp^\top \alpha = \mathbf{0}. \quad (22)$$

Multiplying the VECM (13) from the left by $\alpha_\perp^\top$:

$$\begin{aligned} \alpha_\perp^\top \Delta x_n &= \alpha_\perp^\top \mu + \alpha_\perp^\top \alpha \beta^\top x_{n-1} + \alpha_\perp^\top \varepsilon_n \\ &= \alpha_\perp^\top \mu + \alpha_\perp^\top \varepsilon_n \\ \alpha_\perp^\top x_n &= \alpha_\perp^\top x_{n-1} + \alpha_\perp^\top \mu + \alpha_\perp^\top \varepsilon_n \\ &= \alpha_\perp^\top x_0 + n \alpha_\perp^\top \mu + \sum_{i=1}^n \alpha_\perp^\top \varepsilon_i. \end{aligned} \quad (23)$$

This indicates that in the $(p-r)$ -dimensional direction orthogonal to the adjustment space $\mathcal{R}(\alpha)$, the projection of x_n is a combination of linear drift and accumulated noise).

On the other hand

From VECM (9) and the decomposition (12), left-multiplying $\beta^\top$:

$$\begin{aligned} \beta^\top \Delta x_n &= \beta^\top \mu + \beta^\top \alpha \beta^\top x_{n-1} + \beta^\top \varepsilon_n \\ \beta^\top (x_n - x_{n-1}) &= \beta^\top \mu + (\beta^\top \alpha)(\beta^\top x_{n-1}) + \beta^\top \varepsilon_n \\ \beta^\top x_n &= (I_r + \beta^\top \alpha) \beta^\top x_{n-1} + \beta^\top \mu + \beta^\top \varepsilon_n. \end{aligned} \quad (24)$$

Thus, $\beta^\top x_n$ is an r -dimensional VAR(1) process with coefficient matrix:

$$\Phi := I_r + \beta^\top \alpha. \quad (25)$$

According to Theorem 3, this VAR(1) is stable if:

$$\rho(\Phi) = \rho(I_r + \beta^\top \alpha) < 1. \quad (26)$$

Based on (16), (15), and also $\Pi = \alpha\beta^\top$ and $\beta^\top \alpha$ have the same non-zero eigenvalues(see [24]Chapter3), $\rho(I_r + \beta^\top \alpha) < 1$ holds. Therefore, we can define

$$\mathbb{E}(\beta^\top x_\infty) := \lim_{n \rightarrow \infty} \mathbb{E}(\beta^\top x_n).$$

5.3 Granger Representation

According to Proposition 6.1 (Granger Representation Theorem) in Lütkepohl [22], for our VECM model (13):

$$x_n = \Xi \sum_{i=1}^n \varepsilon_i + \Xi^*(L)\varepsilon_n + x_0^* \quad (27)$$

where

$$\Xi = \beta_{\perp}(\alpha_{\perp}^{\top}\beta_{\perp})^{-1}\alpha_{\perp}^{\top}. \quad (28)$$

$\Xi^*(L)\varepsilon_n$ is an $I(0)$ process and x_0^* contains initial values.

This condition:

$$\det(\alpha_{\perp}^{\top}\beta_{\perp}) \neq 0 \quad (29)$$

is one of our equivalent conditions in (15), and $\mathcal{R}(\beta_{\perp}) \cap \mathcal{R}(\alpha) = \{\mathbf{0}\}$.

In fact, if there exists a non-zero vector v which belongs to both $\mathcal{R}(\beta_{\perp})$ and $\mathcal{R}(\alpha)$: as an element of $\mathcal{R}(\beta_{\perp})$, v is a direction which has no cointegration constraint and belongs to common stochastic trends, where random shocks from ε_n will accumulate to form an $I(1)$ process. But as an element of $\mathcal{R}(\alpha)$, the vector v also receives short-term feedback. So in this direction, the $I(1)$ property cannot be maintained, and we cannot separate the long-term and short-term elements.

5.4 The Physical Interpretation of Parameters

1. Π : Π_{ij} represents the influence of EEG channel j at the current time on EEG channel i at the next time. Thus, Π defines a functional network across the p channels.
2. The matrix β is called the cointegration matrix because its columns contain cointegration vectors $b \in \mathbb{R}^p$ such that $b^{\top}x_n$ is a one-dimensional asymptotically stationary process. All cointegration vectors constitute the cointegration subspace, and the column vectors of β form a basis for this subspace.
3. The matrix α is called the loading matrix, which describes the strength of the feedback or correction applied to $\beta^{\top}x_{n-1}$ so that it is pushed toward its long-term mean.

$$\Delta x_n = \underbrace{\mu + \alpha \mathbb{E}(\beta^{\top}x_{\infty})}_{\text{constant vector}} + \underbrace{\alpha}_{\text{correction}} \underbrace{[\beta^{\top}x_{n-1} - \mathbb{E}(\beta^{\top}x_{\infty})]}_{\text{error}} + \varepsilon_n \quad (30)$$

In the case where $0 < r < p$, although the matrix Π is unique, its decomposition $\Pi = \alpha\beta^{\top}$ is not. For any $r \times r$ nonsingular matrix Q , we can define $\alpha^* = \alpha Q$ and

$\beta^* = \beta(Q^{-1})^\top$, which still yields $\alpha^*(\beta^*)^\top = \Pi$. This corresponds to transforming the basis of the stationary cointegration subspace into another basis. Consequently, we cannot uniquely determine the exact values of α and β solely based on the data; we can only identify the unique cointegration space spanned by the r column vectors of β . In order to obtain uniquely identifiable parameters, we must apply certain normalization restrictions, which will be specified in the following section.

6 Parameter Estimation of VECM model

6.1 Our Models and the log-likelihood

One trial in EEG is usually short, and we may not have sufficient data points to estimate the VECM model 13 parameters robustly. But for each participant in one specific period and condition (see the previous description in 1), because their brain state is roughly the same across different trials, we can assume the same model structure applies to all the trials. Following the method from [9], I pool all the data points from different trials to estimate the model parameters. In fact, this is equivalent to connecting all the data points in the adjacent trials to form a new time series and introducing an exogenous dummy variable in the VECM model to handle the boundary time points of two adjacent trials, at which the differenced values should be discarded and not used for estimation. In this thesis, I built five VECM models, for two periods and two conditions and one baseline_combined, as shown in 2 and Figure 5. They will be discussed later.

Table 2: All the five models

Period	Condition	VECM Model Name
Baseline Period (-2s to 0s)	Recall	baseline_recall
	Manipulation	baseline_manipulation
	Both	baseline_all_combined
Delay Period (2s to 4s)	Recall	delay_recall
	Manipulation	delay_manipulation

For one specific VECM model, assume the data used for estimating the parameters come from M repeated trials, and the m -th trial contains N_m samples. $x_n^{(m)}$ represents the observations at time n in the m -th trial, and $\Delta x_n^{(m)} = x_n^{(m)} - x_{n-1}^{(m)}$. As we already derived $\varepsilon_n \sim N(0, \Sigma)$: Then we can write the **log-likelihood** of the model based on

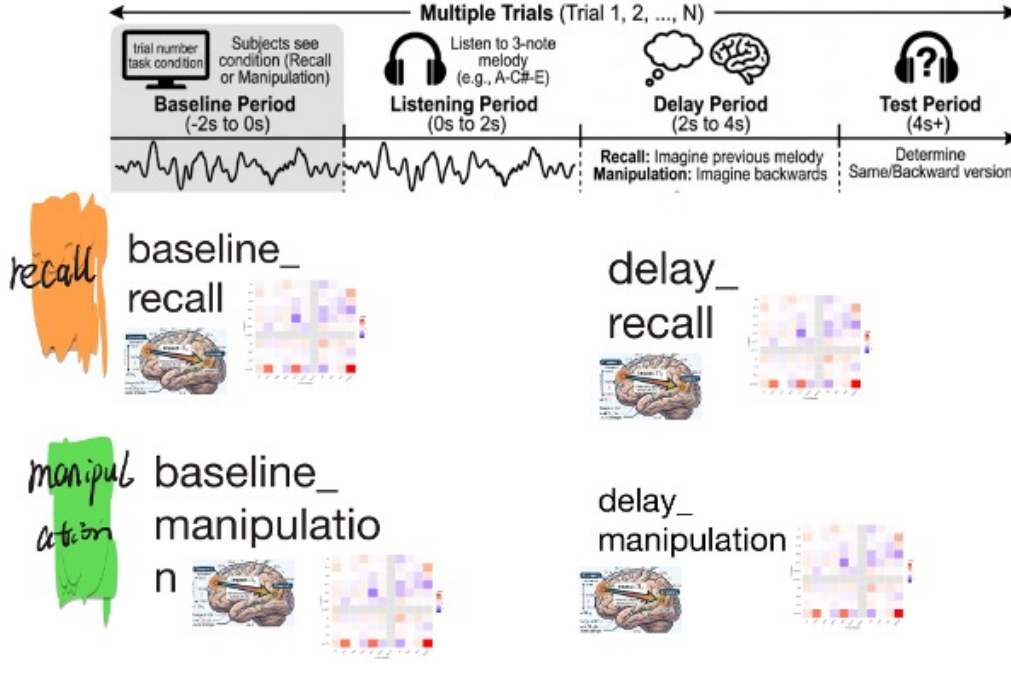


Figure 5: Structure of the four fitted models. The task contains a baseline period and a delay period, and the trials are separated into recall and manipulation conditions. For each condition-period combination, a separate model is fitted and an interaction matrix Π is obtained. Each heatmap visualizes the estimated interaction matrix Π for one condition-period combination. Each colored cell corresponds to one entry Π_{ij} , describing the estimated interaction from channel j to channel i . The final quantity of interest is the change in the interaction matrix from baseline to delay, compared between recall and manipulation.

these pooled data from all trials:

$$\begin{aligned} \ell(\mu, \alpha, \beta, \Sigma; x_{(n=1, \dots, N_m)}^{(m=1:M)}) &\propto -\frac{\log |\Sigma|}{2} \sum_{m=1}^M N_m \\ &\quad - \frac{1}{2} \sum_{m=1}^M \sum_{n=1}^{N_m} \left(\Delta x_n^{(m)} - \alpha \beta^\top x_{n-1}^{(m)} - \mu \right)^\top \Sigma^{-1} \\ &\quad \left(\Delta x_n^{(m)} - \alpha \beta^\top x_{n-1}^{(m)} - \mu \right) \end{aligned} \quad (31)$$

The total sample size is $N_{total} = \sum_{m=1}^M N_m$. For each fixed β , the remaining parameters are estimated based on the standard OLS (or Gaussian MLE) procedure. Substituting their formulas into the original log-likelihood, we can get the profile log-likelihood of β . Then we can just maximize this profile log-likelihood with respect to β :

$$\max_{\mu, \alpha, \beta, \Sigma} \ell(\mu, \alpha, \beta, \Sigma) = \max_{\beta} \max_{\mu, \alpha, \Sigma} \ell(\mu, \alpha, \beta, \Sigma) = \max_{\beta} \ell_p(\beta), \quad (32)$$

where

$$\ell_p(\beta) := \ell(\hat{\mu}(\beta), \hat{\alpha}(\beta), \beta, \hat{\Sigma}(\beta)). \quad (33)$$

6.2 Centering Variable

Define the means $\overline{\Delta x}$ and $\overline{x_{-1}}$ of all time points from all trials as:

$$\overline{\Delta x} = \frac{1}{N_{total}} \sum_{m=1}^M \sum_{n=1}^{N_m} \Delta x_n^{(m)}, \quad \overline{x_{-1}} = \frac{1}{N_{total}} \sum_{m=1}^M \sum_{n=1}^{N_m} x_{n-1}^{(m)} \quad (34)$$

Then define the centered variables by subtracting the means from the expectations .

- $R_{0n} = \Delta x_n - \mathbb{E}[\Delta x_n]$: the random variable of the short-term fluctuation, with $R_{0n}^{(m)} = \Delta x_n^{(m)} - \overline{\Delta x}$: the specific sample
- $R_{1n} = x_{n-1} - \mathbb{E}[x_{n-1}]$: the random variable of the long-term state, with $R_{1n}^{(m)} = x_{n-1}^{(m)} - \overline{x_{-1}}$: the specific sample

Here, all specific residual pairs $\{(R_{0n}^{(m)}, R_{1n}^{(m)})\}_{m=1, \dots, M; n=1, \dots, N_m}$ are pooled for the further derivation.

Then, the covariance matrices s_{ij} and its sample version S_{ij} can be defined as:

$$s_{ij} = \text{Cov}(R_{in}, R_{jn})$$

$$S_{ij} = \frac{1}{N_{total}} \sum_{n=1}^{N_{total}} R_{in} R_{jn}^\top, \quad i, j \in \{0, 1\} \quad (35)$$

where: S_{11} means the sample variance matrix of the long-term state . It is called long-term because x_{n-1} contains all the information from past. S_{00} : the sample variance of the short-term fluctuation . S_{01} : the sample covariance matrix between the long-term state and the short-term fluctuation

6.3 Estimation of Alpha and Mu

For fixed β , we can rewrite the VECM model 9 as:

$$R_{0n} = \alpha(\beta^\top R_{1n}) + \varepsilon_n \quad (36)$$

Both $\alpha \in \mathbb{R}^{p \times r}$ and $\beta \in \mathbb{R}^{p \times r}$ are assumed to have full column rank r .

This is a standard regression problem with $R_{0n} \in \mathbb{R}^p$ as the dependent variable and $\beta^\top R_{1n} \in \mathbb{R}^r$ as the independent variable. By directly applying the standard OLS formula, we obtain the estimate for α :

$$\hat{\alpha}(\beta) = S_{01} \beta (\beta^\top S_{11} \beta)^{-1} \quad (37)$$

and

$$\hat{\mu} = \overline{\Delta x} - \hat{\alpha} \beta^\top \overline{x_{-1}} \in \mathbb{R}^p \quad (38)$$

The discussion about the positive definiteness of S_{11} , S_{00} , and $\beta^\top S_{11} \beta$ can be found in Appendix C.1.

6.4 Estimation of Sigma

The MLE of Σ is the covariance matrix of the sample residuals:

$$\hat{\Sigma}(\beta) = \frac{1}{N_{\text{total}}} \sum_{m=1}^M \sum_{n=1}^{N_m} \hat{\varepsilon}_n^{(m)} (\hat{\varepsilon}_n^{(m)})^\top \quad (39)$$

where $\hat{\varepsilon}_n^{(m)} = R_{0n}^{(m)} - \hat{\alpha}\beta^\top R_{1n}^{(m)}$. After substituting the formula for $\hat{\alpha}$, we get:

$$\hat{\Sigma}(\beta) = S_{00} - S_{01}\beta(\beta^\top S_{11}\beta)^{-1}\beta^\top S_{10} \quad (40)$$

(Detailed derivations can be seen in Appendix C.2).

6.5 The estimation of parameter β

Assuming the cointegration rank r is already determined, we have two equivalent ways to estimate the parameter β .

1. Purely optimization and algebraic derivation

After substituting the estimates of μ , α , and Σ , we can prove that maximizing the profile log-likelihood of β is equivalent to:

$$\max_{\beta} \ell_p(\beta) \iff \min_{\beta} \log |\hat{\Sigma}(\beta)| \iff \min_{\beta} |\hat{\Sigma}(\beta)| \quad (41)$$

And:

$$|\hat{\Sigma}(\beta)| = |S_{00}| \frac{|\beta^\top (S_{11} - S_{10}S_{00}^{-1}S_{01}) \beta|}{|\beta^\top S_{11}\beta|} \quad (42)$$

(The detailed derivations can be seen in Appendix C.3).

Then by setting the normalization condition to be $\beta^\top S_{11}\beta = I_r$, we can convert it into a generalized eigenvalue problem:

$$|\lambda S_{11} - S_{10}S_{00}^{-1}S_{01}| = 0 \quad (43)$$

Let the eigenvalues be $\lambda_1 > \lambda_2 > \dots > \lambda_p$, with corresponding eigenvectors $b_1, b_2, \dots, b_p$. If the cointegration rank is r , the MLE of β is composed of the first r eigenvectors: $\hat{\beta} = [b_1, b_2, \dots, b_r] \in \mathbb{R}^{p \times r}$. and we can also prove $1 \geq \lambda_1 > \lambda_2 > \dots > \lambda_p \geq 0$. Substituting this back, the minimized determinant and the corresponding profile log-likelihood:

$$|\hat{\Sigma}(\hat{\beta})| = |S_{00}| \prod_{i=1}^r (1 - \lambda_i) \quad (44)$$

*Although we only showed this Σ matrix is positive semi-definite in the theoretical derivation (6), empirically, based on our data, $\hat{\Sigma}(\beta)$ is strictly positive definite, which can be verified in the code.

$$\ell_p(\hat{\beta}) \propto -\frac{N_{\text{total}}}{2} \left(\log |S_{00}| + \sum_{i=1}^r \log(1 - \lambda_i) \right) \quad (45)$$

See these lengthy classical algebraic derivations in Appendix D.

2.From a Canonical Correlation Analysis (CCA) perspective

According to Theorem 15.1 in [5], we can formulated the problem as follows:

For any given k th canonical pair ($1 \leq k \leq r$), Let $a, b \in \mathbb{R}^p$, at time n in the m -th trial, we want to find the two linear combinations $a^\top R_{0n}$ and $b^\top R_{1n}$ that maximize their covariance : $a^\top s_{01}b$ subject to the normalization constraints

$$a^\top s_{00}a = 1, \quad b^\top s_{11}b = 1 \quad (46)$$

and $a_i^\top s_{00}a = 0$ for $i = 1, \dots, k-1$.

The maximum correlation is given by: $\lambda_k^{1/2}$, where λ_k is exactly the k -th eigenvalue in 43. and the vector b are also the corresponding eigenvector b_k . Moreover, according to Theorem 15.2 in [5], under the same normalization constraints, given two different eigenvectors b_i and b_j , the corresponding projected variables: $b_i^\top R_{1n}$ and $b_j^\top R_{1n}$ are statistically uncorrelated in the pooled sample.

This indicates that the first cointegration direction (b_1) explains the most canonical correlation between the short-term fluctuations and the long-term state. After excluding the influence associated with the first cointegration direction b_1 , the second cointegration direction b_2 explains the second strongest correlation.

6.6 Replacing OLS with Penalized Regression

6.6.1 Elastic Net

Because the dimension of the iEEG data is high (some subjects have 171 dimensions), although in C.1 we discussed and empirically verified that we do not have a collinearity problem, we still need to penalize the parameters. The reason is that standard OLS tends to give a certain weight to each channels in the error-correction vector $\alpha \in \mathbb{R}^{p \times r}$ for each feature $\beta^\top x_{n-1} \in \mathbb{R}^{r \times 1}$, even if the overall effect of some features or channels are small. So in order to **statistically** find out exactly which channel or feature play an important role and obtain a more stable and robust estimator, we need to use penalized regression. As in [9], I choose to penalize α instead of β because penalized estimate will bring bias and we do not want to distort the stationary subspace formed by the cointegration vectors. Also, penalizing α is easily done by software.

First, we get the estimate $\hat{\beta}$ same as before. Then, we estimate α by minimizing the sum of squared errors with an elastic net penalty:

$$\begin{aligned} \hat{\alpha} = \arg \min_{\alpha \in \mathbb{R}^{p \times r}} & \frac{1}{2N_{\text{total}}} \sum_{m=1}^M \sum_{n=1}^{N_m} \left(R_{0n}^{(m)} - \alpha \hat{\beta}^\top R_{1n}^{(m)} \right)^\top \left(R_{0n}^{(m)} - \alpha \hat{\beta}^\top R_{1n}^{(m)} \right) \\ & + \gamma \left[\frac{1-\omega}{2} \sum_{i=1}^p \|\alpha_{i\cdot}\|_2^2 + \omega \sum_{i=1}^p \|\alpha_{i\cdot}\|_1 \right] \end{aligned} \quad (47)$$

The specific calculations for the Ridge and Lasso penalty terms are:

$$\text{Ridge: } \frac{1}{2} \sum_{i=1}^p \|\alpha_{i\cdot}\|_2^2 = \frac{1}{2} \sum_{i=1}^p \sum_{j=1}^r \alpha_{ij}^2 \Leftrightarrow \text{constrained minimization s.t. } \sum_{i=1}^p \sum_{j=1}^r \alpha_{ij}^2 \leq t \quad (48)$$

$$\text{Lasso: } \sum_{i=1}^p \|\alpha_{i\cdot}\|_1 = \sum_{i=1}^p \sum_{j=1}^r |\alpha_{ij}| \Leftrightarrow \text{constrained minimization s.t. } \sum_{i=1}^p \sum_{j=1}^r |\alpha_{ij}| \leq t \quad (49)$$

Some new parameters in the equation:

- $\gamma \geq 0$: A tuning parameter controlling the overall amount of penalization.
- $t \geq 0$: A parameter in the constrained version corresponds to γ in the penalized version.
- $\omega \in [0, 1]$: A tuning parameter that controls the proportion between the Ridge penalty ($\|\alpha_{i\cdot}\|_2^2$) and the Lasso penalty ($\|\alpha_{i\cdot}\|_1$).
- $\alpha_{i\cdot} \in \mathbb{R}^r$: The i -th row of the loading matrix α .

In the two-dimensional case, considering the geometric shapes of the two constraints above, as discussed in [4] Ch.15, the Lasso constraint is a sharp diamond, and the Ridge constraint is a smooth circle. So, when intersecting with the elliptical contour plot of the standard OLS RSS, Lasso is more likely but not certain to give a sparse solution, which means some channel coefficients in α are 0. Ridge is more likely but not certain to shrink the coefficients equally. The geometric constraint of Elastic Net is between the two and inherits both advantages. So we choose it here.

6.6.2 Equal Hyper-parameter for all four models and channels

For each subject, when determining the Elastic Net hyperparameters γ and ω , the R packages `glmnet` and `cv.glmnet` perform K -fold cross-validation. During this process, I comprehensively consider the overall error across all p channels and all four models (baseline __ recall, baseline __ manipulation, delay __ recall, delay __ manipulation) (in

table 2). To find the best global parameters, I minimize the weighted average of the cross-validation errors:

$$(\hat{\gamma}, \hat{\omega}) = \arg \min_{\gamma, \omega} \sum_{k=1}^4 \frac{N_k}{N_{\text{whole}}} \left(\frac{1}{p} \sum_{j=1}^p \text{CV}_{k,j}(\gamma, \omega) \right) \quad (50)$$

- $\hat{\gamma}$ and $\hat{\omega}$: The optimal hyperparameters finally chosen.
- $k \in \{1, 2, 3, 4\}$: The index representing the four models (or conditions).
- N_k : The number of samples in the k -th model.
- N_{whole} : The total number of samples across all four models ($N_{\text{whole}} = \sum_{k=1}^4 N_k$).
- p : The total number of channels for that participant.
- $\text{CV}_{k,j}(\gamma, \omega)$: The cross-validation error calculated by `cv.glmnet` for the j -th channel in the k -th model, given the parameters γ and ω .

The reason I use equal hyperparameters is that I want to make the results comparable across different conditions and different channels. Because the cognitive question we care about is how each region of interest (combining different channels) changes in the manipulation condition compared to the recall condition (both minus the baseline). If we do not use the same set of parameters, for example, if one channel in the estimated α under one condition is very small, we cannot tell if it is small because the penalty parameter is too large and thus the coefficients get punished to be smaller, or if it is physiologically truly small. So, to get better interpretability, we choose to sacrifice some fitting accuracy and do not select a separate set of hyperparameters for each channel and each condition.

Different subjects have different numbers of electrodes, choices of cointegration rank, and data quality. It is unreasonable to choose the same hyperparameters for all subjects, which may bring a large bias. Even so, because I did normalization for each subject's data in the preprocessing step, we can still try to compare across them, but we need to be careful.

6.7 Convert Some Parameters into Continuous Version

After getting the estimated generalized eigenvector $\hat{\beta}$ and the penalized estimate $\hat{\alpha}$, the intercept μ , the covariance matrix Σ , and the interaction matrix Π are estimated (38) (39) (12) based on these two estimates. Also, because the 400Hz sampling rate is very high here and Δ is relatively small, we can calculate the corresponding parameters of the continuous OU process by dividing by Δ according to (11).

7 Method of determining cointegration rank

7.1 Classical Trace Test

In traditional low-dimensional settings, we usually use the trace test to determine the cointegration rank. $\ell(r_0)$ means the maximum value of the log-likelihood if we set the cointegration rank to be r_0 .

- Null Hypothesis $H_0 : r \leq r_0$
- Alternative Hypothesis $H_a : r = p$

This test is based on likelihood ration test and from equation (45), the test statistics are:

$$-2[\ell(r_0) - \ell(p)] = -N_{\text{total}} \sum_{i=r_0+1}^p \log(1 - \lambda_i) \quad (51)$$

If the null hypothesis holds, i.e., the system has only r_0 really stationary relationships, then the remaining $p - r_0$ eigenvalues should theoretically be 0. (If the long-run combination $b_i^\top R_{1n}$ is not stationary, we cannot use it to predict the short-term R_{0n} , and the correlation for $r > r_0$ is most likely dominated by noise and very low). Then the test statistic will be very small; otherwise, it should be very large. If it is larger than a certain critical value at a given significance level, we can reject the null hypothesis.

The traditional procedure tests the cointegration rank sequentially from $r_0 = 0$ to $p-1$. If at r_0 we cannot reject the null hypothesis, then we can conclude the cointegration rank is r_0 . But the trace test statistics do not follow any standard distribution, and the critical values need to be calculated numerically. If the dimension of the iEEG data is really high (for example, some of our subjects have a dimension of 170+), calculating these values requires huge computational power. Therefore, this method can only be applied to subjects who have a relatively small number of electrodes.

7.2 Scree Plot

From the theoretical derivation of the trace statistics, we can get some insights about the cointegration rank from the scree plot, where ordered values are plotted against the rank. Figure 6 shows a scree plot from subject "CP40" for the baseline_all_combined model.

As discussed in the last section, a larger λ_i means there is a stable linear direction such that the long-term past and the short-term change now have the most stable and stronger statistical correlation. Also, based on our normalization constraints, the correlation explained by different λ_i has no overlap (See CCA perspective above).

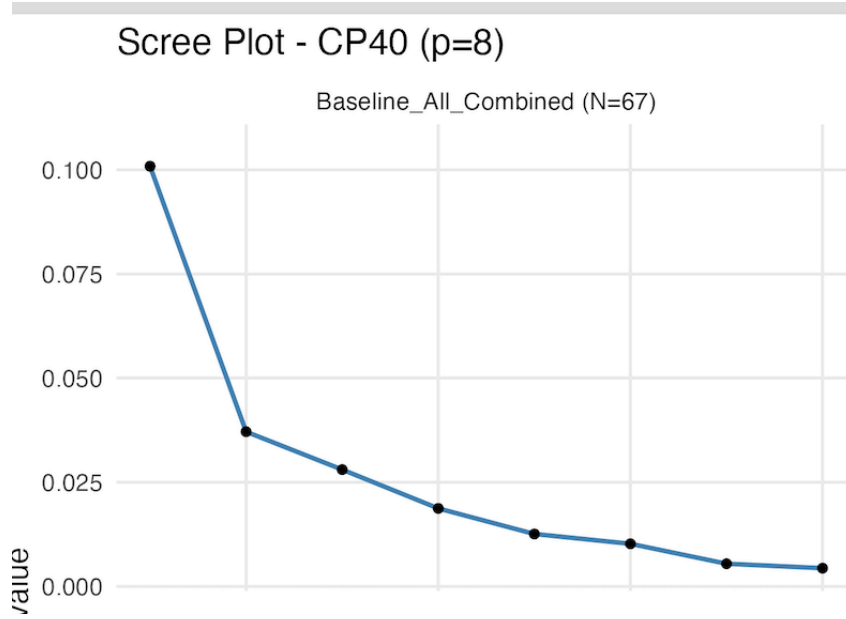


Figure 6: Scree plot for subject CP40 in the baseline_all_combined model. The x-axis shows the eigenvalue index, and the y-axis shows the ordered Johansen eigenvalues. The curve shows how the eigenvalues decrease. The large early drops suggest stronger cointegration directions, while the flatter part later suggests weaker ones.

The main problem of this method is that a small eigenvalue doesn't necessarily mean that the linear combination along the corresponding eigenvector is non-stationary. In fact, consider the simple 1-dimensional case. For clarity, we assume only here that Δx_n and x_{n-1} are already mean-subtracted, we can write $\Delta x_n = \pi x_{n-1} + \varepsilon_n$, where π is the 1-dimensional version of Π . From (8), we have $e^{p\Delta} = \pi + 1$, where p is the 1-dimensional version of P , and let sampling rate $f_s := 1/\Delta$. The square of correlation between x_{n-1} and Δx_n

$$\lambda = \frac{s_{01}^2}{s_{11}s_{00}} = \frac{[\text{Cov}(\pi x_{n-1} + \varepsilon_n, x_{n-1})]^2}{s_{11}\text{Var}(\pi x_{n-1} + \varepsilon_n)} = \frac{\pi^2 s_{11}^2}{s_{11}(\pi^2 s_{11} + \sigma_\varepsilon^2)} = \frac{\pi^2 s_{11}}{\pi^2 s_{11} + \sigma_\varepsilon^2} \quad (52)$$

If x_n is stationary, then $p < 0$ and we have $s_{11} = \text{Var}(x_n) = \text{Var}(x_{n-1})$. Since $x_n = e^{p\Delta}x_{n-1} + \varepsilon_n$, we have $s_{11} = e^{2p\Delta}s_{11} + \sigma_\varepsilon^2 \implies \sigma_\varepsilon^2 = (1 - e^{2p\Delta})s_{11}$. Therefore:

$$\lambda = \frac{(e^{p\Delta} - 1)^2 s_{11}}{(e^{p\Delta} - 1)^2 s_{11} + (1 - e^{2p\Delta})s_{11}} = \frac{1 - e^{p\Delta}}{2} = \frac{1 - e^{p/f_s}}{2} \quad (53)$$

If the original continuous time series of all frequencies is the same, from this equation we can see, when the sampling rate increases (provided the sampling rate is not so low that it loses too much high-frequency information), the generalized eigenvalue will become smaller. However, this does not indicate that those directions are not useful; it is simply because Δ is too small, and we have not observed enough time for mean reversion to occur. In such a small interval, most differenced changes in x_n are likely driven by noise

so it will nearly become a unit-root process.

In fact, if we downsample the actual iEEG data of CP40 ($f_s = 400$ Hz, the dimension is only 8), we can see from Figure 7 that as long as f_s is not too small, overall the eigenvalue will decrease with f_s . and from Figure 6, we are highly likely to choose a small rank.

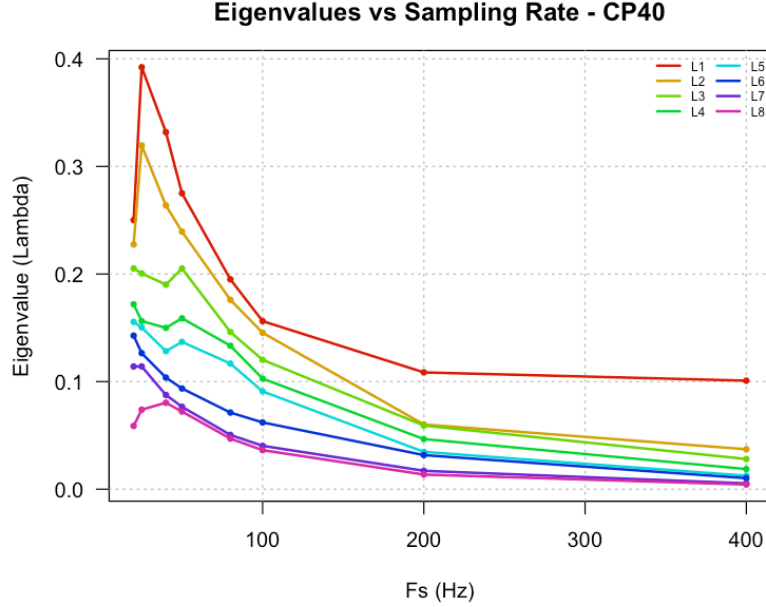


Figure 7: Johansen eigenvalues at different sampling rates for subject CP40. The x-axis shows the sampling rate in Hz, and the y-axis shows the eigenvalue value. Each line represents one ordered eigenvalue, from L1 to L8. The plot shows that the eigenvalues become smaller when the sampling rate increases

But if we use the standard trace test for this subject (as shown in Figure 8), we can reject the null hypothesis $r_0 \leq 7$, so the chosen rank is much larger

Because although the high sampling rate decreases λ , it also increases the total sample size by a lot, which will be taken into the calculation of the trace test statistics. So even a very weak effect can be detected. But I am not saying this method is better. In fact, the cointegration relationship it successfully accepts may be just the correlation between tiny noise because of large N and it may overestimate the rank, not to mention it is not applicable to high-dimensional cases.

In the following section, I will introduce three other methods in [9]: two based on cross-validation, and one based on introducing a penalty term..

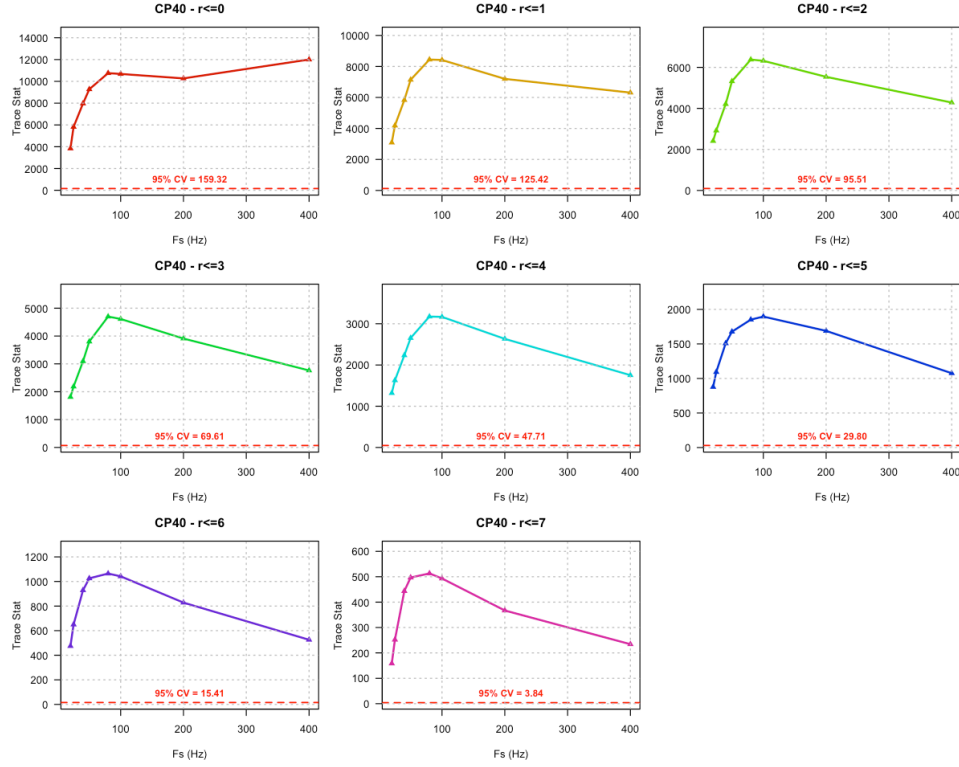


Figure 8: Trace test statistics at different sampling rates for subject CP40. The x-axis shows the sampling rate in Hz, and the y-axis shows the trace statistic. Each panel corresponds to one null hypothesis, from $r \leq 0$ to $r \leq 7$. In each panel, the colored line shows the trace statistic at different sampling rates, and the red dashed line shows the 95% critical value. The trace statistics are far above the critical values in all panels, so the trace test keeps rejecting these ranks.

7.3 Cross-Validation

7.3.1 The definition of Matrix Angle

If we stack a matrix column by column into a long vector,

$$U = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \implies \text{vec}(U) = \begin{pmatrix} a \\ c \\ b \\ d \end{pmatrix}.$$

In the matrix space $\mathbb{R}^{p \times p}$, the Frobenius inner product is used:

$$\langle U, V \rangle_F := \text{tr}(U^\top V) = \sum_{i=1}^p \sum_{j=1}^p u_{ij} v_{ij} = \text{vec}(U)^\top \text{vec}(V). \quad (54)$$

So it is equivalent to the Euclidean inner product of two vectors with length p^2 , and it satisfies the requirements of the definition of an inner product. By the Cauchy–Schwarz

inequality,

$$|\langle U, V \rangle_F| \leq \|U\|_F \|V\|_F. \quad (55)$$

Therefore, when $U \neq 0$ and $V \neq 0$, $\frac{\langle U, V \rangle_F}{\|U\|_F \|V\|_F} \in [-1, 1]$ so we can define the matrix angle:

$$\Theta(U, V) := \arccos\left(\frac{\langle U, V \rangle_F}{\|U\|_F \|V\|_F}\right) \in [0, \pi]. \quad (56)$$

A smaller $\Theta(U, V)$ means that U and V are similar in their overall direction and share a similar overall structure. This angle definition does not depend on the absolute values of the matrix elements. When $U = 0$ or $V = 0$, we set $\Theta(U, V) = \pi/2$, which means the zero matrix does not carry direction information.

7.3.2 Matrix Angle Methodology and ideal results

In an ideal situation, if we already know the real value Π , we can then plot $\Theta(\hat{\Pi}_r, \Pi)$ with respect to r . When r increases from 0 to the true rank r_0 , the matrix angle between $\hat{\Pi}_r$ and the true Π will decrease. But if r is larger than r_0 , the angle will not decrease but will fluctuate around a certain level (dominated by tiny noise and sampling error). But because the true Π is unknown, we need to use another alternative matrix to represent a common reference matrix.

One possible solution is to use the full rank estimate $\hat{\Pi}_p$, but this will produce a curve that decreases to 0 when $r = p$ because these two matrices are equal. So as in [9] I decide to use cross-validation to construct an "observable true matrix".

If we split our data into K folds (in our cognitive task, splitting by trials is more reasonable because it avoids putting data from the same trial into both the training set and the test set), for each fold k :

$\hat{\Pi}_p^{(-k)}$: estimated using the training data *excluding the k -th fold* under the *full rank* setting (as the "approximate true matrix Π on the training side"). This matrix contains both the underlying functional network of iEEG channels and the specific stochastic noise in the training trials.

$\hat{\Pi}_r^{(k)}$, where $r = 0, 1, \dots, p$: estimated using the *k -th fold* validation data under the setting with a *rank constraint of r* .

And then I plot how $\Theta(\hat{\Pi}_r^{(k)}, \hat{\Pi}_p^{(-k)})$ changes with r for each k and find that when r increases but is less than r_0 , because the model still misses some true cointegration vectors, the angle will decrease and the K folds will have little variance. After r_0 , the model has captured all important structures, but because the noise between the training set and test set is different, the angle will not continue to decrease to 0 but fluctuate. And because different folds have their own unique noise, the variance of the angles across different K folds will increase.

7.3.3 Test set Mean Squared Error (MSE)

After splitting the training set and test set by trials, for the k -th fold cross-validation, the test set is denoted as I_k (contains N_k samples), and the corresponding training set is denoted as $\mathcal{T}_{-k}$ (contains N_{-k} samples).

$\hat{\alpha}_r^{(-k)}$ and $\hat{\beta}_r^{(-k)}$ are the estimated loading and cointegration matrices on the training set. The prediction squared error of a single test point can be defined as:

$$\begin{aligned} e_n^{(k)}(r) &= \Delta x_n^{(k)} - \hat{\alpha}_r^{(-k)} (\hat{\beta}_r^{(-k)})^\top x_{n-1}^{(k)} - \hat{\mu}_r^{(-k)} \\ &= \Delta x_n^{(k)} - \hat{\alpha}_r^{(-k)} (\hat{\beta}_r^{(-k)})^\top x_{n-1}^{(k)} - \left(\bar{\Delta} x^{(-k)} - \hat{\alpha}_r^{(-k)} (\hat{\beta}_r^{(-k)})^\top \bar{x}_{-1}^{(-k)} \right) \\ &= \widetilde{\Delta} x_n^{(k)} - \hat{\alpha}_r^{(-k)} (\hat{\beta}_r^{(-k)})^\top \widetilde{x}_{n-1}^{(k)}. \end{aligned} \quad (57)$$

$$\widetilde{\Delta} x_n^{(k)} := \Delta x_n^{(k)} - \bar{\Delta} x^{(-k)}, \quad \widetilde{x}_{n-1}^{(k)} := x_{n-1}^{(k)} - \bar{x}_{-1}^{(-k)}. \quad (58)$$

(note this is different from the previous centering method for R_{0n} and R_{1n}):

And for all K folds all the test points:

$$\text{MSE}(r) = \frac{1}{K} \sum_{k=1}^K \left[\frac{1}{N_k} \sum_{n \in I_k} e_n^{(k)}(r)^\top e_n^{(k)}(r) \right]. \quad (59)$$

7.3.4 MSE methodology and ideal results

Fixing the k -th training fold, we can get $S_{ij}^{(-k)}$ subject to the normalization restriction $(\hat{\beta}_r^{(-k)})^\top S_{11}^{(-k)} \hat{\beta}_r^{(-k)} = I_r$ and the corresponding $b_1^{(-k)}, \dots, b_p^{(-k)}$.

The estimate for rank r is thus:

$$\hat{\beta}_r^{(-k)} = [b_1^{(-k)}, \dots, b_r^{(-k)}] \in \mathbb{R}^{p \times r}, \quad \hat{\alpha}_r^{(-k)} = S_{01}^{(-k)} \hat{\beta}_r^{(-k)} \in \mathbb{R}^{p \times r}. \quad (60)$$

When the rank increases from r to $r+1$, the newly added direction is $a_{r+1}^{(-k)} := S_{01}^{(-k)} b_{r+1}^{(-k)}$, and we have the following nested relationship:

$$\hat{\alpha}_{r+1}^{(-k)} (\hat{\beta}_{r+1}^{(-k)})^\top = \hat{\alpha}_r^{(-k)} (\hat{\beta}_r^{(-k)})^\top + a_{r+1}^{(-k)} (b_{r+1}^{(-k)})^\top. \quad (61)$$

Substituting (61) into (57), we get:

$$e_n^{(k)}(r+1) = e_n^{(k)}(r) - a_{r+1}^{(-k)} (b_{r+1}^{(-k)})^\top \widetilde{x}_{n-1}^{(k)}. \quad (62)$$

We can define the change in the squared residual for a single test point and expand

it (considering $(b_{r+1}^{(-k)})^\top \tilde{x}_{n-1}^{(k)}$ is a scalar):

$$\begin{aligned} \Delta Q_n^{(k)}(r+1) &:= e_n^{(k)}(r+1)^\top e_n^{(k)}(r+1) - e_n^{(k)}(r)^\top e_n^{(k)}(r) \\ &= -2 \underbrace{\left[(b_{r+1}^{(-k)})^\top \tilde{x}_{n-1}^{(k)} \right]}_{\text{projection of past long-term state on new cointegration direction}} \underbrace{\left[(e_n^{(k)}(r))^\top a_{r+1}^{(-k)} \right]}_{\text{projection of old residual on new short-term error correction direction}} \\ &\quad + \underbrace{\left[(b_{r+1}^{(-k)})^\top \tilde{x}_{n-1}^{(k)} \right]^2}_{\text{projection strength on new cointegration direction}} \underbrace{\left\| a_{r+1}^{(-k)} \right\|_2^2}_{\text{overall strength of new response direction}}. \end{aligned} \quad (63)$$

$$\Delta \text{MSE}(r+1) = \frac{1}{K} \sum_{k=1}^K \left[\frac{1}{N_k} \sum_{n \in I_k} \Delta Q_n^{(k)}(r+1) \right]. \quad (64)$$

This decomposition for a single centered test point (63) indicates:

The first part (the term containing -2) shows that the more the newly added cointegration direction aligns with this point, and the more the newly added error-correction direction aligns with the residual left by the rank- r model, the bigger the benefit of adding this rank. The second part shows the cost to pay for adding this rank.

Averaging over all test points, when $r < r_0$, there are always some points that can align well with the newly added beta direction (if they do not align, both the cost and benefit are close to 0 and cancel out). The residuals left by the smaller rank- r model for these points usually contain unexplained fixed structures, i.e., correlations, shared by both the test set and the training set, so they can often be captured by the newly added error-correction direction. At this time, the benefit will be larger than the cost.

When r continues to increase and becomes larger than r_0 , the unexplained residuals—except for model misspecification such as missing lag terms or nonlinear effects—are mostly caused by random noise in the **test set**. They do not have much overlap with the newly added error-correction direction trained from the **training set** (which is also driven by noise at this time). So this term will be very small, and after averaging over different test points, the benefit approaches 0.

But because these noise-driven error-correction directions are also very weak themselves (recall the strength of the last generalized eigenvalues is close to 0), the cost term will also approach 0. So it enters a basically flat period.

7.3.5 Average Cross-Validated Loglikelihood and Ideal Result

We can also consider the cross-validated version of (31)

$$\ell(r) = \frac{1}{K} \sum_{k=1}^K \left\{ -\frac{1}{2} \log |\hat{\Sigma}_r^{(-k)}| - \frac{1}{2N_k} \sum_{n \in I_k} e_n^{(k)}(r)^\top (\hat{\Sigma}_r^{(-k)})^{-1} e_n^{(k)}(r) \right\} \quad (65)$$

For the first term, according to (40), when r increases, because $1 - \lambda_i$ is always less than 1, $\hat{\Sigma}_r^{(-k)}$ will decrease and the first term will always increase, which represents the benefit of adding a rank.

For the second term, I use the inverse of $\hat{\Sigma}_r^{(-k)}$ from the training set to calculate the weighted test set error. If the noise is large in one direction of the $\hat{\Sigma}_r^{(-k)}$, then the tolerance for the test set error in this direction will become larger.

When r increases but less than r_0 , both the MSE of a single test point and $\hat{\Sigma}_r^{(-k)}$ are decreasing, so the second term remains unchanged and the $\ell(r)$ value as a whole will increase.

When r increases larger than r_0 , $\hat{\Sigma}_r^{(-k)}$ continues to decrease, but as discussed in the previous section, the MSE basically will not change. Therefore, the second term will increase, which may cancel out the benefit of the first term. and the $\ell(r)$ value will be likely flat.

7.4 Rank Selection Criteria(rsc):penalty term

As discussed in [9], the rank chosen can also be the number of generalized eigenvalues larger than

$$\theta = \frac{4}{N_{total} - p} \sum_{m=1}^M \sum_{n=1}^{N_m} \left(R_{0n}^{(m)} - \hat{\Pi}_p R_{1n}^{(m)} \right)^\top \left(R_{0n}^{(m)} - \hat{\Pi}_p R_{1n}^{(m)} \right)$$

For our model, $N_{total} \gg p$, so θ can be understood as 4 times the average unexplained component after the cointegration model exhausts its maximum number of parameters(full-rank). This artificially sets a threshold for the small tail eigenvalues.

and this method $\Longleftrightarrow$

$$\hat{\Pi} = \arg \min_{\Pi} \left[\sum_{m=1}^M \sum_{n=1}^{N_m} \left(R_{0n}^{(m)} - \Pi R_{1n}^{(m)} \right)^\top \left(R_{0n}^{(m)} - \Pi R_{1n}^{(m)} \right) + \theta \text{rank}(\Pi) \right] \quad (66)$$

If we denote the first term inside the bracket as RSS, when the rank r increases by k , the change of the term inside the bracket is $\Delta \text{RSS} = \text{RSS}(r+k) - \text{RSS}(r) + k\theta$. So, only when the decrease of the total residual ΔRSS is larger than $k\theta$, can we say it is worth increasing the rank by k .

At first, I felt strange about this formula because θ is an average indicator, but the decrease ΔRSS is related to the sample size N . Inspired by (61), the process of increasing the rank by k may be regarded as adding k new features or covariates $[b_{r+1}, \dots, b_{r+k}]^\top x_{n-1} \in R^{k \times 1}$, and this makes me recall a somewhat similar standard question in regression analysis. For a long regression $Y = X_1\beta_1 + X_2\beta_2 + \epsilon$ and a short regression $Y = X_1\beta_1 + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \sigma^2 I_N)$, if the null hypothesis $\beta_2 = 0$ holds, the

statistic $(\text{RSS}_{\text{short}} - \text{RSS}_{\text{long}})/\sigma^2$ has a chi-square distribution which does not depend on the sample size N , as shown in [4]. But if $\beta_2 \neq 0$ holds (which means adding a covariate can increase explanatory power), this distribution may depend on the sample size N .

In the context of this RSC formula, when $r < r_0$ (similar to $\beta_2 \neq 0$ in the above regression question), the decrease ΔRSS may easily exceed $k\theta$. And because the distribution of ΔRSS depends on the sample size N , a larger N may make it easier to exceed $k\theta$ (or a larger sample size makes it easier to detect a tiny effect). When $r \geq r_0$ (similar to $\beta_2 = 0$, because those new features doesn't play an important role), the distribution of ΔRSS may not depend on N , and only depends on k and the average noise level. So it is reasonable to compare it with $k\theta$. As shown in Figure 9, because the num-

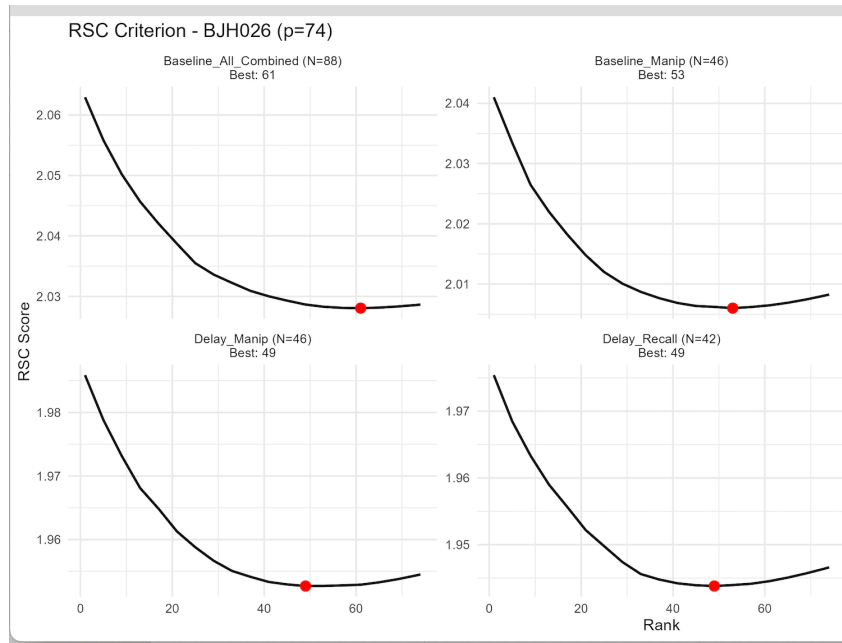


Figure 9: RSC rank selection for subject BJH026. The x-axis shows the tested rank, and the y-axis shows the RSC score. Each panel corresponds to one model, with the number of trials shown in the title. The red point marks the rank with the lowest RSC score, which is the selected rank for that model.

ber of trials in the "baseline_all_combined" model equals the total number of trials in the recall and manipulation models, RSC methods clearly chooses a higher rank in this "baseline_all_combined" model than the others. This pattern exists in all subjects.

7.5 Choose the same cointegration rank for all four models

If we choose different cointegration ranks for all four models, it may be difficult to compare their cointegration vectors later. In order to investigate whether the four models share the same rank, I decide to run all four rank selection methods for all four models

of one participant (COG022) with the highest number of trials.

As shown in 10, 11, 12, and 13, all four models show similar patterns, such as the distribution of generalized eigenvalues, the optimal rank chosen by RSC, and the turning points of the cross-validated plots. Although there may be some differences within a small range, we will show in the later sensitivity analysis section 8.4 that as long as we choose a roughly proper rank, the final results are not sensitive to such small rank changes. Therefore, I decide to use the same rank for all four models.

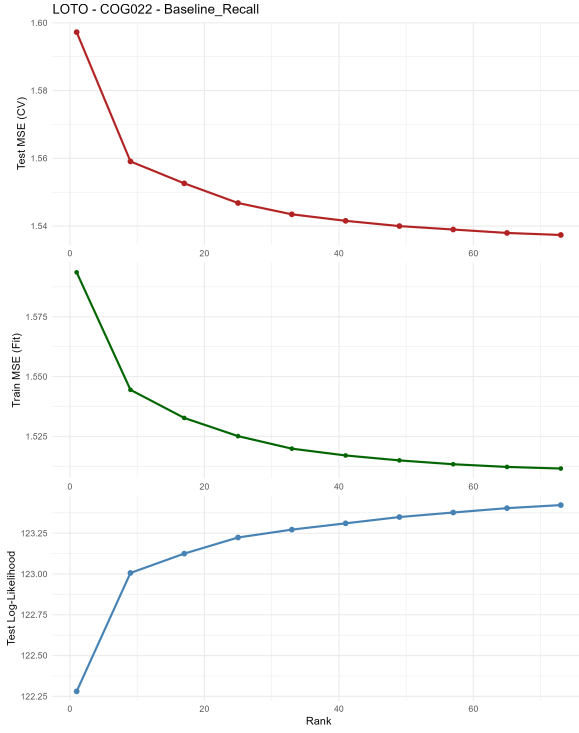
Because the cross-validated matrix angle and MSE method rely heavily on the sample size, for each specific condition and period, some subjects may only have 30 trials and thus leads to an unstable estimate. In order to solve this, for each subject, I decide to combine all the trials (both the recall and manipulation) of the baseline period to estimate the rank in the cross-validated methods. This is reasonable because in the baseline period, subjects haven't started to listen to a specific melody, and the brain states are basically the same (except they know they will do a manipulation or a recall task). The rank selection results of the `baseline_all_combined` models are also very similar to the other four models (see Figure 14).

About the specific number of K , for the cross-validated MSE method, I choose Leave-one-out cross-validation (LOTO) because it is more important to make sure the sample size of the training set is large enough. However, for the matrix angle method, because we need to compare the matrix Π trained from K different testing folds with its corresponding training folds, it is also important to make sure the sample size of the test set is not too small (but still the training folds should have a larger size). I test $K=5$ and $K=4$. As the above figure shows, their angle plots show similar patterns.

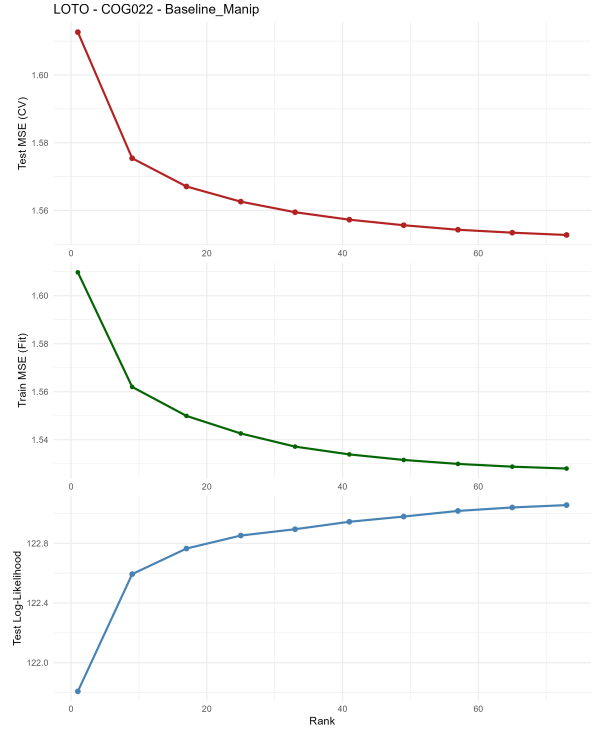
7.6 The turning points of the cross-validated matrix angle and MSE methods

One practical problem is that not all the cross-validated curves will show a clear and single turning point; sometimes there will be three turning points. Take SLCH018 (the dimension is $p = 74$) for example. As shown in Figure 15, when the rank is 10, the curve of cross-validated matrix angle and test mse shows the first significant drop (the first turning point); after that, the curve continues dropping, and when the rank reaches 40 (the second turning point), the curve decreases slower. Until the rank reaches 58-60 (the third turning point), the curve enters into a flat period. In addition, the evaluation based on the rank selection criteria in Figure 15c shows the best rank in the `Baseline_All_Combined` model is 49, which tends to choose a larger rank.

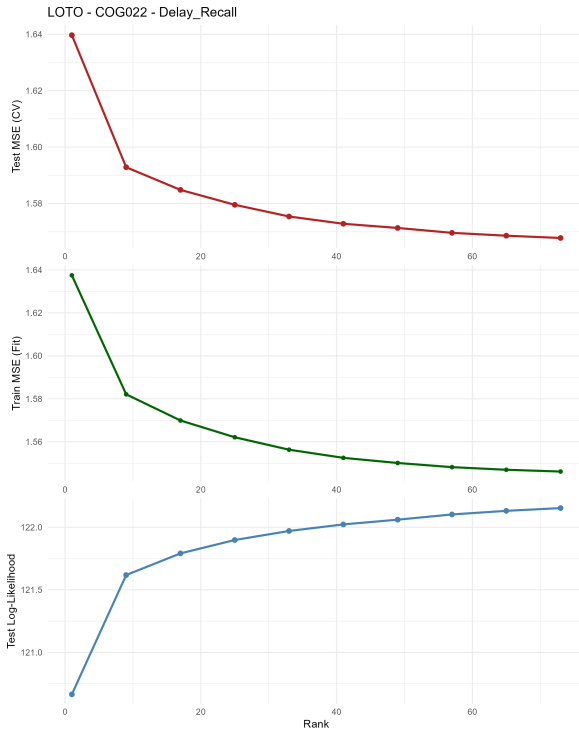
The significant drop around the first turning point indicates that the model has



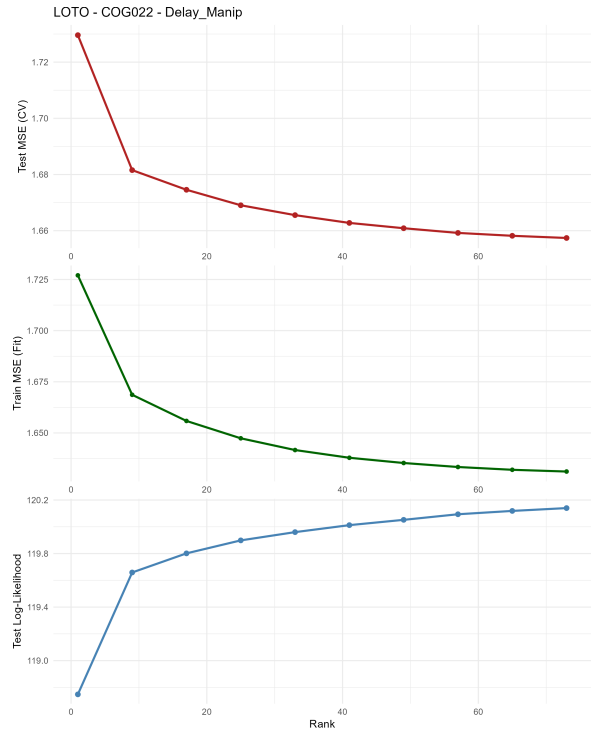
(a) Baseline Recall



(b) Baseline Manipulation



(c) Delay Recall



(d) Delay Manipulation

Figure 10: Cross-validated MSE results for the four COG022 models. Each subfigure is one condition-period model. The x-axis is the rank. In each subfigure, the top panel shows the test MSE from cross-validation, the middle panel shows the training MSE, and the bottom panel shows the test log-likelihood. The four models show similar patterns: the curves change quickly at small ranks and then become flatter at larger ranks.

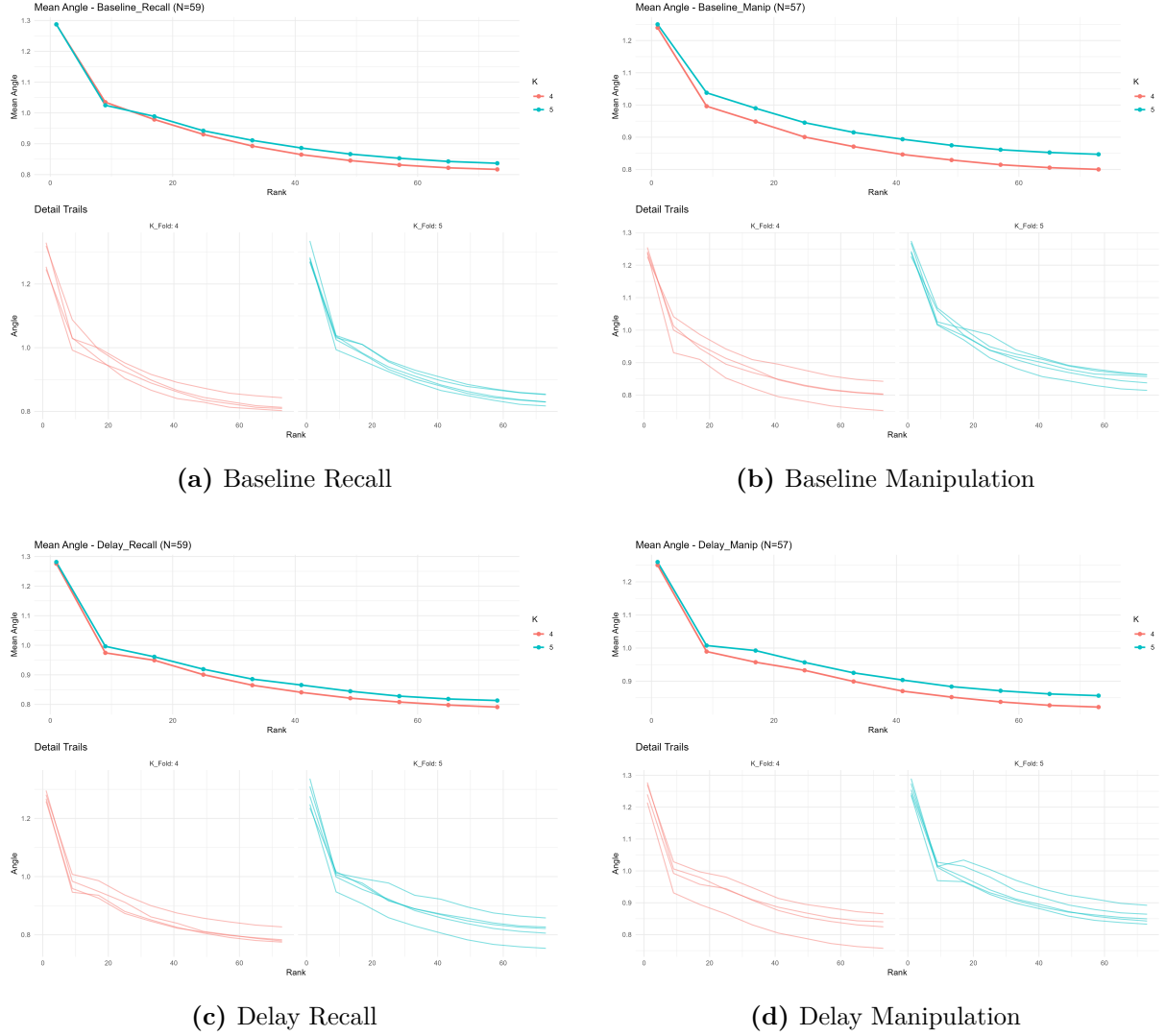


Figure 11: Cross-validated matrix angle results for the four COG022 models. Each subfigure is one condition-period model. The x-axis is the rank, and the y-axis is the matrix angle. A smaller angle means the estimated interaction matrices are more similar. The top panel shows the mean angle for $K = 4$ and $K = 5$, and the lower panels show the fold-level curves. The four models show similar decreasing patterns.

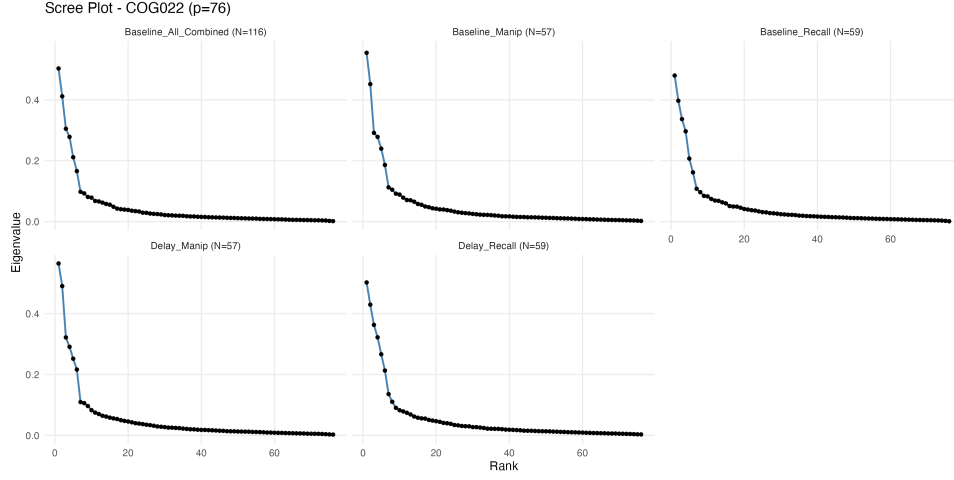


Figure 12: Scree plots for the five COG022 models. Each panel is one model. The x-axis is the ordered eigenvalue index, and the y-axis is the ordered Johansen eigenvalue. In all five models, the first few eigenvalues are much larger, and the curve becomes flatter later.

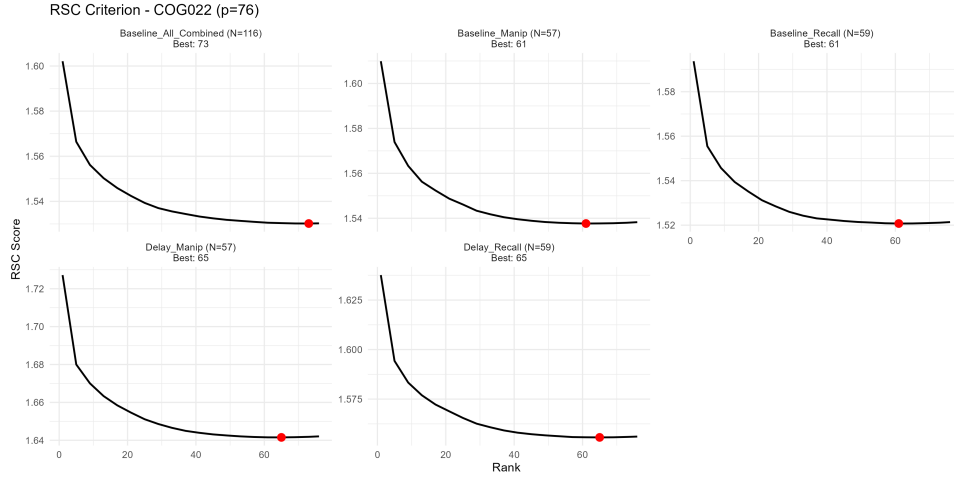


Figure 13: RSC rank selection for the five COG022 models. Each panel is one model. The x-axis is the tested rank, and the y-axis is the RSC score. The red point marks the rank with the lowest RSC score, which is the selected rank for that model. The selected ranks are close across the four condition-period models.

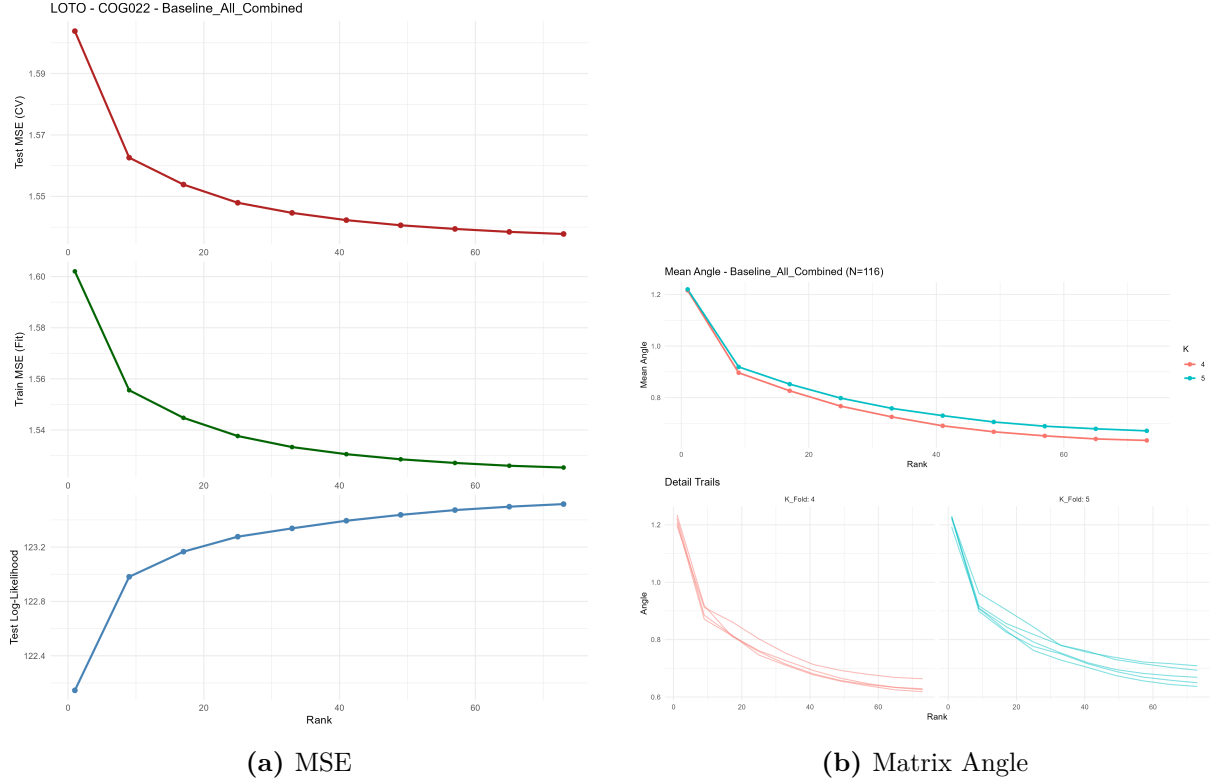


Figure 14: Cross-validated results for the baseline_all_combined model of COG022. The left panel shows the MSE-based results, and the right panel shows the matrix angle results. In both panels, the x-axis is the rank. This combined baseline model uses both recall and manipulation trials from the baseline period, and its rank selection pattern is similar to the four separate condition-period models.

captured the most explainable (explaining the correlation between Δx_n and x_{n-1}) cointegration directions, but because the curve continues to drop and has not reached an absolute flat period, it means that there still remain some weak but stationary cointegration directions that are not fully captured by the model.

In order to figure out which turning point we should choose, for some subjects, I did a sensitivity analysis (see Section 8.4), and found that the final results are most stable if we choose the third turning point as the final rank.

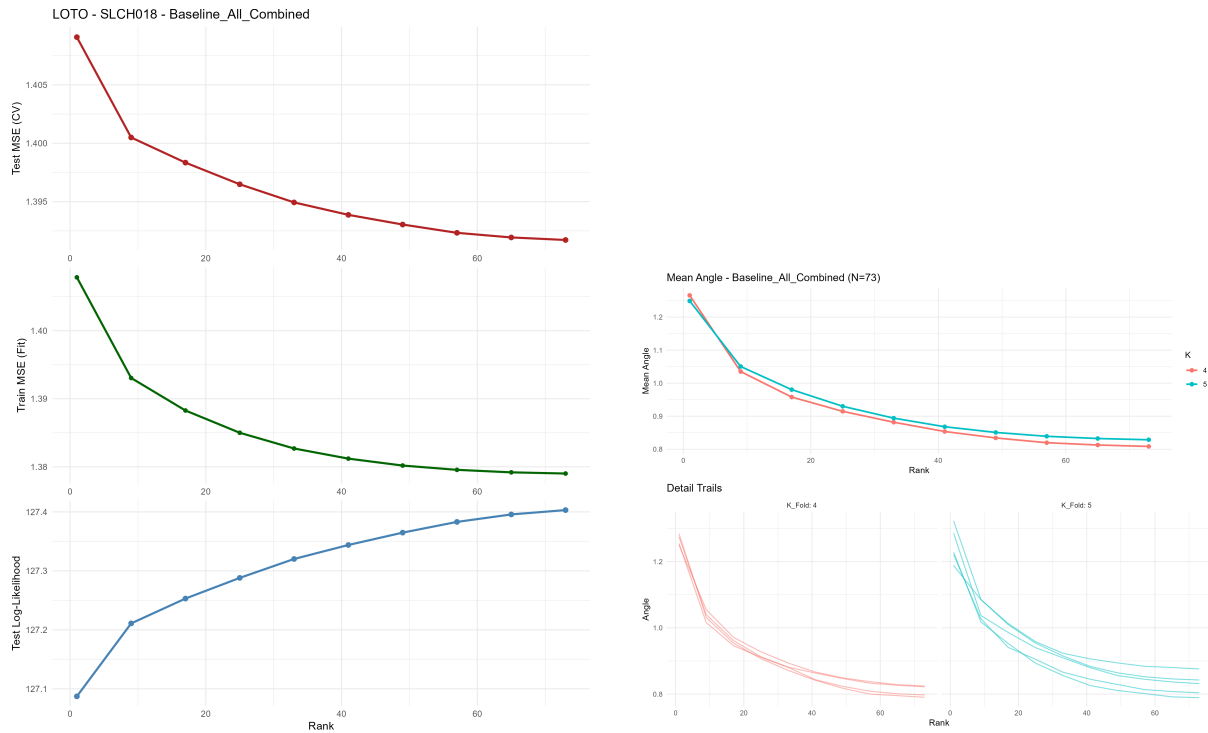
7.7 The discussion of non-ideal results of cross-validated matrix angle and MSE methods

For some subjects, whether using the cross-validated MSE or the matrix angle method, the curve basically steadily drops until the end of the rank. In some subjects' matrix angle plots, 5 curves from different folds tightly stick together and decrease, and the variance is very small (for example, in the previous SLCH018 rank selection plot Figure 15b). This may be because under any rank, the model has not fully learned the complete network structure. No matter which fold is used as the test fold, the dominant structure in the estimated interaction matrix Π is the shared cointegration direction, and thus is very similar. Another possible reason is that although the model has fully learned the complete network structure, the noise difference between different trials is really small.

The underlying reason for this phenomenon may be that we apply 0.1Hz high-pass filtering in the pre-processing step, which makes the data in all channels more stationary and decreases the variation across different trials. We will discuss this more in Section 9.2.

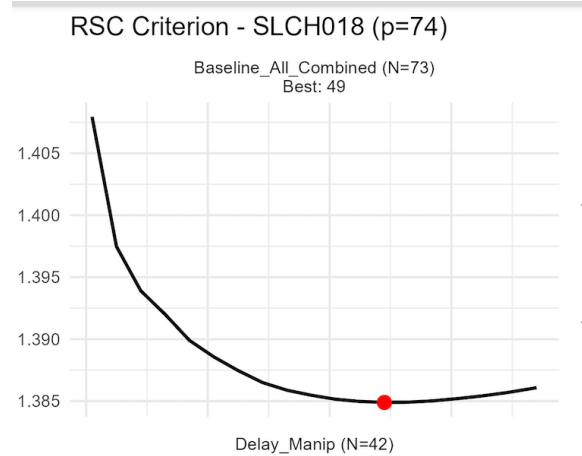
What if we use the data that is not filtered by the 0.1Hz high-pass filter? As shown in Figure 16b, the matrix angle of the 5 folds diverges at just rank = 1, and they don't have a process of closely decreasing together. This indicates that we cannot find any cointegration direction in the unfiltered dataset and confirmed our guess in the preprocessing section to some extent: it is highly likely that the slow drift below 0.1Hz is not induced by the same working memory task across different trials, and they vary significantly between different trials, thus leading to no stable structures. Even if the angle is decreasing, it does not have any turning point, and the decrease magnitude is also less than that of the filtered data. This is because the decrease here may be purely due to the parameter increase and the full-rank setting in the training folds.

It is more evident in the cross-validated MSE plot 16a. In the unfiltered dataset, the test set MSE increases from rank 1 and continues to increase with the rank (although



(a) Cross-validated MSE Curve of Baseline_All_Combined Model for SLCH018

(b) Cross-validated Matrix Angle Curve of Baseline_All_Combined Model for SLCH018



(c) Rank Selection Criteria Results of Model Baseline_All_Combined for SLCH018

Figure 15: Rank selection results for the baseline_all_combined model of SLCH018. In all three panels, the x-axis is the tested cointegration rank. The MSE panel shows the cross-validated test MSE, training MSE, and test log-likelihood; the matrix angle panel shows the cross-validated matrix angle; and the RSC panel shows the RSC score, with the red point marking the lowest score. These plots are used to compare the turning points discussed in the text.

the training set MSE still decreases). From the previous Equation (63), if the test set MSE wants to decrease, the benefit of increasing the rank must exceed the cost. The phenomenon here means that the benefit of increasing the rank by one is very small. The old residuals left by the lower-rank model cannot be projected onto the newly added short-term error-correction direction as a whole. They scatter in many different directions, and thus they are basically 0 after averaging. This indicates that the old residuals contain no information that can be explained by the cointegration direction.

At the same time, as the rank increases, the model has to pay for the cost term in (63). Although the strength of the newly added error-correction vector may not be large, it still does not decrease to 0. Figure 17 (the scree plot of eigenvalues) may serve as an indication of this small but non-zero strength. So as the rank increases, the accumulated cost increases, which leads to a continuous increase in the test set MSE.

When the rank increases further, the benefit and the cost basically remain unchanged due to the same reason as in the ideal case, and the MSE enters a flat period.

Of course, for those subjects which don't show a significant slow drift below 0.1Hz, whether they use 0.1Hz high-pass filtering or not, they have similar EEG voltage forms, and the corresponding cointegration rank estimation results are also very similar.

8 Results Interpretation

8.1 Interaction Matrix

We will discuss why we don't choose α or β to explain in detail in section 9.1 .

As discussed, the formula for $\Delta x_{i,n}$ is:

$$\Delta x_{i,n} = \Pi_{i1}x_{1,n-1} + \cdots + \Pi_{ij}x_{j,n-1} + \cdots$$

Here, Π_{ij} represents how channel j at the current time affects the change of channel i at the next time step.

First, a common mistake is to think that $\Pi_{ij} > 0$ means channel j excites the activity of channel i . But in our study, bipolar referencing for the iEEG signals is used. Here, E_{1n} and E_{2n} represent the raw voltages of the two adjacent electrodes forming the bipolar pair at time n . Even if Δx_{in} increases and thus $x_{in} = E_{1n} - E_{2n}$ increases, we cannot tell whether this is because the first raw electrode E_1 went up or the second raw electrode E_2 went down. Because of this, we cannot interpret the positive or negative sign of the voltage change Δx_n as biological excitation or inhibition.

Second, the sign problem. $\Pi_{ij} > 0$ means channel i tends to change in the same direction as the sign of the current voltage of channel j . A more positive value means

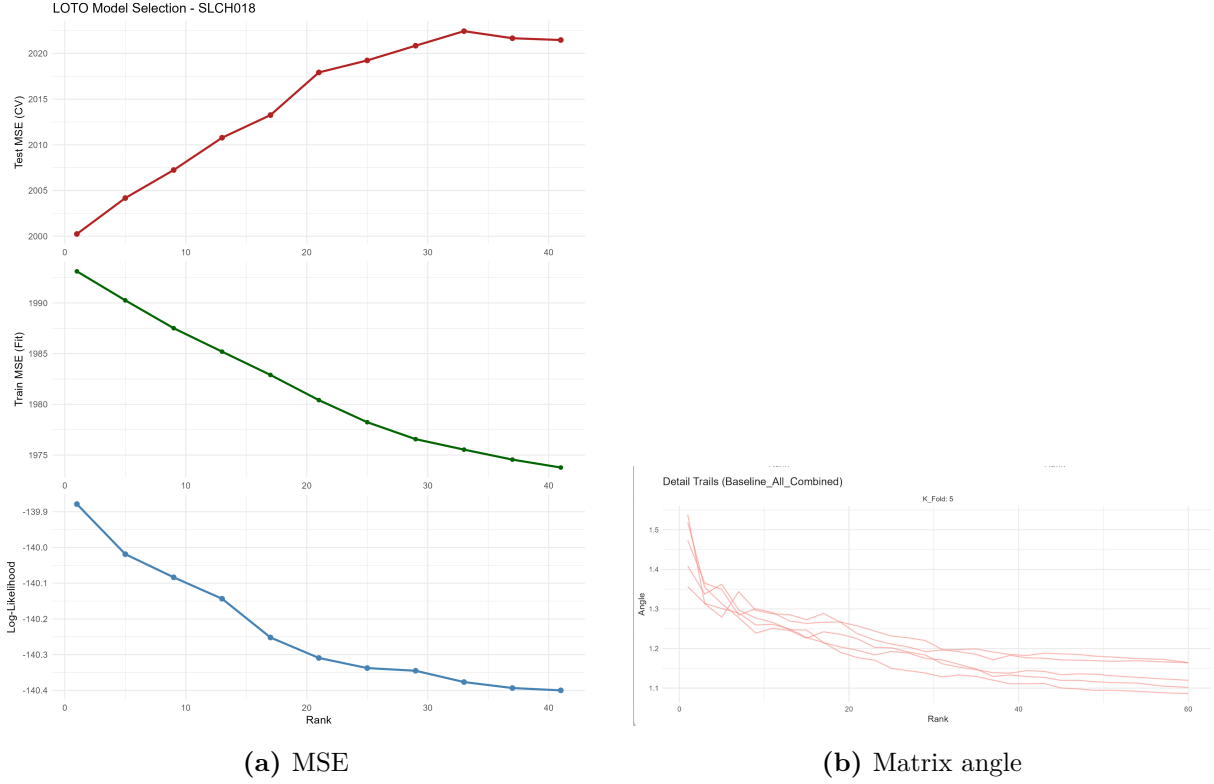


Figure 16: Cross-validated results for SLCH018 without 0.1Hz high-pass filtering. The left panel shows that the test MSE increases with rank, even though the training MSE decreases. The right panel shows that the matrix angle curves from different folds separate from rank 1. This suggests that the unfiltered data does not give a stable cointegration structure across trials.

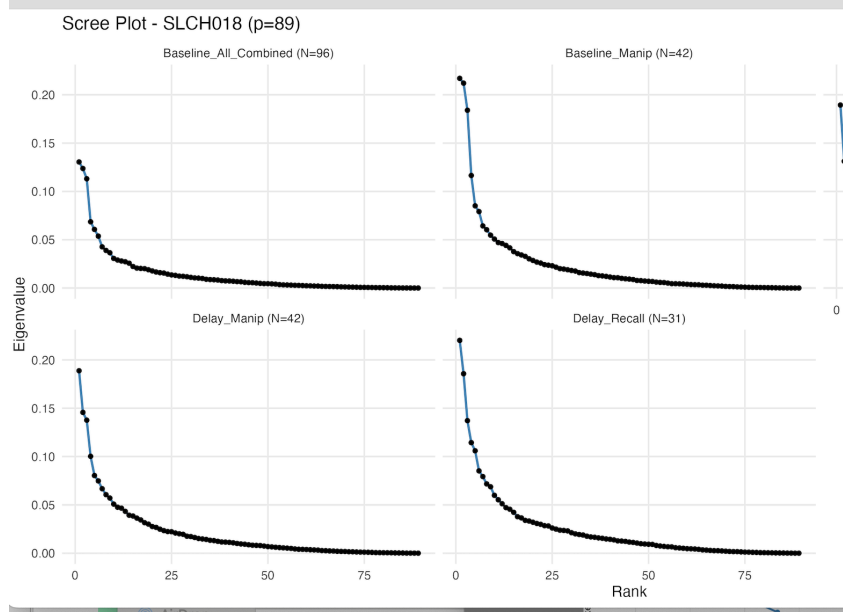


Figure 17: Scree plot for SLCH018 without 0.1Hz high-pass filtering. The x-axis shows the ordered eigenvalue index, and the y-axis shows the ordered Johansen eigenvalues. The plot shows that many eigenvalues are small but still nonzero, which means that adding more ranks can still add model cost even when it does not improve test prediction.

a higher voltage magnitude change. If $\Pi_{ij} < 0$, it means channel i tends to change in the opposite sign direction of current channel j . However, because the voltage sign of channel j is changing at every time, this explanation is very confusing.

Therefore, it is better to use the absolute value of Π_{ij} . A larger absolute value just means channel j has a larger influence on the change of the voltage of the channel i . But even with this, we can only say it causes a larger voltage change after bipolar referencing. We absolutely cannot say this means a larger impact on the original electrical activity.

To see why, consider an example where bipolar voltage $x_{j,n-1} = 1$, and $x_{i,n-1} = 0$ (for instance, raw electrodes $E_{1,n-1} = 3$ and $E_{2,n-1} = 3$). If we exclude influence from other channels, random error term and only focus on the influence of j on i :

- **Baseline** ($\Pi_{ij} = 1$): The bipolar voltage changes by $\Delta x_{in} = 1 * 1 = 1$. Thus, the new bipolar voltage is $x_{in} = 1$. This could happen if:

$$(E_{1,n-1} = 3, E_{2,n-1} = 3) \xrightarrow{\text{big raw change}} (E_{1n} = 60, E_{2n} = 59) \implies x_{in} = 60 - 59 = 1$$

- **Delay** ($\Pi_{ij} = 10$): The bipolar voltage changes by $\Delta x_{in} = 1 * 10 = 10$. Thus, the new bipolar voltage is $x_{in} = 10$. This could happen if:

$$(E_{1,n-1} = 3, E_{2,n-1} = 3) \xrightarrow{\text{small raw change}} (E_{1n} = 30, E_{2n} = 20) \implies x_{in} = 30 - 20 = 10$$

As shown above, the smaller coefficient ($|\Pi_{ij}| = 1$) in the baseline period could mean a much larger change in the raw electrical activity. Because the raw voltage values are lost after the subtraction, the coefficient alone does not tell us the true biological impact.

Therefore, when we talk about the **"interaction"** between channel j and i or the impact of channel j on channel i in our study, it only means : channel j at the current time causes a larger magnitude of **bipolar** voltage change in channel i at the next time step. In the following text, unless stated otherwise, **"interaction"** is strictly limited to this meaning .

8.2 Region of Interest

Because some subjects have been implanted with more than 170 electrodes, it is really difficult to interpret the original large 170×170 matrix, and the differences of the interaction between electrodes are mostly not so large. Also, in order to better compare our results with other literatures, which may use a different connectivity analysis method, I decide to follow the same way as [16] and divide the electrodes into 11 different regions of interest (A1, STG, STS, MTG, HPC, PHC, parietal, VLPFC, DLPFC, M1, and ACC) as shown in Figure 18. Then we convert the original channel-channel interaction into ROI-ROI interaction.

Assume we have two target regions of interest: the sender brain region is denoted as A (From_Region, which has the same meaning as the column index j in Π_{ij} , which exerts influences), and the receiver brain region is denoted as B (To_Region). Let S_A be all the N_A channels that are assigned to brain region A , and S_B and N_B have the equivalent meaning for brain region B . Define the interaction between brain region A and region B as:

$$W_{A \rightarrow B} = \frac{1}{N_A \times N_B} \sum_{j \in S_A} \sum_{i \in S_B} |\Pi_{ij}|$$

where Π_{ij} may come from all our four models (baseline_recall, delay_recall, baseline_manipulation, delay_manipulation). If A and B come from the same region ($W_{A \rightarrow A}$), then it is calculated as the average connection strength between all different channels in the same region.

Then, in order to subtract the influence of the baseline period, we calculate:

$$\Delta W_{A \rightarrow B}^{\text{Recall}} = W_{A \rightarrow B}^{\text{Delay_Recall}} - W_{A \rightarrow B}^{\text{Base_Recall}}$$

$$\Delta W_{A \rightarrow B}^{\text{Manip}} = W_{A \rightarrow B}^{\text{Delay_Manip}} - W_{A \rightarrow B}^{\text{Base_Manip}}$$

Finally, we define the **interaction contrast** between manipulation and recall:

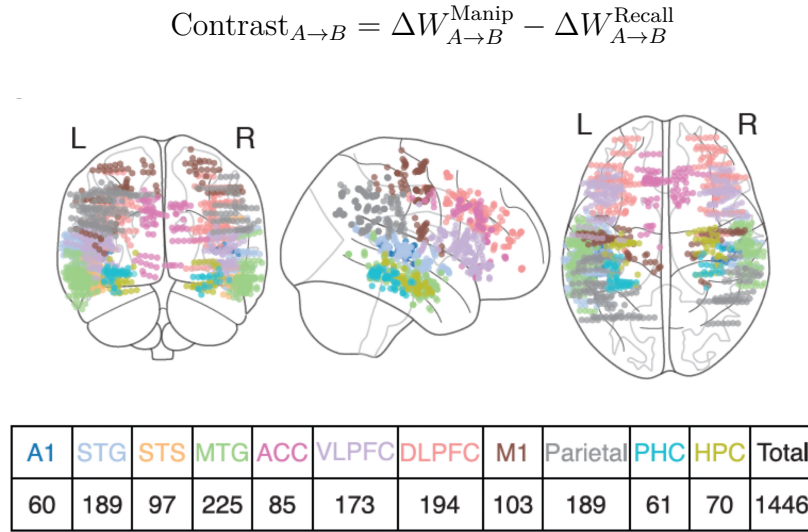


Figure 18: Brain regions of interest (ROIs) used in the analysis. The colored dots show electrode locations assigned to different ROIs, and the table gives the number of electrodes in each ROI. The colors in the brain plots match the ROI labels in the table. This figure is from [16].

Figures 19 and 20 show the ROI-ROI interaction contrast matrix of subjects BJH046 and BJH052. For example, in the results of BJH052, the connection from M1 to parietal has a dark red color, and the value is around 6, which means: after subtracting the influence of their respective baseline periods, compared to the recall condition, the manipulation causes the average interaction from M1 to parietal to increase by 6 units. Because we already z-scored all the channel data based on all its trials, this 6 means the influence of the fluctuations of M1 at the current time on the parietal’s voltage fluctuation change at the next time step increases by 6 standard deviation levels. But for BJH046, the interaction from M1 to parietal is shown as -4.

This may be because subjects have different numbers and locations of electrodes in that region. To investigate possible general patterns, we compare the interaction contrast across subjects.

8.3 Cross-subject Comparison and Statistical Test

Although in the preprocessing procedure we already z-scored all the channel data to make the data between different subjects comparable, we should still be very careful about their absolute magnitude, because in the Lasso regression each subject received a different hyperparameter, and the data quality of each subject may be very different. For example, if ten subjects have shown the same direction, say, a moderate or weak

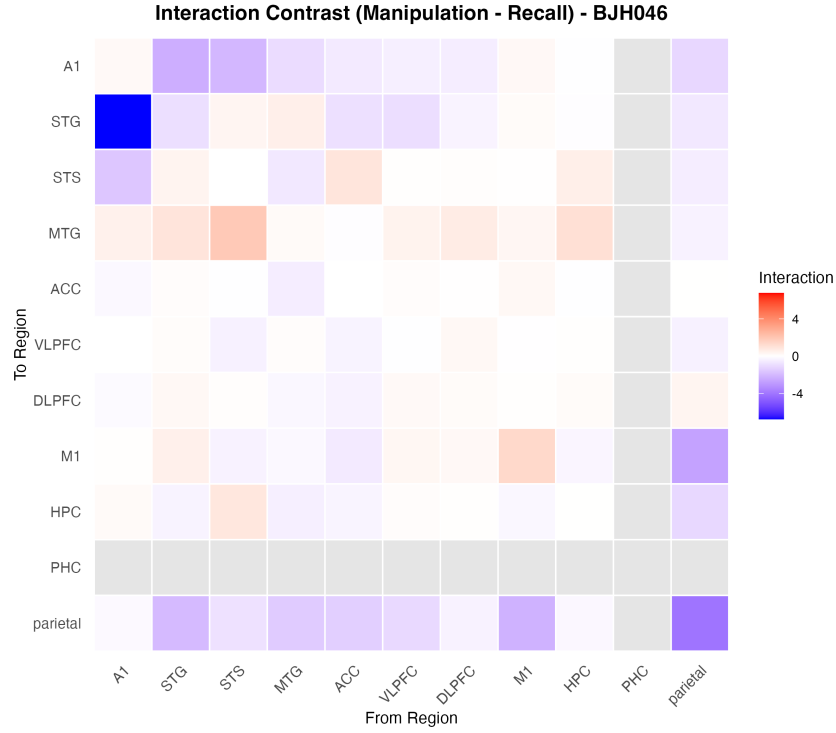


Figure 19: Region-level interaction contrast matrix for subject BJH046. The x-axis shows the source region, and the y-axis shows the target region, so each cell represents the contrast from the x-axis region to the y-axis region. The contrast is defined as $(|\Pi_{\text{Delay Manipulation}}| - |\Pi_{\text{Baseline Manipulation}}|) - (|\Pi_{\text{Delay Recall}}| - |\Pi_{\text{Baseline Recall}}|)$, aggregated from the channel-level $|\Pi_{ij}|$ values. Red cells have positive contrast values, meaning the baseline-to-delay change is larger in manipulation than in recall. Blue cells have negative contrast values, meaning the baseline-to-delay change is smaller in manipulation than in recall. Darker colors indicate larger absolute contrast values, while colors close to white indicate values close to zero. Gray cells indicate unavailable region pairs because this subject has no electrodes in one of the corresponding regions.

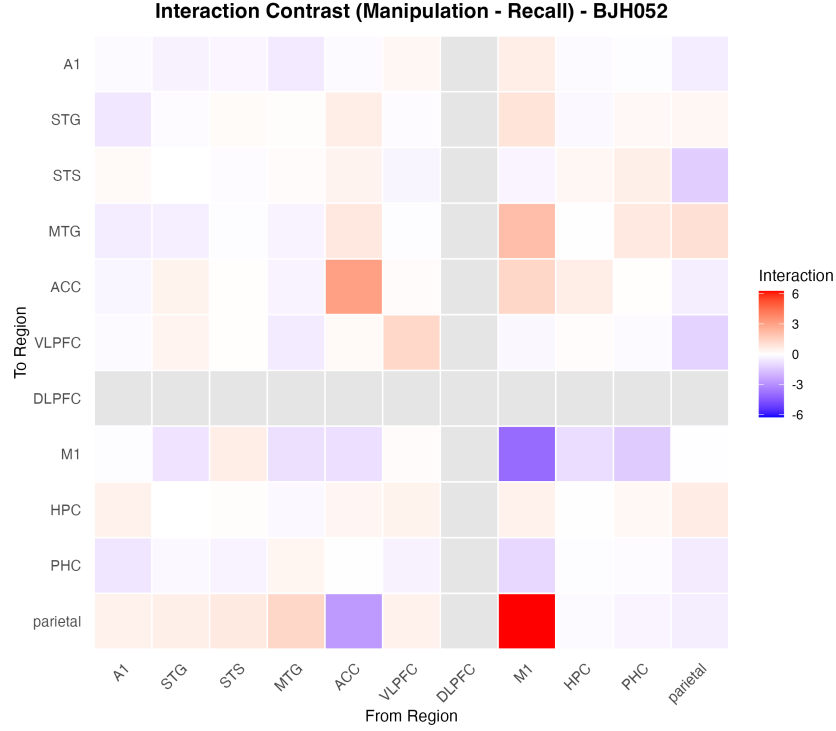


Figure 20: Region-level interaction contrast matrix for subject BJH052. The axes, colors, and contrast definition are the same as in Figure 19.

interaction contrast decrease, and only one subject shows a huge interaction contrast increase, if we calculate this based on the original magnitude level, we may not get the **universal** rule. But we should still acknowledge such a big increase and give it a relatively high influence. In order to balance the magnitude and the consistency across the majority of subjects, I decide to use the Wilcoxon Signed-Rank Test to test this interaction contrast ($\text{Contrast}_{A \rightarrow B}$).

As defined in [15], for a fixed region A and B, assume C_i (which represents the $\text{Contrast}_{A \rightarrow B}$ from the i -th subject out of n subjects that have electrodes in both region A and region B) is a random variable $C_i = \theta + e_i$. We assume $e_1, \dots, e_n$ are independent and identically distributed (i.i.d.) and their probability density function is symmetric about 0.

Because we care about whether there is a difference between the two conditions, the hypothesis test is two-sided:

$$H_0 : \theta = 0 \quad \text{versus} \quad H_A : \theta \neq 0$$

The Wilcoxon Signed-Rank Test converts the distance between the original data into its relative order, i.e., rank. Let $R_{|C_i|}$ be the rank of $|C_i|$ among all the absolute values

$|C_1|, \dots, |C_n|$ sorted from smallest to largest. The test statistic W is defined as:

$$W = \sum_{i=1}^n \text{sign}(C_i) R_{|C_i|}$$

If the null hypothesis H_0 holds, for example, if the original data is -0.1, 0.15, 0.2, -0.25 (which intuitively neither significantly increases nor decreases), the absolute values are sorted to give ranks 1, 2, 3, 4. The corresponding signs are -, +, +, -, so the sum $W = -1 + 2 + 3 - 4 = 0$. Compared to another dataset 0.1, 0.15, 0.05, -0.25 (which clearly shows an overall increase), where the signs are +, +, +, -, and $W = 2 + 3 + 1 - 4 = 2$. So the appropriate rule for rejecting H_0 is whether W deviates significantly from 0.

And then I applied the above Wilcoxon Signed-Rank Test to the 11×11 ROI pairs. In the original results which are not corrected by the multiple hypothesis test, only very few connections reach $p \leq 0.05$. The most significant connections are shown as follows:

Table 3: Edges with the smallest p-values, not corrected by multiple testing)

From Region	To Region	N Subjects	Mean Contrast	Median Contrast	Exact p-value
HPC	HPC	20	0.5506	0.2658	0.0014
HPC	STG	15	-0.2788	-0.2116	0.0215
DLPFC	M1	5	0.2896	0.2377	0.0625

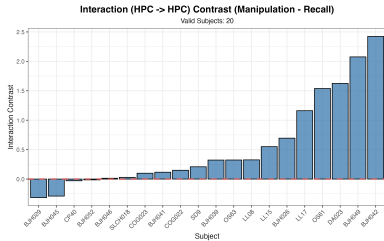


Figure 21: HPC → HPC

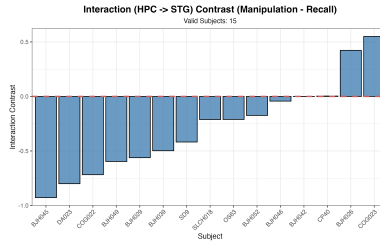


Figure 22: HPC → STG

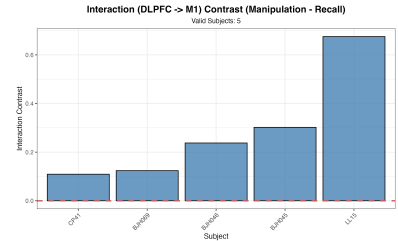


Figure 23: DLPFC → M1

Figure 24: Subject-level ROI-ROI interaction contrasts for the three ROI pairs listed in Table 3. Each panel shows one ROI pair. The x-axis shows subjects with valid electrodes in both regions, and the y-axis shows the ROI-level contrast value. Subjects are ordered by the contrast value, and the number of valid subjects is shown above each plot.

As shown in the table, under the rule $p \leq 0.05$, we only observe the interaction increase of HPC → HPC in the manipulation task compared to the recall task, and the interaction decrease of HPC → STG. But after introducing any kind of multiple comparison test, all the results for the connections are not significant.

I guess this is because some regional interactions have too few subjects (for example, DLPFC → M1 only has 5 subjects), and the statistical power is heavily influenced by such a small sample size.

Therefore, based on these results, we cannot find any significant changes of interaction between different ROIs during the manipulation task compared to the recall task.

8.4 Sensitivity Analysis for Rank

In order to make sure our final results are not highly sensitive to the rank parameter, and because of the limited computational power, I did a sensitivity analysis for only two subjects (SLCH018 and BJH046).

Figures 25 and 26 show how the ROI-ROI interaction contrast changes with the rank. These graphs are sorted according to the standard deviation of the contrast in the high-rank region, prioritizing showing the reader the most unstable connections in the high-rank region. We can see that for SLCH018, as long as the rank parameter is around the third turning point, i.e., entering the flat area in the cross-validated matrix angle or test MSE curve, the interaction contrast is not sensitive to the specific value of the rank (even for the most unstable connections of the high rank). However, around the first and second turning points, the final results are very sensitive to the specific selection of the rank. If we sometimes choose a rank slightly deviating from the center of the turning point, the contrast may change from positive to negative.

But for BJH046, although the region around the first turning point still shows large deviations, the most unstable connections around the third turning point also show some fluctuations.

In order to investigate the stability of the interactions for all connections, I select five rank values (the center rank, center -4 , center -2 , center $+2$, and center $+4$) around the first turning point. Then, I calculate the standard deviation of the interaction contrast across these five ranks for each ROI-ROI connection (11*11 in total) and plot the distribution of these standard deviations.

The same calculation is done for the second and third turning points.

Figure 27 and 28) show that around the third turning point, the standard deviations concentrate near 0 and thus the contrast fluctuations across different ranks for most ROI-ROI connections are very small. Although there is some variation between different individuals (for example, the distribution of BJH046 is relatively wider, which means its final results are relatively more sensitive to the rank parameter), the first turning point still has the widest distribution, the second turning point is narrower, and the third turning point behaves the best for both subjects. Therefore, it is safe to choose the larger rank.

In fact, this also corresponds to the results selected by the rank selection criteria, which tends to choose a larger rank. Also, even if the larger rank we choose contains some excessive noise directions, the subsequent Lasso and Ridge regression can play the

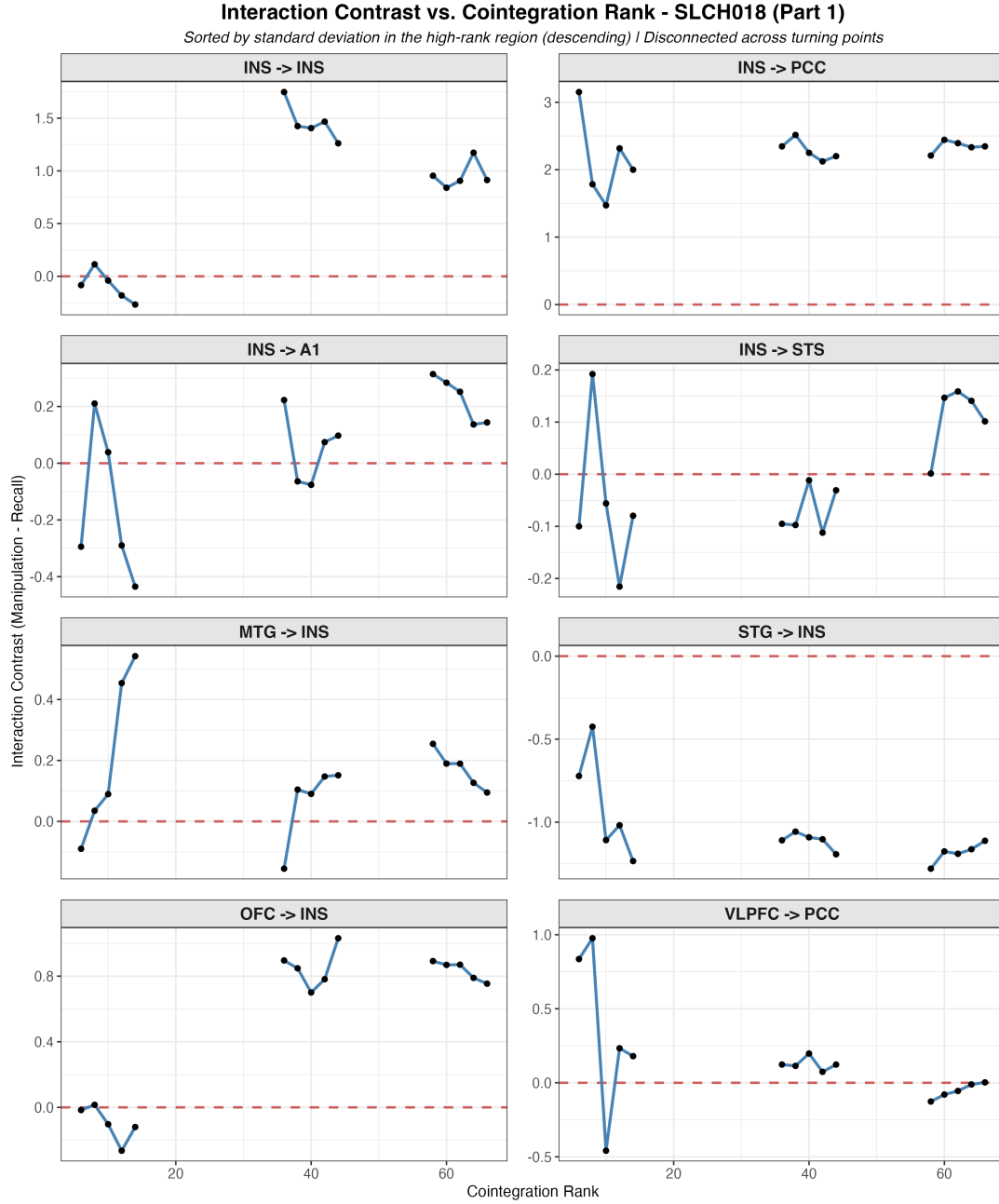


Figure 25: Sensitivity of ROI-ROI interaction contrasts to the choice of cointegration rank for SLCH018. Each panel shows one ROI-ROI contrast across selected ranks around the turning points, and the red dashed line marks zero. The panels shown here are the most unstable ROI pairs in the high-rank region. For SLCH018, the contrast is much more stable around the third turning point than around the first and second turning points, where small rank changes can lead to large changes or sign flips.

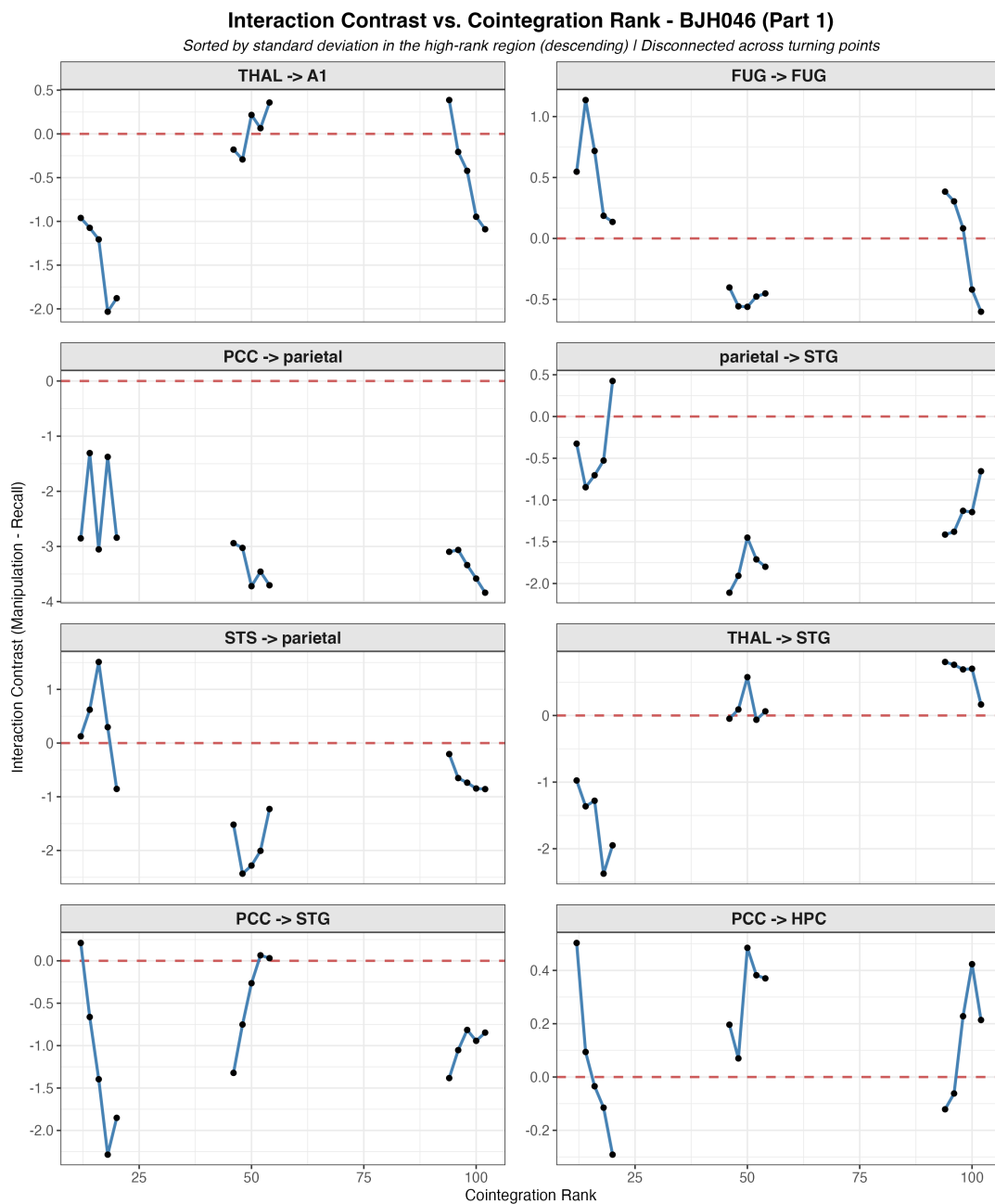


Figure 26: Sensitivity of ROI-ROI interaction contrasts to the choice of cointegration rank for BJH046. The axes, zero line, and panel sorting rule are the same as in Figure 25. For BJH046, the first turning point is still clearly unstable, while some contrasts around the third turning point also show visible fluctuations.

role of pressing the unimportant error-correction coefficients to 0 and help to increase the robustness.

9 Discussion

9.1 Limitations of Coefficient Interpretation

As we discussed before, the interaction matrix represents the effect of the voltage of one electrode at the current time on the voltage change of another electrode at the next time step. It basically copies the mathematical meaning. In my view this interpretation is too narrow, and cannot be extrapolated to a broader physiological meaning.

Because β and α are not unique and the normalization is applied artificially, we cannot interpret their precise meanings. But let's think about why in econometrics we can interpret β and α . Consider the following example,

$$X_t = (H_t, R_t, I_t)',$$

where H_t is the housing price, R_t is the rent, and I_t is the income. We estimate the cointegration rank is $r = 2$. From the perspective of ordinary people, the most interesting variable may be I_t , so we can apply some normalization on β such that

$$\beta' = \begin{bmatrix} 1 & 0 & -10 \\ 0 & 1 & -0.2 \end{bmatrix},$$

Then

$$\beta' X_{t-1} = \begin{bmatrix} H_{t-1} - 10I_{t-1} \\ R_{t-1} - 0.2I_{t-1} \end{bmatrix}.$$

The first relation can be interpreted as the housing price-to-income ratio. This represents: if the housing price is too high relative to the income (i.e., $H_{t-1} \gg 10I_{t-1}$), then the housing price will decrease in the next period. We know clearly the econometric meaning of these coefficients.

If we change to another perspective, then we have other questions we are interested in and apply different normalization methods. For example, maybe real estate agent care more about the rent. then he can set

$$\beta' = \begin{bmatrix} 1 & 50 & 0 \\ 0 & -0.3 & 1 \end{bmatrix},$$

and interpret the first relationship as the price-to-rent ratio. If this is too low, maybe people will buy more house rather than rent and the price of housing will increase in the next period.

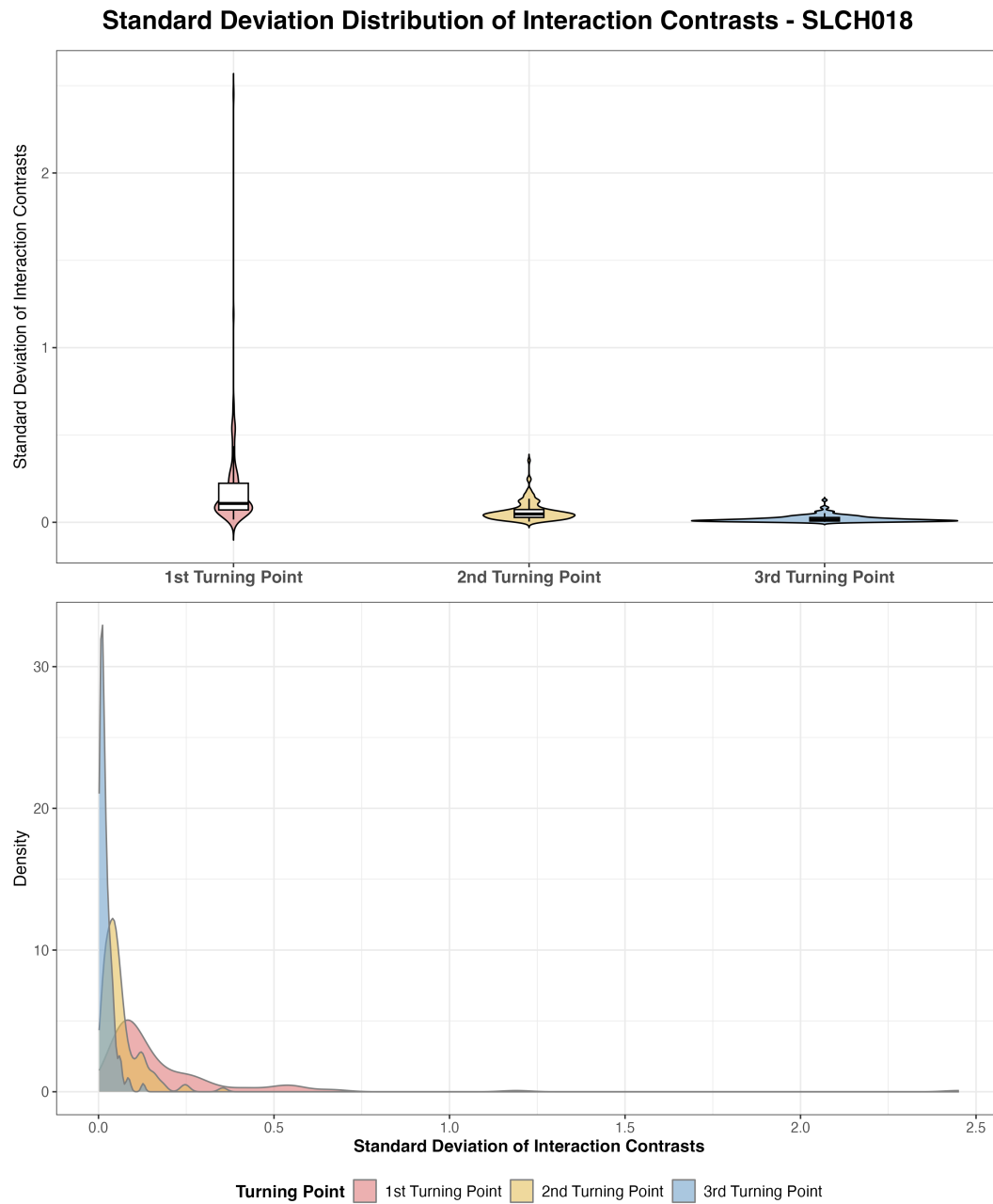


Figure 27: Distribution of the standard deviations of ROI-ROI interaction contrasts for subject SLCH018, computed across five nearby rank values around each turning point. The upper panel shows violin plots with boxplots, and the lower panel shows the corresponding density curves. The first turning point has the widest distribution, the second is narrower, and the third is most concentrated near 0. This shows that the interaction contrasts are most stable around the third turning point.

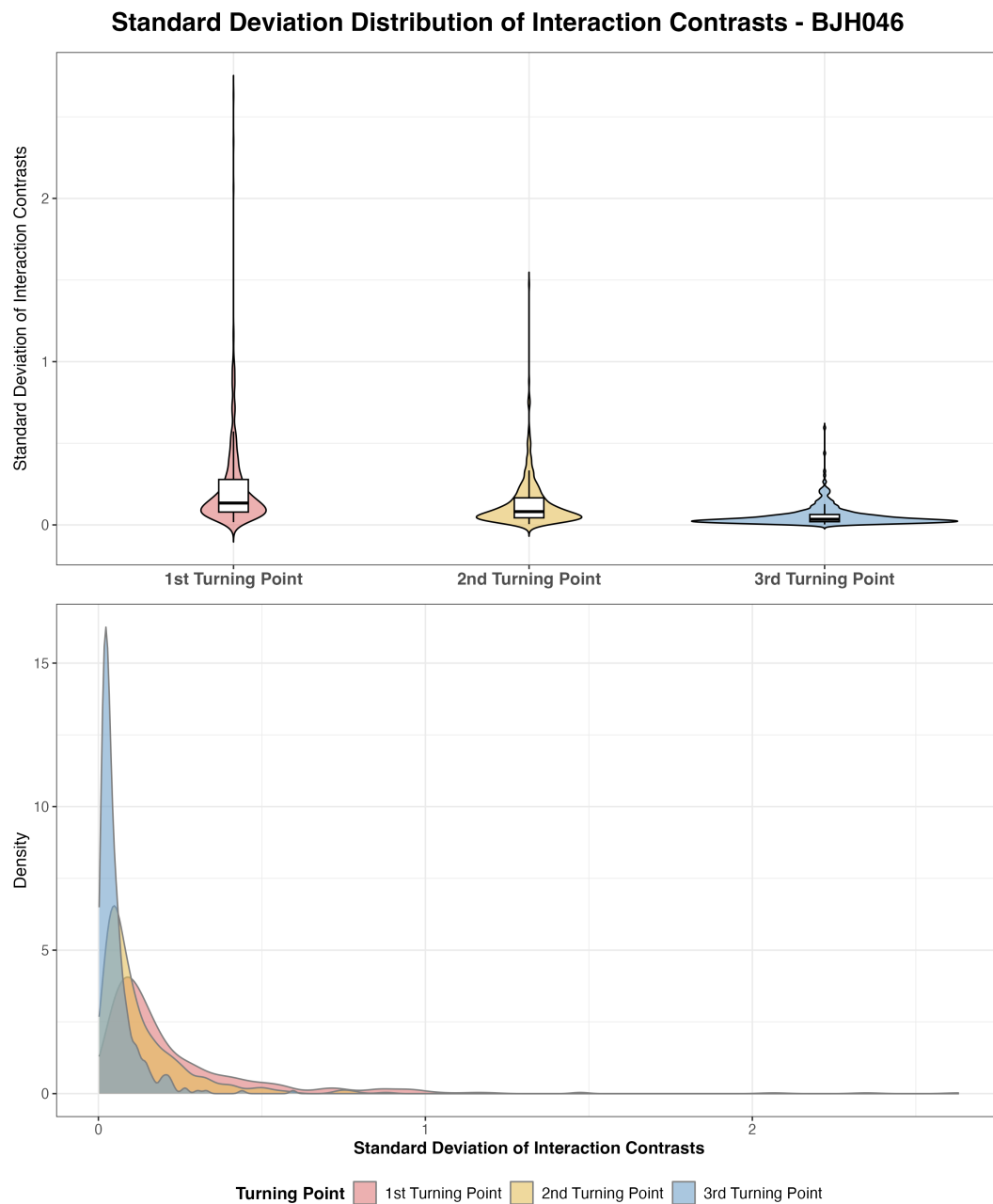


Figure 28: The same distribution for subject BJH046 as in Figure 27. The third turning point still behaves best, but the distribution is wider than for SLCH018, especially around the first turning point, and some variation remains even around the third turning point.

The key is: the reason why we can interpret it in economics is not because α and β are naturally unique, but because the **prior knowledge** in economics makes us believe that there may exist some certain relationship between housing price and income, and then we estimate the parameters, which have very clear meanings. iEEG is different, because we do not have any reliable prior knowledge to make us believe that intracranial electrode A and electrode B have a clear long-term relationship.

When we write a certain cointegration vector as

$$\beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3,$$

it usually can only mean that "there is a statistically stable linear combination here", but we cannot interpret the coefficients as naturally as in economics.

Also, the time scale in economics is more natural. The time between the "current period" and the "next period" sounds long enough for the economic cycle to complete the error correction. However, in iEEG, the time between the "current sampling point" and the "next sampling point" may be very short. We are not sure whether the brain has enough time to finish such an error correction process. If we increase the sampling interval, we will lose lots of valuable high frequency information.

9.2 Limitations after 0.1Hz High-Pass Filtering

According to [3], for an AR(1) process $X_t = \phi X_{t-1} + \epsilon_t$, where $\epsilon_t \sim \text{i.i.d.} N(0, \sigma^2)$, the spectral density function is:

$$f(\omega) = \frac{\sigma^2}{2\pi(1 - 2\phi \cos \omega + \phi^2)}$$

Here, ω represents a kind of frequency.

When $\phi \rightarrow 1$, $f(0) \rightarrow \infty$, which means when the process is close to a random walk $I(1)$, its energy mostly concentrates in the low-frequency part. But if $0 < \phi < 1$ (which means it is an stationary process $I(0)$), $f(\omega)$ takes finite values at any frequency. Because $\cos \omega \in [-1, 1]$, the denominator is between $(1 - \phi)^2$ and $(1 + \phi)^2$. If $0 < \phi < 1$, this denominator will never become 0, and thus at any ω , $f(\omega)$ is finite.

According to the Granger representation (27), if the cointegration rank is between 0 and p , the time series contain both the $I(1)$ and $I(0)$ components. It can be proven that the frequency ω here is roughly the same as the ω in the formula $X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-i\omega n}$, $\omega \in [-\pi, \pi)$ in the filtering section (see Chapter 17 in [10]), so applying a 0.1Hz high-pass filtering is equivalent to removing the low-frequency energy of the $I(1)$ random walk part. The remaining part is mostly the energy of the $I(0)$ components[†]. This is also the reason

[†]Of course, the data still contains other non-stationary components, because in this thesis, we only

why all our rank selection methodology chooses a relatively large cointegration rank for each subject.

In summary, if we want to effectively use the cointegration method in EEG data analysis, my view is that:

First, we need special experimental instruments or techniques that can clearly distinguish between non-cognitive noises and true brain signals induced by cognitive tasks for those EEG signals below 0.1Hz.

Second, we should not include a large number of channels in the beginning. Instead, we can first pick only a few electrodes which have a very clear physiological meaning, and restrict the cointegration matrix β based on the prior knowledge and our interest. In this way, the cointegration vector β will have a more interpretable meaning.

consider the unit root strictly equal to 1, rather than other roots located on the unit circle. Other cases are too complex and beyond the scope of this thesis.

A Derivations and Discussions in Hilbert Transform

A.1 Hilbert Transform for Cosine Function

In the time domain ,the Hilbert Transform is defined as : $x_H(t) = x(t) * \frac{1}{\pi t}$. By using convolution theorem and the fourier transform of $\frac{1}{\pi t}$. we can get its form in the frequency domain $X_H(f) = -i \operatorname{sgn}(f) X(f)$, (see details in [10] ,Recall the core idea of Fourier transform is to represent a signal as a linear combination of different complex basis vectors. For a complex basis signal which has a single frequency $e^{i\omega_0 t}$, $\omega_0 > 0$, its coordinate (the coefficient of the Fourier transform) is just 1. After applying the Hilbert transform to this basis signal, the coefficient becomes $-i \operatorname{sgn}(\omega_0) = -i \cdot 1$. But the transformed signal still has only one frequency. So, in the time domain, $\mathcal{H}\{e^{i\omega_0 t}\} = -i e^{i\omega_0 t}$.

And then by using Euler's formula and linearity of Hilbert Transform, we can calculate directly : $\mathcal{H}\{\cos(\omega_0 t)\} = \frac{1}{2}\mathcal{H}\{e^{i\omega_0 t}\} + \frac{1}{2}\mathcal{H}\{e^{-i\omega_0 t}\} = \frac{1}{2}(-ie^{i\omega_0 t}) + \frac{1}{2}(ie^{-i\omega_0 t}) = \sin(\omega_0 t)$.

A.2 Hilbert Transform for Narrow band and Broadband Signal

Consider a narrow band signal with a special form $x(t)$:

$$x(t) = \sum_{n=1}^N a_n \cos(\omega_n t + \theta_n)$$

, and any frequency ω_n can be written as

$$\omega_n = \omega_c + \Delta\omega_n$$

where $|\Delta\omega_n| \ll \omega_c$. ω_c is the central frequency(or called carrier frequency)

then:

$$\begin{aligned} x(t) &= \sum_{n=1}^N a_n \cos((\omega_c + \Delta\omega_n)t + \theta_n) \\ &= \sum_{n=1}^N a_n \cos(\omega_c t + (\Delta\omega_n t + \theta_n)) \\ &= \sum_{n=1}^N a_n [\cos(\omega_c t) \cos(\Delta\omega_n t + \theta_n) - \sin(\omega_c t) \sin(\Delta\omega_n t + \theta_n)] \\ &= \underbrace{\left[\sum_{n=1}^N a_n \cos(\Delta\omega_n t + \theta_n) \right]}_{I(t)} \cos(\omega_c t) - \underbrace{\left[\sum_{n=1}^N a_n \sin(\Delta\omega_n t + \theta_n) \right]}_{Q(t)} \sin(\omega_c t) \end{aligned}$$

Notice that $I(t)$ and $Q(t)$ only depend on $\Delta\omega_n$ and thus change very slowly. Their spectra do not overlap with ω_c ($\Delta\omega_n \ll \omega_c$, for all $n = 1, 2, \dots, N$). And this satisfies the assumption

of Bedrosian's theorem[12]. Then:

$$\begin{aligned}\mathcal{H}\{I(t) \cos(\omega_c t)\} &= I(t) \sin(\omega_c t) \\ \mathcal{H}\{Q(t) \sin(\omega_c t)\} &= -Q(t) \cos(\omega_c t)\end{aligned}$$

This shows that for narrowband waves, the Hilbert transform mainly influence the central carrier, rather than the slow changing envelope.

For a broadband signal, the $\Delta\omega_n$ is no longer tiny compared to ω_c , which means the spectrum of envelope may overlap with the high-frequency carrier. So we cannot use Bedrosian's theorem to separate these two terms and get clear physical meaning.

B Proof in the section of Continuous SDE

B.1 The decomposition of P

Because $\text{rank}(P) = r$, we can find r linearly independent vectors $v_1, \dots, v_r$ from the column vectors of P , which constitute the basis of the column space of P . Then if we arrange the r basis vectors column by column, so that α :

$$\alpha = \begin{bmatrix} | & & | \\ v_1 & \cdots & v_r \\ | & & | \end{bmatrix}_{p \times r}. \quad (67)$$

It's easy to see $\text{rank}(\alpha) = r$.

For the j -th column p_j , there exists a vector of coefficients $w^{(j)} = [w_1^{(j)}, w_2^{(j)}, \dots, w_r^{(j)}]^\top$, such that

$$p_j = \sum_{k=1}^r v_k w_k^{(j)} = \alpha w^{(j)}. \quad (68)$$

If we arrange $w^{(1)}, \dots, w^{(p)}$ column by column to get a coefficient matrix, denoted as $\beta^\top$:

$$\beta^\top = \begin{bmatrix} | & & | \\ w^{(1)} & \cdots & w^{(p)} \\ | & & | \end{bmatrix}_{r \times p}, \quad (69)$$

then we have $P = \alpha\beta^\top$.

Because for any matrices A and B , $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$, we have:

$$r = \text{rank}(P) \leq \min(\text{rank}(\alpha), \text{rank}(\beta^\top)) = \min(r, \text{rank}(\beta^\top)).$$

Also, $\text{rank}(\beta^\top) \leq \min(r, p) = r$. Combining these two, $\text{rank}(\beta^\top) = r$.

B.2 Equivalence of Three Conditions

We first prove $\det(\beta^\top \alpha) \neq 0 \iff \mathcal{R}(\alpha) \cap \mathcal{R}(\beta_\perp) = \{0\}$.

1) ($\Rightarrow$): $\forall x \in \mathcal{R}(\alpha) \cap \mathcal{R}(\beta_\perp)$, there exist a, b such that $x = \alpha a = \beta_\perp b$. Multiplying the left by $\beta^\top$, we have $\beta^\top \alpha a = \beta^\top \beta_\perp b = 0$. Because $\det(\beta^\top \alpha) \neq 0$, a must equal 0 $\Rightarrow x = 0$. So $\mathcal{R}(\alpha) \cap \mathcal{R}(\beta_\perp) = \{0\}$.

2) ($\Leftarrow$): We prove its contrapositive. If $\det(\beta^\top \alpha) = 0$, there must exist $a \neq 0$ such that $\beta^\top \alpha a = 0$. Set $x = \alpha a$, then $\beta^\top x = 0$. And $x \neq 0$ because α is of full column rank. And, according to [24, Chapter 3, Section 3.6.2], "the null space of A is the orthogonal complement of the range space of $A^\top$," so $x \in \mathcal{N}(\beta^\top) = \mathcal{R}(\beta_\perp)$. It is obvious that $x \in \mathcal{R}(\alpha)$. So $x \in \mathcal{R}(\alpha) \cap \mathcal{R}(\beta_\perp)$, where $x \neq 0$.

A very similar reasoning process can be used to prove another equivalence.

B.3 The proof of Theorem 1

First introduce a helper transformation $Y_t = g(t, x_t) = e^{-Pt}x_t$, where $g : [0, \infty) \times \mathbb{R}^p \rightarrow \mathbb{R}^p$. According to **Theorem 4.2.1 (The general Itô formula)** in [14]: for the k -th component in the vector Y_t ,

$$dY_k = \frac{\partial g_k}{\partial t} dt + \sum_i \frac{\partial g_k}{\partial x_i} dX_i + \frac{1}{2} \sum_{i,j} \frac{\partial^2 g_k}{\partial x_i \partial x_j} dX_i dX_j. \quad (70)$$

Because $\frac{d}{dt}(e^{-Pt}) = -Pe^{-Pt}$, we have $\frac{\partial g}{\partial t} = -Pe^{-Pt}x_t$. The function $g(t, x)$ is linear with respect to x , so the Jacobian matrix is $\nabla_x g = e^{-Pt}$ and $\frac{\partial^2 g_k}{\partial x_i \partial x_j} = 0$. Combining these terms with the original equation $dx_t = P(x_t - m)dt + DdW_t$, we get:

$$\begin{aligned} dY_t &= \frac{\partial g}{\partial t} dt + \nabla_x g \cdot dx_t \\ &= -Pe^{-Pt}x_t dt + e^{-Pt} [P(x_t - m)dt + DdW_t] \\ &= -Pe^{-Pt}x_t dt + \underbrace{e^{-Pt}P}_{=Pe^{-Pt}} x_t dt - e^{-Pt}Pm dt + e^{-Pt}DdW_t \\ &= -e^{-Pt}Pm dt + e^{-Pt}DdW_t. \end{aligned}$$

Integrating the above equation over the interval $[0, t]$:

$$\begin{aligned}
e^{-Pt}x_t - x_0 &= \int_0^t -e^{-Ps}Pm ds + \int_0^t e^{-Ps}DdW_s \\
&= \left(\int_0^t -e^{-Ps}P ds \right) m + \int_0^t e^{-Ps}DdW_s \\
&= [e^{-Ps}]_0^t m + \int_0^t e^{-Ps}DdW_s \\
&\quad \left(\text{Note: according to Chapter 6, page 128 in [8], } \frac{d}{ds}e^{-Ps} = -e^{-Ps}P \right) \\
&= (e^{-Pt} - I)m + \int_0^t e^{-Ps}DdW_s.
\end{aligned}$$

Multiplying both sides of the equation by e^{Pt} from the left:

$$\begin{aligned}
x_t - e^{Pt}x_0 &= e^{Pt}(e^{-Pt} - I)m + e^{Pt} \int_0^t e^{-Ps}DdW_s \\
&= (I - e^{Pt})m + \int_0^t e^{P(t-s)}DdW_s \\
&= m - e^{Pt}m + \int_0^t e^{P(t-s)}DdW_s \\
&\Rightarrow x_t = m + e^{Pt}(x_0 - m) + \int_0^t e^{P(t-s)}DdW_s.
\end{aligned}$$

B.4 Condition of Stability

If we make a stronger assumption, for example, that the real parts of all eigenvalues of the matrix P are negative, we can obtain a stable solution. In fact, if we take the expectation of the final solution derived above, because the Itô integral is a martingale, we have: $\mathbb{E} \left[\int_0^t e^{P(t-s)}D dW_s \mid \mathcal{F}_0 \right] = 0$. Taking the expectation again: $\mathbb{E} \left[\int_0^t e^{P(t-s)}D dW_s \right] = 0$. So, the expectation of x_t is: $\mathbb{E}[x_t] = m + e^{Pt}(x_0 - m)$. In order for $\mathbb{E}[x_t] \rightarrow m$ as $t \rightarrow \infty$, we must have:

$$\lim_{t \rightarrow \infty} e^{Pt}(x_0 - m) = 0.$$

According to **Perko [17] (Theorem 2, p. 57)**, the necessary and sufficient condition for the above limit equality is that the real parts of **all** eigenvalues of P are negative. Furthermore, if this condition holds, the process x_t has a stationary distribution. Its stationary covariance matrix $\Sigma = \lim_{t \rightarrow \infty} \text{Cov}(x_t)$ satisfies: $P\Sigma + \Sigma P^\top = -DD^\top$. For a detailed discussion, see **Gardiner [19], Chapter 4, Section 4.4.6 (Multivariate Ornstein-Uhlenbeck Process)**.

If some eigenvalues of P are 0, we say it is null recurrent. In this case, it will show some characteristics of random walk.

B.5 From SDE to VAR

Based on the solution of the O-U process, set the current time $t = t_n$, the starting time $t_0 = t_{n-1}$, and $\Delta = t_n - t_{n-1}$:

$$x_{t_n} = m + e^{P(t_n - t_{n-1})}(x_{t_{n-1}} - m) + \int_{t_{n-1}}^{t_n} e^{P(t_n - s)} D dW_s \quad (71)$$

Denoting x_{t_n} as x_n :

$$\begin{aligned} x_n &= m + e^{P\Delta}(x_{n-1} - m) + \int_{t_{n-1}}^{t_n} e^{P(t_n - s)} D dW_s \\ &= (I_p - e^{P\Delta})m + e^{P\Delta}x_{n-1} + \varepsilon_n \end{aligned}$$

Comparing with the VAR model $x_n = \mu + Ax_{n-1} + \varepsilon_n$, we obtain: $A = e^{P\Delta}$, $\mu = (I_p - A)m$, and $\varepsilon_n = \int_{t_{n-1}}^{t_n} e^{P(t_n - s)} D dW_s$.

The increments of Brownian motion $W_{t+\Delta} - W_t$ are independent across different time intervals, and if the time intervals have the same length, they also have the same distribution. For any $t \geq 0$ and $\Delta > 0$, the increments are $W_{t+\Delta} - W_t \sim \mathcal{N}(\mathbf{0}, \Delta I_d)$. When calculating ε_n , we only use the information in the current time interval (t_{n-1}, t_n) . Since we set the sampling intervals to equal lengths, the terms ε_n are independent and identically distributed (i.i.d.) and are independent of past.

Similar to the last subsection, $E[\varepsilon_n] = \mathbf{0}_p$ because of the martingale property. According to the Itô Isometry, its covariance matrix is:

$$\begin{aligned} \Sigma &= E[\varepsilon_n \varepsilon_n^\top] = \int_{t_{n-1}}^{t_n} (e^{P(t_n - s)} D) (e^{P(t_n - s)} D)^\top ds \\ &= \int_{t_{n-1}}^{t_n} e^{P(t_n - s)} D D^\top e^{P^\top(t_n - s)} ds \end{aligned}$$

Let $v = t_n - s$, then $ds = -dv$. When $s = t_{n-1}$, $v = \Delta$; when $s = t_n$, $v = 0$.

$$\Sigma = \int_{\Delta}^0 e^{Pv} D D^\top e^{P^\top v} (-dv) = \int_0^{\Delta} e^{Pu} D D^\top e^{P^\top u} du \quad (72)$$

B.6 Delta towards 0

When $\Delta \rightarrow 0$:

(1)

$$A = e^{P\Delta}$$

If we use the Taylor expansion for $e^{P\Delta}$:

$$e^{P\Delta} = I_p + P\Delta + \frac{1}{2!}(P\Delta)^2 + \frac{1}{3!}(P\Delta)^3 + \dots$$

$$\frac{1}{\Delta}(A - I_p) = P + \frac{1}{2!}P^2\Delta + O(\Delta^2)$$

If $\Delta \rightarrow 0$, $O(\Delta) \rightarrow 0$ and $O(\Delta^2) \rightarrow 0$. Then we have:

$$P \approx \frac{1}{\Delta}(A - I_p)$$

(2) Using a similar argument, we can derive as follows:

$$\mu = (I_p - A)m = (I_p - e^{P\Delta})m \approx -P\Delta m = \Psi_{cont}\Delta. \text{ Thus, } \Psi_{cont} \approx \frac{1}{\Delta}\mu.$$

Because $e^{Pu} \rightarrow I_p$ for $u \in [0, \Delta]$,

$$\Sigma = \int_0^\Delta e^{Pu} DD^\top e^{P^\top u} du \approx \int_0^\Delta I_p (DD^\top) I_p^\top du = (DD^\top)\Delta, \rightarrow DD^\top \approx \frac{1}{\Delta}\Sigma. \quad (73)$$

In fact, given $\mathbb{E}[dW_t] = \mathbf{0}$ and $\mathbb{E}[dW_t dW_t^\top] = I_d dt$, the instantaneous covariance of the diffusion term in the O-U is exactly expressed in $DD^\top$

$$\begin{aligned} \text{Cov}(DdW_t) &= \mathbb{E}[(DdW_t)(DdW_t)^\top] \\ &= D\mathbb{E}[dW_t dW_t^\top] D^\top \\ &= D(I_d dt) D^\top \\ &= (DD^\top) dt. \end{aligned}$$

B.7 Zero Eigenvalues of Π

First, we have $\Pi = \alpha\beta^\top$ ($\alpha, \beta \in \mathbb{R}^{p \times r}$) and $\det(\beta^\top \alpha) \neq 0$.

For $\lambda \neq 0$,

$$\det(\lambda I_p - \alpha\beta^\top) = \lambda^p \det(I_p - \lambda^{-1}\alpha\beta^\top).$$

According to a well-known equality in linear algebra [27],

$$\det(I_m + AB) = \det(I_n + BA),$$

we have:

$$\det(I_p - \lambda^{-1}\alpha\beta^\top) = \det(I_r - \lambda^{-1}\beta^\top \alpha).$$

Therefore,

$$\begin{aligned} \det(\lambda I_p - \Pi) &= \lambda^p \det(I_r - \lambda^{-1}\beta^\top \alpha) \\ &= \lambda^{p-r} \det(\lambda I_r - \beta^\top \alpha). \end{aligned}$$

So the eigenvalues of Π are: 0 (algebraic multiplicity $p - r$), r eigenvalues of $\beta^\top \alpha$

Because $\det(\beta^\top \alpha) \neq 0$, the r eigenvalues are non-zero. So Π has exactly $p - r$ zero eigenvalues.

C Derivations in the Estimation Procedure

C.1 positive definite of S_{11} in the estimation procedure

First, by definition, S_{11} is positive semi-definite.

The matrix S_{11} is not positive definite (i.e., it is singular) :

$$\begin{aligned}
 &\iff \exists a \neq 0, \text{ s.t. } a^\top S_{11} a = 0 \\
 &\iff \exists a \neq 0, \text{ s.t. } a^\top R_{1n}^{(m)} = 0 \quad (\forall m, n) \\
 &\iff \exists a \neq 0, \text{ s.t. } a^\top (x_{n-1}^{(m)} - \bar{x}_{-1}) = 0 \quad (\forall m, n) \\
 &\iff \exists a \neq 0, \text{ s.t. } a^\top x_{n-1}^{(m)} = C \quad (\forall m, n), \text{ where } C \text{ is a constant.}
 \end{aligned} \tag{74}$$

This implies the existence of perfect co-linearity among different EEG channels. In practical EEG experiments, this condition is basically impossible to satisfy. Each electrode site has unique background noise, and there are also random variations across different trials. Therefore, we can assume that S_{11} is strictly positive definite (and the same reasoning applies to S_{00}). And this property can also be verified empirically in the code.

Assuming β has full column rank and S_{11} is positive definite, for any non-zero vector $x \in \mathbb{R}^r$, the vector $z = \beta x$ must be non-zero. Consequently, we have $x^\top (\beta^\top S_{11} \beta) x = (\beta x)^\top S_{11} (\beta x) = z^\top S_{11} z > 0$, which ensures that $\beta^\top S_{11} \beta$ is positive definite.

C.2 The derivation for the estimation of Σ

$$\begin{aligned}
 \hat{\Sigma}(\beta) &= \frac{1}{N_{\text{total}}} \sum_{m=1}^M \sum_{n=1}^{N_m} \left(R_{0n}^{(m)} - \hat{\alpha} \beta^\top R_{1n}^{(m)} \right) \left(R_{0n}^{(m)} - \hat{\alpha} \beta^\top R_{1n}^{(m)} \right)^\top \\
 &= \frac{1}{N_{\text{total}}} \sum_{m=1}^M \sum_{n=1}^{N_m} \left[R_{0n}^{(m)} (R_{0n}^{(m)})^\top - \hat{\alpha} \beta^\top R_{1n}^{(m)} (R_{0n}^{(m)})^\top - R_{0n}^{(m)} (R_{1n}^{(m)})^\top \beta \hat{\alpha}^\top + \hat{\alpha} \beta^\top R_{1n}^{(m)} (R_{1n}^{(m)})^\top \beta \hat{\alpha}^\top \right] \\
 &= S_{00} - \hat{\alpha} \beta^\top S_{10} - S_{01} \beta \hat{\alpha}^\top + \hat{\alpha} (\beta^\top S_{11} \beta) \hat{\alpha}^\top \\
 &= S_{00} - \hat{\alpha} \beta^\top S_{10} - S_{01} \beta \hat{\alpha}^\top + [S_{01} \beta (\beta^\top S_{11} \beta)^{-1}] (\beta^\top S_{11} \beta) \hat{\alpha}^\top \\
 &= S_{00} - \hat{\alpha} \beta^\top S_{10} - S_{01} \beta \hat{\alpha}^\top + S_{01} \beta \hat{\alpha}^\top \\
 &= S_{00} - \hat{\alpha} \beta^\top S_{10} \\
 &= S_{00} - S_{01} \beta (\beta^\top S_{11} \beta)^{-1} \beta^\top S_{10}
 \end{aligned} \tag{75}$$

C.3 Equivalence between maximizing the likelihood and minimizing the determinant

After substituting $\hat{\Sigma}(\beta)$ and $\hat{\alpha}(\beta)$ into the log-likelihood function, we can get the profile log-likelihood:

$$\ell_p(\beta) \propto -\frac{N_{\text{total}}}{2} \log |\hat{\Sigma}(\beta)| - \underbrace{\frac{1}{2} \sum_{m=1}^M \sum_{n=1}^{N_m} (\hat{\varepsilon}_n^{(m)})^\top (\hat{\Sigma}(\beta))^{-1} \hat{\varepsilon}_n^{(m)}}_{Q(\beta)} \quad (76)$$

where $\hat{\varepsilon}_n^{(m)} = R_{0n}^{(m)} - \hat{\alpha}\beta^\top R_{1n}^{(m)}$ is the estimated residual vector based on the current β . we can derive $Q(\beta)$ as follows:

$$\begin{aligned} Q(\beta) &\stackrel{c=\text{tr}(c)}{=} \sum_{m=1}^M \sum_{n=1}^{N_m} \text{tr} \left((\hat{\varepsilon}_n^{(m)})^\top (\hat{\Sigma}(\beta))^{-1} \hat{\varepsilon}_n^{(m)} \right) \\ &\stackrel{\text{tr}(AB)=\text{tr}(BA)}{=} \sum_{m=1}^M \sum_{n=1}^{N_m} \text{tr} \left((\hat{\Sigma}(\beta))^{-1} \hat{\varepsilon}_n^{(m)} (\hat{\varepsilon}_n^{(m)})^\top \right) \\ &\stackrel{\text{linear}}{=} \text{tr} \left((\hat{\Sigma}(\beta))^{-1} \left[\sum_{m=1}^M \sum_{n=1}^{N_m} \hat{\varepsilon}_n^{(m)} (\hat{\varepsilon}_n^{(m)})^\top \right] \right) \\ &\stackrel{\text{definition}}{=} \text{tr} \left((\hat{\Sigma}(\beta))^{-1} \cdot N_{\text{total}} \hat{\Sigma}(\beta) \right) \\ &= N_{\text{total}} \cdot \text{tr}(I_p) \\ &= N_{\text{total}} \cdot p \end{aligned} \quad (77)$$

Substituting $Q(\beta)$ back into Equation 76, we get the profile log likelihood:

$$\ell_p(\beta) \propto -\frac{N_{\text{total}}}{2} \log |\hat{\Sigma}(\beta)| - \frac{N_{\text{total}}p}{2} \quad (78)$$

Therefore, maximizing the profile log-likelihood is completely equivalent to minimizing the determinant of the residual covariance matrix:

$$\max_{\beta} \ell_p(\beta) \iff \min_{\beta} \log |\hat{\Sigma}(\beta)| \iff \min_{\beta} |\hat{\Sigma}(\beta)| \quad (79)$$

Because the matrix

$$\hat{\Sigma}(\beta) = S_{00} - (S_{01}\beta)(\beta^\top S_{11}\beta)^{-1}(\beta^\top S_{10}) \quad (80)$$

has the form of $A - BD^{-1}C$, we can construct the following $(p+r) \times (p+r)$ block matrix $\mathcal{M}$:

$$\mathcal{M} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} S_{00} & S_{01}\beta \\ \beta^\top S_{10} & \beta^\top S_{11}\beta \end{pmatrix} \quad (81)$$

The residual covariance matrix $\hat{\Sigma}(\beta)$ is exactly the Schur complement of block D in $\mathcal{M}$. According to Seber [25] (Chapter 14.1), we have:

$$|\mathcal{M}| = |A - BD^{-1}C| \cdot |D| = |\hat{\Sigma}(\beta)| \cdot |\beta^\top S_{11} \beta| \quad (82)$$

We also have an alternative expansion for the determinant:

$$\begin{aligned} |\mathcal{M}| &= |A| \cdot |D - CA^{-1}B| \\ &= |S_{00}| \cdot |\beta^\top S_{11} \beta - (\beta^\top S_{10}) S_{00}^{-1} (S_{01} \beta)| \\ &= |S_{00}| \cdot |\beta^\top (S_{11} - S_{10} S_{00}^{-1} S_{01}) \beta| \end{aligned} \quad (83)$$

Equation (82) and Equation (83) must be equal:

$$|\hat{\Sigma}(\beta)| \cdot |\beta^\top S_{11} \beta| = |S_{00}| \cdot |\beta^\top (S_{11} - S_{10} S_{00}^{-1} S_{01}) \beta| \quad (84)$$

Thus

$$|\hat{\Sigma}(\beta)| = |S_{00}| \frac{|\beta^\top (S_{11} - S_{10} S_{00}^{-1} S_{01}) \beta|}{|\beta^\top S_{11} \beta|} \quad (85)$$

D supplementary

D.1 Estimate for beta

Here are supplementary derivations for $\hat{\beta}$ and $|\hat{\Sigma}(\hat{\beta})|$ and relations to CPP code in my master's thesis. The proof is divided into four steps:

First, we use standard linear algebra to turn the objective function into an optimization problem about the eigenvalues of a compressed matrix.

Second, we examine the properties of the compressed matrix H to find out how its eigenvalues relate to those of the original matrix C .

Third, we use the Schur complement to set further bounds on the eigenvalues.

Fourth, we use standard linear algebra to go from the regular eigenvalue equation back to the generalized eigenvalue problem.

The key theorem for the second and third steps is the Min-Max theorem [1].

D.1.1 First Step: Conversion of the Objective Function

Algebraic Abstraction If we ignore the constant term $|S_{00}|$, we can rewrite the objective function as:

$$J(\beta) = \frac{|\beta^\top (S_{11} - S_{10} S_{00}^{-1} S_{01}) \beta|}{|\beta^\top S_{11} \beta|} \quad (86)$$

Let's define matrices B and A as follows:

$$B := S_{11}, \quad A := S_{10} S_{00}^{-1} S_{01} \quad (87)$$

The objective function can then be simplified to:

$$J(\beta) = \frac{|\beta^\top (B - A)\beta|}{|\beta^\top B\beta|} \quad (88)$$

As discussed in the main text, B is positive definite, and it is easy to see that A is also a symmetric matrix.

Right-multiplying β by any invertible matrix $U \in \mathbb{R}^{r \times r}$ does not change the value of $J(\beta)$. Let $\tilde{\beta} = \beta U$. For the numerator:

$$\begin{aligned} |\tilde{\beta}^\top (B - A)\tilde{\beta}| &= |U^\top \beta^\top (B - A)\beta U| \\ &= |U^\top| \cdot |\beta^\top (B - A)\beta| \cdot |U| \\ &= |U|^2 \cdot |\beta^\top (B - A)\beta| \end{aligned} \quad (89)$$

For the denominator:

$$|\tilde{\beta}^\top B\tilde{\beta}| = |U|^2 \cdot |\beta^\top B\beta| \quad (90)$$

Plugging these back into the objective function, the $|U|^2$ terms cancel out:

$$J(\tilde{\beta}) = \frac{|U|^2 |\beta^\top (B - A)\beta|}{|U|^2 |\beta^\top B\beta|} = J(\beta) \quad (91)$$

This indicates that the objective function depends only on the subspace spanned by the column vectors of β , not on the specific vectors themselves.

Because of this, we can introduce a normalization constraint: $\beta^\top B\beta = I_r$. Since $\beta^\top B\beta$ is also positive definite, we can choose $U = (\beta^\top B\beta)^{-1/2}$ (where $U = U^\top$) and set $\bar{\beta} = \beta U$. This gives:

$$\bar{\beta}^\top B\bar{\beta} = U^\top (\beta^\top B\beta) U = (\beta^\top B\beta)^{-1/2} (\beta^\top B\beta) (\beta^\top B\beta)^{-1/2} = I_r \quad (92)$$

Since we already proved that $J(\beta)$ is invariant to right-multiplication by an invertible matrix, under this constraint, the denominator becomes $|\beta^\top B\beta| = |I_r| = 1$. The numerator simplifies to:

$$\beta^\top (B - A)\beta = \beta^\top B\beta - \beta^\top A\beta = I_r - \beta^\top A\beta \quad (93)$$

Therefore, our original optimization problem is completely equivalent to:

$$\min_{\beta} |I_r - \beta^\top A\beta| \quad \text{s.t.} \quad \beta^\top B\beta = I_r \quad (94)$$

The steps above follow a very direct logic. The next part mainly focuses on converting this B -orthogonal constraint into a standard orthogonal constraint using a new matrix Q .

Standardizing the Constraint Since B is positive definite ($B \succ 0$), according to [24] Ch.4.7.1 it has a unique symmetric positive definite square root matrix $B^{1/2}$ and its inverse $B^{-1/2}$. Let's introduce an auxiliary matrix $Q \in \mathbb{R}^{p \times r}$ and define the following linear transformation:

$$Q := B^{1/2}\beta \iff \beta = B^{-1/2}Q \quad (95)$$

Substituting this into the original constraint:

$$\beta^\top B \beta = (B^{-1/2}Q)^\top B (B^{-1/2}Q) = Q^\top (B^{-1/2} B B^{-1/2}) Q = Q^\top I_p Q = Q^\top Q \quad (96)$$

Therefore, the generalized orthogonal constraint $\beta^\top B \beta = I_r$ is converted into a standard orthogonal constraint $Q^\top Q = I_r$.

Now, substitute $\beta = B^{-1/2}Q$ into the quadratic term $\beta^\top A \beta$ in the objective function:

$$\beta^\top A \beta = (B^{-1/2}Q)^\top A (B^{-1/2}Q) = Q^\top (B^{-1/2} A B^{-1/2}) Q \quad (97)$$

Let's define a "compressed matrix" C :

$$C := B^{-1/2} A B^{-1/2} \quad (98)$$

Since both A and $B^{-1/2}$ are symmetric matrices, it is easy to prove that C is also a real symmetric matrix. The objective function now becomes:

$$J(\beta) = |I_r - Q^\top C Q| \quad (99)$$

Define an $r \times r$ matrix $H := Q^\top C Q$. Because C is symmetric, H is also symmetric. Therefore, there exists an orthogonal matrix $O \in \mathbb{R}^{r \times r}$ that can diagonalize H :

$$O^\top H O = \text{diag}(\mu_1, \dots, \mu_r) \quad (100)$$

where μ_i are the eigenvalues of the matrix H .

Using the similarity invariance of determinants, the objective function can be expanded into a product of eigenvalues:

$$\begin{aligned} |I_r - H| &= |O^\top (I_r - H) O| \\ &= |I_r - \text{diag}(\mu_1, \dots, \mu_r)| \\ &= \prod_{i=1}^r (1 - \mu_i) \end{aligned} \quad (101)$$

D.1.2 Second Step: Geometric Intuition and Eigenvalue Bounds

Now we move to the second step. As we can see, the key to the objective function lies in the eigenvalues of matrix H . Let's first build some geometric intuition.

Geometric Intuition: The True Meaning of the Compressed Matrix $H = Q^\top CQ$

Let $\mathcal{U} := \text{span}(Q)$ (column space), and define

$$P_{\mathcal{U}} := QQ^\top$$

as the orthogonal projection matrix onto the subspace $\mathcal{U}$.

We can easily verify its orthogonality: take any vector $u \in \mathcal{U}$ (where $u \in \mathbb{R}^p$) and any vector $x \in \mathbb{R}^p$. Since $u \in \text{span}(Q)$, we can express it as $u = Qy$, where $y \in \mathbb{R}^r$. Because Q satisfies the orthogonal constraint $Q^\top Q = I_r$, this r -dimensional coordinate vector y is completely unique (if we had two vectors such that $Qy_1 = Qy_2$, left-multiplying both sides by $Q^\top$ immediately gives $y_1 = y_2$). Then:

$$u^\top (x - P_{\mathcal{U}}x) = (Qy)^\top (x - QQ^\top x) = y^\top (Q^\top x - Q^\top QQ^\top x) = y^\top (Q^\top x - (Q^\top Q)Q^\top x) = 0.$$

Therefore, for any $u \in \mathcal{U}$, we have $u^\top (x - P_{\mathcal{U}}x) = 0$. This means the residual $x - P_{\mathcal{U}}x$ is orthogonal to $\mathcal{U}$, confirming that $P_{\mathcal{U}}$ is indeed the orthogonal projection onto $\mathcal{U}$.

Now, consider a composite mapping where the linear operator C acts on a vector in $\mathbb{R}^p$, and the result is projected back into $\mathcal{U}$:

$$\mathbb{R}^p \ni u \mapsto P_{\mathcal{U}}Cu \in \mathcal{U}.$$

Take any $u = Qy \in \mathcal{U}$, then:

$$\begin{aligned} P_{\mathcal{U}}Cu &= QQ^\top C(Qy) \\ &= Q(Q^\top CQ)y \\ &= QHy, \end{aligned} \tag{102}$$

where $H := Q^\top CQ$.

Equation (102) shows that, under the basis Q , the effect of the operator $u \mapsto P_{\mathcal{U}}Cu$ on the coordinate vector y is given exactly by the matrix H . Therefore, H precisely captures the effect of "applying C and then compressing the result back into the subspace $\mathcal{U}$ ". This is exactly what we mean by "compression matrix".

Here is a purely intuitive, non-rigorous geometric argument: an eigenvalue essentially relates to the resulting vector when C acts on u . Now, this transformed vector gets orthogonally projected into a subspace. Because of this projection, its length shrinks. Therefore, it is highly likely that the eigenvalues of H are smaller than those of C .

Next, we will prove this rigorously using the Min-Max theorem.

We will do this by: 1. Using the Rayleigh quotient to redefine this "shrinkage". 2. Establishing the relationship for the k -th largest eigenvalue. 3. Mapping any k -dimensional subspace $\mathcal{W}$ in $\mathbb{R}^r$ to a subspace $\mathcal{M}$ in $\mathbb{R}^p$, like $u = Qy$. This allows us to characterize the relationship between the eigenvalues of H and C from a subspace perspective, proving that $\mu_k \leq \lambda_k$.

Bounding Eigenvalues Using the Min-Max Theorem To firmly establish the numerical relationship between the eigenvalues μ_k of the compressed matrix $H = Q^\top C Q$ and the eigenvalues λ_k of the original matrix C , we will use the Courant-Fischer-Weyl (Min-Max)[1] theorem for a rigorous proof.

1. Definitions and Preliminaries

- $C \in \mathbb{R}^{p \times p}$ is a real symmetric matrix (defined earlier as $B^{-1/2} A B^{-1/2}$), with its eigenvalues sorted in descending order: $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p$.
- $Q \in \mathbb{R}^{p \times r}$ is an orthogonal column matrix satisfying $Q^\top Q = I_r$.
- $H = Q^\top C Q \in \mathbb{R}^{r \times r}$ is the symmetric compressed matrix, with its eigenvalues sorted in descending order: $\mu_1 \geq \mu_2 \geq \dots \geq \mu_r$.

2. Expressing λ_k with the Min-Max Theorem According to the Min-Max theorem, for a p -dimensional symmetric matrix C , its k -th largest eigenvalue λ_k can be written as:

$$\lambda_k = \max_{\substack{\mathcal{V} \subset \mathbb{R}^p \\ \dim(\mathcal{V})=k}} \min_{\substack{x \in \mathcal{V} \\ \|x\|=1}} x^\top C x \quad (103)$$

This formula tells us that λ_k is the maximum value of the "minimum Rayleigh quotient" across all possible k -dimensional subspaces $\mathcal{V}$. This means that if we look at **any** specific k -dimensional subspace $\mathcal{V}$, the minimum Rayleigh quotient inside it can never exceed this eigenvalue ceiling λ_k :

$$\min_{\substack{x \in \mathcal{V} \\ \|x\|=1}} x^\top C x \leq \lambda_k \quad (104)$$

3. Expressing μ_k with the Min-Max Theorem Similarly, for the r -dimensional symmetric matrix H , its k -th largest eigenvalue μ_k ($1 \leq k \leq r$) satisfies:

$$\mu_k = \max_{\substack{\mathcal{W} \subset \mathbb{R}^r \\ \dim(\mathcal{W})=k}} \min_{\substack{y \in \mathcal{W} \\ \|y\|=1}} y^\top H y \quad (105)$$

When $k = 1$, the subspace $\mathcal{W}$ is just a 1-dimensional line, meaning we are simply looking for the highest possible value across all directions. So, the largest eigenvalue μ_1 is just the global maximum of the Rayleigh quotient: $\mu_1 = \max_{\|y\|=1} y^\top H y$.

On the other hand, when $k = r$, the subspace $\mathcal{W}$ is the entire r -dimensional space itself. Since there is only one such space, the outer "maximum" choice is fixed, and we are simply left finding the absolute lowest value overall. Thus, the smallest eigenvalue μ_r is just the global minimum of the Rayleigh quotient: $\mu_r = \min_{\|y\|=1} y^\top H y$.

4. Subspace Mapping and Connecting the Inequalities To build a bridge between μ_k and λ_k , we map any k -dimensional subspace $\mathcal{W}$ from $\mathbb{R}^r$ into $\mathbb{R}^p$. Let's define the mapped subspace $\mathcal{M}$ as:

$$\mathcal{M} := \{Qy \mid y \in \mathcal{W}\} \subset \mathbb{R}^p \quad (106)$$

Because of the orthogonal constraint $Q^\top Q = I_r$ (which also means Q has full column rank), this mapping has the following perfect properties:

- **Preserves dimension:** $Q^\top Q = I$ makes the mapping $y \mapsto Qy$ injective. If $Qy = 0$, left-multiplying both sides by $Q^\top$ gives $Q^\top Qy = 0 \Rightarrow Iy = 0 \Rightarrow y = 0$. So, different y vectors won't collapse into the same vector, meaning the dimension won't drop. Therefore, $\dim(\mathcal{M}) = \dim(\mathcal{W}) = k$.
- **Preserves length:** For any unit vector $y \in \mathcal{W}$ (where $\|y\| = 1$), its mapped version $x = Qy$ has the exact same length: $\|x\|^2 = y^\top Q^\top Qy = y^\top I_r y = \|y\|^2 = 1$.
- **Preserves Rayleigh quotients:** The value of the objective function stays exactly the same: $y^\top Hy = y^\top (Q^\top CQ)y = (Qy)^\top C(Qy) = x^\top Cx$.

Because of these properties, the inner minimization problem in Equation (105) can be equivalently rewritten as minimizing over the subspace $\mathcal{M}$:

$$\min_{\substack{y \in \mathcal{W} \\ \|y\|=1}} y^\top Hy = \min_{\substack{x \in \mathcal{M} \\ \|x\|=1}} x^\top Cx \quad (107)$$

Since $\mathcal{M}$ is a completely valid k -dimensional subspace in $\mathbb{R}^p$, we know from Equation (104) that this minimum value must be less than λ_k :

$$\min_{\substack{y \in \mathcal{W} \\ \|y\|=1}} y^\top Hy \leq \lambda_k \quad (108)$$

This inequality holds true no matter which subspace $\mathcal{W} \subset \mathbb{R}^r$ we pick. Therefore, even if we check all possible $\mathcal{W}$ to find the maximum value on the left side (which gives us μ_k), it still cannot break the ceiling:

$$\mu_k \leq \lambda_k \quad (109)$$

D.1.3 Third Step: Determining the Range of the Eigenvalues μ

1. Building the joint matrix S and proving it is positive semi-definite First, let's consider the joint vector made from the difference residuals $R_{0n}^{(m)}$ and the lagged residuals $R_{1n}^{(m)}$. For the n -th observation in the m -th trial, we define the stacked vector $z_n^{(m)} \in \mathbb{R}^{2p}$:

$$z_n^{(m)} := \begin{pmatrix} R_{0n}^{(m)} \\ R_{1n}^{(m)} \end{pmatrix} \quad (110)$$

Based on this, we construct the joint moment matrix S . Using our previous formula for moment matrices, S can be explicitly written as the sum of the outer products of all samples:

$$S := \begin{pmatrix} S_{00} & S_{01} \\ S_{10} & S_{11} \end{pmatrix} = \frac{1}{N_{total}} \sum_{m=1}^M \sum_{n=1}^{N_m} z_n^{(m)} (z_n^{(m)})^\top \quad (111)$$

It is easy to see that S is positive semi-definite ($S \succeq 0$)

2. Using the Schur complement for non-negative definiteness According to Seber (2008) [25] regarding the properties of non-negative definite matrices (Proposition 14.5), if a partitioned symmetric matrix S is positive semi-definite and its main diagonal block S_{00} is positive definite, then its Schur complement with respect to S_{00} must also be positive semi-definite.

The Schur complement of S with respect to S_{00} is defined as:

$$S/S_{00} = S_{11} - S_{10}S_{00}^{-1}S_{01} \quad (112)$$

From the theorem mentioned above, we know $S/S_{00} \succeq 0$. Let's recall our definitions for matrix A and B :

$$B = S_{11}, \quad A = S_{10}S_{00}^{-1}S_{01} \quad (113)$$

Substituting these into the Schur complement inequality gives:

$$B - A \succeq 0 \quad \Longleftrightarrow \quad A \preceq B \quad (114)$$

3. Proving matrix A is positive semi-definite On the other hand, let's examine the properties of matrix A itself. For any non-zero vector $u \in \mathbb{R}^p$:

$$u^\top A u = u^\top (S_{10}S_{00}^{-1}S_{01})u = (S_{10}^\top u)^\top S_{00}^{-1}(S_{10}^\top u) \quad (115)$$

Let $w = S_{10}^\top u$. Because S_{00}^{-1} is a positive definite matrix, its quadratic form $w^\top S_{00}^{-1}w \geq 0$ is always true. Therefore:

$$A \succeq 0 \quad (116)$$

4. The final bounds for the eigenvalues Putting it all together, for any vector $z \in \mathbb{R}^p$, we have:

$$0 \leq z^\top A z \leq z^\top B z.$$

Let $z = \beta y$ (where $y \in \mathbb{R}^r$), and apply our constraint $\beta^\top B \beta = I_r$. We get:

$$0 \leq y^\top (\beta^\top A \beta) y \leq y^\top (\beta^\top B \beta) y = y^\top y.$$

Notice that $H(Q) = Q^\top C Q = \beta^\top A \beta$. So, for any $y \neq 0$, we have:

$$0 \leq \frac{y^\top H(Q) y}{y^\top y} \leq 1.$$

Let $x = \frac{y}{\|y\|}$, which means $\|x\| = 1$. Substitute $y = \|y\|x$ into the Rayleigh quotient:

$$\frac{y^\top H(Q) y}{y^\top y} = \frac{(\|y\|x)^\top H(Q) (\|y\|x)}{(\|y\|x)^\top (\|y\|x)} = \frac{\|y\|^2 (x^\top H(Q) x)}{\|y\|^2 (x^\top x)} = x^\top H(Q) x$$

Combining with Min-Max theorem, all eigenvalues of $H(Q)$ must fall within the range $0 \leq \mu_i(Q) \leq 1$.

Because of this, every term $(1 - \mu_i(Q))$ in Equation (94) is non-negative and strictly decreases as $\mu_i(Q)$ increases from 0 to 1. This turns the abstract goal of minimizing $\min_{Q^\top Q = I_r} \prod_{i=1}^r (1 - \mu_i(Q))$ into a strict, highly actionable objective—push the eigenvalues $\mu_i(Q)$ as high as possible until it hits λ_i

D.1.4 Fourth Step: Reverting to the Generalized Eigenvalue Problem and the Final Optimal Solution

1. From standard to generalized eigenvalue problem Using our previous notation, let the eigenvalues of the symmetric matrix C be sorted in descending order as $\lambda_1 \geq \dots \geq \lambda_p$, with corresponding orthonormal eigenvectors $w_1, \dots, w_p$ (meaning $Cw_i = \lambda_i w_i$ and $w_i^\top w_j = \delta_{ij}$).

Recall how we constructed matrix C (Equation 98):

$$B^{-1/2} A B^{-1/2} w = \lambda w \quad (117)$$

To return to our original variables, we use the substitution $w = B^{1/2} v$. Plugging this in:

$$\begin{aligned} B^{-1/2} A B^{-1/2} (B^{1/2} v) &= \lambda (B^{1/2} v) \\ B^{-1/2} A (B^{-1/2} B^{1/2}) v &= \lambda B^{1/2} v \\ B^{-1/2} A v &= \lambda B^{1/2} v \end{aligned} \quad (118)$$

and left-multiply both sides by $B^{1/2}$

$$\begin{aligned} B^{1/2} (B^{-1/2} A v) &= B^{1/2} (\lambda B^{1/2} v) \\ (B^{1/2} B^{-1/2}) A v &= \lambda (B^{1/2} B^{1/2}) v \\ A v &= \lambda B v \end{aligned} \quad (119)$$

This shows that through the inverse transformation, the eigenvalues λ and eigenvectors w of C correspond perfectly to the solutions of the original generalized eigenvalue problem $A v = \lambda B v$.

And we have:

$$v = B^{-1/2} w \quad (120)$$

2. Final proof for the eigenvalue bounds $\lambda \in [0, 1]$ Combining Equations (114) and (116), we established a key ordering between the matrices:

$$0 \preceq A \preceq B \quad (121)$$

For the generalized eigenvalue problem $Av = \lambda Bv$, since B is positive definite, we can left-multiply by $v^\top$ to get the Rayleigh quotient:

$$\lambda = \frac{v^\top Av}{v^\top Bv} \quad (122)$$

Based on the matrix ordering, we can determine the upper and lower bounds for λ :

- **Lower bound:** Since $A \succeq 0$, the numerator $v^\top Av \geq 0$. And because $B \succ 0$, the denominator $v^\top Bv > 0$. Therefore, $\lambda \geq 0$.
- **Upper bound:** Since $B - A \succeq 0$, for any vector v we have $v^\top (B - A)v \geq 0$, which expands to $v^\top Bv \geq v^\top Av$. Dividing both sides by the positive term $v^\top Bv$ gives $1 \geq \lambda$.

In summary, the generalized eigenvalues λ must fall in the range $0 \leq \lambda \leq 1$.

3. Finding the optimal solution Let

$$W_r := [w_1, \dots, w_r] \in \mathbb{R}^{p \times r}.$$

Clearly, $W_r^\top W_r = I_r$, and

$$W_r^\top C W_r = \begin{pmatrix} w_1^\top C w_1 & \cdots & w_1^\top C w_r \\ \vdots & \ddots & \vdots \\ w_r^\top C w_1 & \cdots & w_r^\top C w_r \end{pmatrix} = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_r \end{pmatrix} =: \Lambda_r, \quad (123)$$

where we used the facts that $Cw_j = \lambda_j w_j$ and $w_i^\top w_j = \delta_{ij}$.

On the other hand, for any valid matrix Q satisfying $Q^\top Q = I_r$, let the eigenvalues of $Q^\top C Q$ be $\mu_1 \geq \cdots \geq \mu_r$. Then:

$$|I_r - Q^\top C Q| = \prod_{i=1}^r (1 - \mu_i).$$

Since we previously proved the individual upper bounds $0 \leq \mu_i \leq \lambda_i \leq 1$ (for $i = 1, \dots, r$), we get:

$$|I_r - Q^\top C Q| = \prod_{i=1}^r (1 - \mu_i) \geq \prod_{i=1}^r (1 - \lambda_i) = |I_r - \Lambda_r| = |I_r - W_r^\top C W_r|.$$

Therefore, choosing $Q_* := W_r$ allows the objective function to reach its global minimum. This makes Q_* the optimal solution for our problem.

4. Establishing B -orthonormality We can use the orthonormality of the standard eigenvectors $\{w_i\}$ to derive the properties of the original generalized eigenvectors $\{v_i\}$. We know that the eigenvectors of C satisfy:

$$w_i^\top w_j = \delta_{ij} \quad (124)$$

Using our substitution $w = B^{1/2}v$, let's plug $w_i = B^{1/2}v_i$ and $w_j = B^{1/2}v_j$ into the equation above:

$$\begin{aligned} (B^{1/2}v_i)^\top (B^{1/2}v_j) &= \delta_{ij} \\ v_i^\top (B^{1/2})^\top B^{1/2}v_j &= \delta_{ij} \end{aligned} \quad (125)$$

Since B is a symmetric positive definite matrix, its square root $B^{1/2}$ is also symmetric, meaning $(B^{1/2})^\top = B^{1/2}$. Because $B^{1/2}B^{1/2} = B$, the equation simplifies to:

$$\begin{aligned} v_i^\top B^{1/2}B^{1/2}v_j &= \delta_{ij} \\ v_i^\top Bv_j &= \delta_{ij} \end{aligned} \quad (126)$$

This shows that the generalized eigenvectors $\{v_i\}$ (obtained from the inverse transformation of the standard eigenvectors) form an orthonormal basis when weighted by matrix B .

5. The optimal cointegration matrix From the transformation $Q = B^{1/2}\beta$, we can choose the optimal cointegration matrix as:

$$\hat{\beta} = B^{-1/2}Q_* = B^{-1/2}W_r = [B^{-1/2}w_1, \dots, B^{-1/2}w_r].$$

So, the column vectors of $\hat{\beta}$ are exactly the generalized eigenvectors corresponding to the r largest generalized eigenvalues (from Equations 123 and 119), as seen in Equation 120. Under the normalization constraints in Equations 94 and 126, this $\hat{\beta}$ provides a representative solution for the maximum likelihood estimation.

D.2 Explicit Formula for the Determinant Term

From the main text, we know the determinant of the estimated covariance matrix is given by:

$$|\hat{\Sigma}(\beta)| = |S_{00}| \frac{|\beta^\top (S_{11} - S_{10}S_{00}^{-1}S_{01})\beta|}{|\beta^\top S_{11}\beta|} \quad (127)$$

Let's recall our previous matrix definitions: $B = S_{11}$ and $A = S_{10}S_{00}^{-1}S_{01}$. Because we set the normalization constraint where the denominator becomes 1 (as established in Equation 94, where $\beta^\top B\beta = I_r$), the denominator entirely drops out as $|I_r| = 1$.

Looking at the numerator, the first term $\beta^\top B \beta$ becomes I_r . For the second term, based on our exact definitions of Q and C (see Equation 98), $\beta^\top A \beta$ is simply $Q^\top C Q$. Therefore, the complex fraction simplifies perfectly down to:

$$|I_r - Q^\top C Q| = \prod_{i=1}^r (1 - \mu_i) \quad (128)$$

When we plug in our optimal solution, these eigenvalues μ_i become our generalized eigenvalues λ_i . This gives us the clean product form:

$$|\hat{\Sigma}(\hat{\beta})| = |S_{00}| \prod_{i=1}^r (1 - \lambda_i) \quad (129)$$

Since we already strictly proved that $0 \leq \lambda_i \leq 1$, every $(1 - \lambda_i)$ term we multiply is a fraction between 0 and 1.

I used large language models as an auxiliary tool during the development of parts of the proofs above. They were used for brainstorming and suggesting relevant references. I then independently checked the relevant theorems, verified the mathematical arguments, and wrote the final proof in my own words.

D.3 Mapping Theoretical Derivations to the C++ Implementation

The following section maps the mathematical derivations from our proof directly to the actual computational implementation in the `johansen.cpp` file. The C++ code presented here is part of the `ca4eeg` package (Version 0.1.0), created by Marie Levakova for cointegration analysis of EEG data.

1. Constructing the Moment Matrices A and B The optimization process begins by calculating the foundational moment matrices from the residuals. As shown in the derivation preceding Equation 94:

$$B := S_{11}, \quad A := S_{10} S_{00}^{-1} S_{01} \quad (130)$$

In the C++ code, this corresponds to computing the cross-products using the `S` function and then inverting S_{00} :

```
// Lines 48-52 in johansen.cpp
S00 = S(R0,R0);
S11 = S(R1,R1);
S01 = S(R0,R1);
S10 = S(R1,R0);
S00_1 = inv(S00);
```

2. Constructing the Compressed Matrix C To transform the generalized eigenvalue problem into a standard one, we defined a "compressed matrix" C . Referencing Equation 98:

$$C := B^{-1/2}AB^{-1/2} \quad (131)$$

The implementation calculates the inverse square root of B (stored as N12) and then chains the matrix multiplications to form `solveMat`:

```
// Lines 58-59 in johansen.cpp
N12 = powmat(S11, -0.5);
solveMat = conv_to<mat>::from(N12)*S10*S00_1*S01*conv_to<mat>::from(
N12);
```

3. Solving the Eigenvalue Problem The theoretical step requires diagonalizing the matrix C to find its eigenvalues. As indicated by the matrix Λ_r in Equation 123:

$$W_r^T C W_r = \text{diag}(\lambda_1, \dots, \lambda_r) \quad (132)$$

The code uses the Armadillo library's `eig_gen` function to perform this decomposition and then sorts the results in descending order:

```
// Lines 66-74 in johansen.cpp
eig_gen(eigval_cx, eigvec_cx, solveMat);
vec eigval = real(eigval_cx);
mat eigvec = real(eigvec_cx);
eigvec = eigvec.cols(sort_index(eigval,"descend"));
eigval = eigval(sort_index(eigval,"descend"));
```

4. Reverting to the Optimal Solution $\hat{\beta}$ Finally, we transform the standard eigenvectors back to our original variables. Referencing the mapping in Equation 120:

$$\hat{\beta} = B^{-1/2}W_r \quad (133)$$

The C++ code executes this by multiplying $B^{-1/2}$ (N12) by the first r columns of the sorted eigenvector matrix:

```
// Line 87 in johansen.cpp
mat b_hat = conv_to<mat>::from(N12) * eigvec.cols(0,r-1);
```

E References

References

- [1] Wikipedia contributors, “Min-max theorem,” *Wikipedia, The Free Encyclopedia*. [Online]. Available: https://en.wikipedia.org/wiki/Min-max_theorem#Min-max_theorem [Accessed: 2026-03-19]
- [2] Montgomery, D. C., Peck, E. A., & Vining, G. G. (2021). *Introduction to Linear Regression Analysis*. John Wiley & Sons.
- [3] Neusser, K. (2016). *Time Series Econometrics*. Springer. Chapter 6.
- [4] P. Ding, *Linear Model and Extensions*, arXiv preprint arXiv:2401.00649, 2024.
- [5] Härdle, W. K., & Simar, L. (2012). *Applied Multivariate Statistical Analysis* (3rd ed.). Springer.
- [6] Higham, N. J. (2008). *Functions of Matrices: Theory and Computation*. Society for Industrial and Applied Mathematics (SIAM).
- [7] Wang, X.-J. (2025). *Theoretical Neuroscience: Understanding Cognition*. CRC Press.
- [8] Hirsch, M. W., Smale, S., & Devaney, R. L. (2013). *Differential equations, dynamical systems, and an introduction to chaos* (3rd ed.).
- [9] Levakova, M., Christensen, J. H., & Ditlevsen, S. (2022). Classification of brain states that predicts future performance in visual tasks based on co-integration analysis of EEG data. *Royal Society Open Science*, 9, 220621.
- [10] Puthusserypady, S. (2021). *Applied Signal Processing*. Boston–Delft: Now Publishers.
- [11] Madsen, H. (2007). *Time Series Analysis*. Chapman & Hall/CRC Texts in Statistical Science.
- [12] Wikipedia contributors, “Hilbert transform – Bedrosian’s theorem,” *Wikipedia, The Free Encyclopedia*. [Online]. Available: https://en.wikipedia.org/wiki/Hilbert_transform#Bedrosian's_theorem [Accessed: 2026-03-08].
- [13] Østergaard, J., Rahbek, A., & Ditlevsen, S. (2017). Oscillating systems with cointegrated phase processes. *Journal of Mathematical Biology*, 75, 845-883.
- [14] Øksendal, B. (2003). *Stochastic Differential Equations: An Introduction with Applications* (6th ed.). Springer.

- [15] Kloeke, J., & McKean, J. (2024). *Nonparametric Statistical Methods Using R* (2nd ed.). Chapman and Hall/CRC.
- [16] Quiroga-Martinez, D. R., He, T., Fernández-Rubio, G., et al. (2025). Brain dynamics of mental manipulation. *bioRxiv*. <https://doi.org/10.1101/2025.10.20.683458>
- [17] Perko, L. (2001). *Differential Equations and Dynamical Systems* (3rd ed.). Springer.
- [18] Luck, S. J. (2014). *An introduction to the event-related potential technique* (2nd ed.). MIT press.
- [19] Gardiner, C. W. (2004). *Handbook of Stochastic Methods: for Physics, Chemistry and the Natural Sciences* (3rd ed.). Springer.
- [20] Shumway, R. H., & Stoffer, D. S. (2025). *Time Series Analysis and Its Applications: With R Examples* (5th ed.). Springer.
- [21] Lachaux, J. P., Rudrauf, D., & Kahane, P. (2003). Intracranial EEG and human brain mapping. *Journal of Physiology-Paris*, 97(4-6), 613-628.
- [22] Lütkepohl, H. (2005). *New Introduction to Multiple Time Series Analysis*. Springer.
- [23] Hamilton, J. D. (1994). *Time Series Analysis*. Princeton University Press.
- [24] Gentle, J. E. (2024). *Matrix Algebra: Theory, Computations and Applications in Statistics* (3rd ed.). Springer, Cham. doi:10.1007/978-3-031-42144-0.
- [25] Seber, G. A. F. (2008). *A Matrix Handbook for Statisticians*. John Wiley & Sons.
- [26] Wikipedia contributors, “Fourier transform – Uncertainty principle,” *Wikipedia, The Free Encyclopedia*. [Online]. Available: https://en.wikipedia.org/wiki/Fourier_transform#Uncertainty_principle [Accessed: 2026-03-07].
- [27] Wikipedia contributors, “Weinstein–Aronszajn identity,” *Wikipedia, The Free Encyclopedia*. [Online]. Available: https://en.wikipedia.org/wiki/Weinstein%E2%80%93Aronszajn_identity [Accessed: 2026-03-09].
- [28] Wikipedia contributors, “Filter design – The uncertainty principle,” *Wikipedia, The Free Encyclopedia*. [Online]. Available: https://en.wikipedia.org/wiki/Filter_design#The_uncertainty_principle [Accessed: 2026-03-07].